



Facility Manual

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UNITED STATES MARINE CORPS
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BOX 555151
CAMP PENDLETON CA 92055-5151

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From: Air Traffic Control Facility Officer
To: All Facility Personnel

Subj: AIR TRAFFIC CONTROL FACILITY MANUAL

Ref: (a) NAVAIR 00-80T-114

1. Purpose. This manual is promulgated under the provisions of reference (a).
2. Cancellation. All previous editions of this manual are obsolete.
3. Effective Date. This manual is effective the date signed.
4. Scope. This manual is applicable to the air traffic controllers operating at Marine Corps Air Station (MCAS) Camp Pendleton. It is not intended to cover every contingency that may arise, nor every rule of safety and good practice. To achieve maximum value, the contents of all pertinent directives must be studied and understood.
5. Action. The Air Traffic Control Facility Officer is responsible for the administration, enforcement, and currency of this manual.


M. A. RODRIGUEZ

DISTRIBUTION

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MCIWEST

RECORD OF CHANGES

Log completed change action to this manual.

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LIST OF ABBREVIATIONS AND ACRONYMS

ADC	Arrival/Departure Control
AFLCS	Airfield Lighting Control System
AGL	Above Ground Level
ALS	Approach Lighting System
AOM	Airfield Operations Manual
ARP	Airport Reference Point
ATCF	Air Traffic Control Facility
ATCFO	Air Traffic Control Facility Officer
ATCMD	Air Traffic Control Maintenance Division
CALA	Combat Aircraft Loading Area
CD	Clearance Delivery
CEB	Controller Evaluation Board
CFR	Code of Federal Regulations
CWT	Consolidated Wake Turbulence
DOD	Department of Defense
DZ	Drop Zone
ELP	Emergency Landing Pad
FAA	Federal Aviation Administration
FACDIR	Facility Directive
FC	Final Control
FD	Flight Data – In Tower Branch
FOD	Foreign Objects and Debris, or Foreign Object Damage
FWO	Facility Watch Officer
GCA	Ground Controlled Approach
GC	Ground Control
HERO	Hazards of Electromagnetic Radiation to Ordnance
HOLF	Helicopter Outlying Field
IMC	Instrument Meteorological Conditions
LOA	Letter of Agreement
LC	Local Control
LQS	Local Qualification Standard
LTG	Lesson Topic Guide

LZ	Landing Zone
MCAS	Marine Corps Air Station
MCB	Marine Corps Base
MOU	Memorandum of Understanding
MSL	Mean Sea Level
MTF	MACCS Training Form
NAATSEA	Naval Airspace and ATC Standards and Evaluation Agency
NITEX	Night Vision Device Training
NORAD	North American Air Defense Command
NORDO	No Radio
NWS	National Weather Service
ODO	Opposite Direction Operations
OJT	On-the-Job Training
OJTI	On-the-Job Training Instructor
OTS	Out of Service, or Over-the-Shoulder
PAPI	Precision Approach Path Indicator
PEB	Procedures Evaluation Board
PI	Primary Instructor
RATCF	Radar Air Traffic Control Facility
RD	Flight Data – In Radar Branch
RTS	Returned to Service
RS	Radar Supervisor
SCT	Southern California TRACON
SID	Standard Instrument Departure Procedure
STARS	Standard Terminal Automation Replacement System
SUA	Special Use Airspace
SVFR	Special VFR
TRACON	Terminal Radar Approach Control Facility
TTL	Training Team Leader
TS	Tower Supervisor
TSO	Training and Standardization Officer
VMC	Visual Meteorological Conditions
WCLC	Watch Checklist Complete

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Chapter 1. Facility Introduction

Section 1. General

1-1-1. PURPOSE

This manual provides facility personnel with administrative and operational guidance governing the operation of the Marine Corps Air Station Camp Pendleton Air Traffic Control Facility.

1-1-2. APPLICABILITY

This manual applies to all air traffic control personnel assigned to MCAS Camp Pendleton. This manual is not intended to supersede any portion of FAAO JO 7110.65 (series), or applicable Navy directives that govern air traffic control procedures. All members of the ATC facility will adhere to the provisions contained within this manual and shall exercise their best judgment should they encounter situations not specifically covered.

1-1-3. CANCELLATION

MCAS Camp Pendleton Air Traffic Control Facility Manual, March 2024.

1-1-4. PROCEDURES FOR CHANGE

- a. The ATCFO is responsible for keeping this manual updated with current procedures, changes, and directives.
- b. Recommended changes shall be submitted through the TSO to the ATCFO for final approval.
- c. Procedures Evaluation Board (PEB)
 - 1. PEB's shall be conducted as outlined in the NAVAIR 00-80T-114.
 - 2. The TSO or Training Chief shall be the chairperson of the PEB.
- d. This manual shall be reviewed and updated annually. In reviewing this manual, all applicable letters of agreement shall be reviewed for accuracy and conformance with current policy. In conjunction with the annual review, Branch Chiefs will incorporate any necessary updates or changes to the simulator training program. Additions or deletions to this manual will be issued as facility directives, facility memorandums and facility manual change notices.

1-1-5. CREW BRIEFS

- a. FWOs and/or Crew Chiefs shall conduct a thorough crew briefing at least 30 minutes prior to assumption of the watch. The oncoming FWO shall receive a recorded briefing from the standing FWO to ensure an accurate pass down prior to the crew brief commencing.
- b. This briefing shall include:
 - 1. Current and forecasted weather
 - 2. Current and projected airport conditions and anticipated operations
 - 3. Equipment status
 - 4. New facility directives and current NOTAMS
 - 5. Special operations

6. Operating positions and training assignments

REFERENCE-

NAVAIR 00-80T-114 Para 4.7.5, Watch Team Briefing

1-1-6. FACDIR BINDERS AND INFORMATION BOARD

a. Hard copies of current facility directives are maintained in the facility manual binder within each branch. Digital files are also maintained in the ATC Digital Library located on Microsoft Teams, channel name 'ATC Publications'. These directives are published to outline details of any new procedures not yet incorporated into the Facility Manual.

b. The information board is in the brief room. This board contains information of general interest to all personnel such as duty assignments, administrative notes, etc.

1-1-7. UNIFORM OF THE DAY

The uniform of the day is prescribed by the MCAS Commanding Officer.

1-1-8. FACILITY FIELD DAY

Field day shall be accomplished weekly unless otherwise directed. Crew Chiefs will ensure that proper field day is completed prior to close of business (Tower, Radar, brief room, admin offices, head across from radar room, front entry way of building 2399).

1-1-9. VISITORS

The control tower and radar facilities are designated restricted areas. No photographs are allowed without permission of the MCAS Commanding Officer. Visitors may be allowed with the consent of the ATCFO or designated representative. Visitors will sign the visitor log located in the ATC brief room and always remain with their sponsors to preclude any interference with ATC operations.

1-1-10. TELEPHONE USAGE

The telephones in both the radar room and tower are to be used for official business only. Under no circumstances will these phone numbers be given out to anyone who does not work in ATC. Telephonic threat forms are in the TS / RS binders. Any request for information concerning any incident will be directed to the Public Affairs Office (PAO) at 725-5011. At no time will operational information be released without approval of the FWO, nor should personal information be given out. This includes work schedules, personal phone numbers, etc.

1-1-11. USE OF PEDS

The use of Portable Electronic Devices (PEDs) is not authorized while signed on position. All devices shall be in non-transmitting mode in the tower cab and radar room or turned completely off. Government-issued PEDs are authorized, but use must be kept to a minimum as to not become a distraction.

REFERENCE-

NAVAIR 00-80T-114 Para 4.9.2, Portable Electronic Device Usage in ATC Operational Spaces.

1-1-12. LEAVE AND LIBERTY

a. Leave requests will be submitted to the SNCOIC via the Crew Chief/designated representative through Marine Online at least five working days in advance of the leave date and shall be accompanied by a HARP form. All leave requests shall be discussed with the Crew Chief for potential staffing or training impacts.

b. Civilian leave requests will be submitted to the ATCFO or TSO via DAI five working days in advance. For emergency or sick leave, notify the ATCFO or TSO as soon as able.

c. Students shall take extra consideration when submitting leave while training at percentages of 75% and higher. These leave requests will be approved by the ATCFO on a case-by-case basis. If the leave period will significantly negatively impact the students training or qualification timeline, the student should expect to defer the leave period to a later date. This shall not apply to emergency leave, Christmas, or religious holidays.

d. Liberty is in accordance with the watch schedule or as authorized by the crew officer.

e. During normal working hours, emergency leave will be handled by the Crew Chief or designated representative.

1-1-13. WATCH SCHEDULE

The facility operates on a rotating crew schedule. The watch schedule, crew assignments, and civilian work schedules will be maintained on the facility Teams page and distributed electronically.

1-1-14. WATCH CONDUCT

a. Food and drink are authorized in the tower and radar room provided the following is adhered to:

1. Drinks shall be in spill proof containers.

2. Food is authorized in the tower and radar room only if personnel shortages dictate.

b. Smoking is only authorized in designated areas at least 50 feet from building entrances and air intake ducts and will not be in areas commonly used by nonsmokers. The term “smoking” includes the use of e-cigarettes and all new and emergent tobacco products such as cigarettes, cigars, little cigars, pipe tobacco, roll-your-own tobacco, and electronic nicotine delivery systems.

c. Use of television is not permitted in either control room.

1-1-15. AIR TRAFFIC CONTROL AWARDS

a. Controller of the Quarter

1. Recognition of skilled professionals is essential. The primary avenue for recognizing outstanding performance within the ATC facility is the Controller of the Quarter board.

2. A Controller of the quarter board will convene at the beginning of April, July and October, and mid-November. The ATC SNCOIC is responsible for assembling the board and forwarding a recommendation to the ATCFO for final selection. Examples of achievements that warrant nomination are contained in the reference. The following procedures apply to the nomination and selection process:

- (a) All air traffic control personnel assigned to this command during the quarter are eligible.
 - (b) Period of consideration is the first through the last working day of the quarter. Nominations shall be submitted five working days prior to the convening date of the board.
 - (c) Any air traffic controller may submit a nomination; however, it shall be submitted via the Crew Chief/Crew Officer.
 - (d) Nominations shall be made in writing and forwarded via the chain of command. All nominations shall contain a detailed summary of the nominee's actions.
 - (e) Selection will be based on the action summary and individual training record.
- b. Selectees from each quarter will compete for the distinction of Controller of the Year, MCAS Camp Pendleton. The Controller of the Quarter Board, fourth quarter will make the Controller of the Year recommendation to the ATCFO to be considered for the Admiral Robert B. Pirie Naval Air Traffic Controller of the Year Award.

REFERENCE-

NAVAIR 00-80T-114 Para 4.4.4, Annual Awards

1-1-16. MEDICAL REQUIREMENTS

All controllers are required to maintain ATC medical standards. Medical up-chits shall be obtained and provided to the training department upon initial check-in to the ATC Facility and annually during the controller's birth month.

REFERENCE-

NAVAIR 00-80T-114 Para 4.6.2.3, Physical Qualification and Examination

1-1-17. OFF-DUTY EDUCATION AND EMPLOYMENT

On a case-by-case basis, the ATCFO may authorize controllers to attend off-duty education classes and / or work a part-time job. Requests should be forwarded through the chain of command to the SNCOIC.

1-1-18. FACILITY HEADSET POLICY

- a. Prior to starting OJT, controllers will be issued a headset. The serial number will be recorded and maintained by the training department. Upon checkout (DeFAP, PCA, PCS, resignation), controllers will turn in their headset to the training department.
- b. Controllers shall be held accountable for the proper care and maintenance of this equipment.

Chapter 2. Administration

Section 1. General

2-1-1. COMMAND VISION / MISSION

MCAS Camp Pendleton provides operating forces and tenant organizations with continuous and effective aviation service and support; ensuring their ability to respond to present and emerging national defense requirements. We utilize dedicated professionalism, a can-do work ethic and innovative technology to attain the best possible support services, better enabling our customers to accomplish their mission.

2-1-2. AIR TRAFFIC CONTROL MISSION

The mission of the Air Traffic Control Facility is to provide for the safe, orderly, and expeditious movement of air traffic, including aircraft movement on the ground, to and from the airport, and within assigned airspace, and to interface with the National Airspace System.

2-1-3. GUIDING PRINCIPLES

- a. Our mission should reflect in all that we do.
- b. Our controllers are our most important assets. We provide a professional working environment that ensures the dignity of our controllers, encourages them to develop their skills, promotes job pride and satisfaction, and rewards initiative and improvement.
- c. Listen to your seniors *and* subordinates. Encourage participation in decision-making. Promote and recognize teamwork in our efforts.
- d. Don't be afraid of change. Look for better ways to get the job done. If current policies and processes don't make sense, ask questions, and make suggestions for improvement.
- e. Focus on quality; don't settle for good enough. Provide the best service you can, and train to produce the best controllers possible to support the Air Station and the MOS. If something is broken, start fixing it.
- f. Provide training opportunities; support the concept of "train for the future."
- g. Personally lead, and reward excellence.

2-1-4. CUSTOMER SERVICE

Provide the best possible service and information to the aircrews we serve. No controller in this facility, regardless of rank, shall be anything but respectful, courteous, and professional. If pilots / aircrews are abusive or have unrealistic requests, attempt to professionally approve or deny the request, then notify the chain of command.

2-1-5. FUNCTIONS AND RESPONSIBILITIES

The Marine Corps Air Station provides arrival and departure control, GCA radar service, and tower control service to aircraft operating within Class D and E airspace. In addition, flight planning and approval services are conducted by Marine Corps personnel. Southern California TRACON (SCT), by letter of agreement, has delegated the airspace known as Lima and Golf (see Appendix G) to MCAS Camp Pendleton. This airspace is controlled by GCA and provides IFR approach

control and departure control services to aircraft operating to / from the air station and Fallbrook Airpark.

2-1-6. LOCATION AND TOPOGRAPHY

MCAS Camp Pendleton is located six miles north of the city of Oceanside, California, and situated on the 028 degree radial at 5 nautical miles from the Oceanside VORTAC. The latitude and longitude for the Airport Reference Point (ARP) is 33°18'04.53" north, 117°21'18.61" west. The field elevation is 78 feet above sea level. The airfield is located within mountainous terrain.

2-1-7. CLIMATOLOGY

The airport is situated in a valley, oriented northeast to southwest approximately 7 miles long and 2 ½ to 3 miles wide with surrounding foothills at an elevation of 100 to 500 feet. The weather in the Camp Pendleton area is mainly controlled by the Pacific Ocean to the west and the Santa Margarita mountains to the east. The ocean tends to moderate temperatures in the lower layers of the atmosphere due to the sea breeze, while the mountains to the east act as a barrier between the desert and coastal area thus giving a drying effect to the easterly flow. The relationship between these air masses creates a semi-permanent low-level inversion over the local area that creates conditions conducive to the formation of low stratus and fog. However, during the winter these conditions are occasionally interrupted by frontal passage and highly modified air masses.

2-1-8. CLASS D AIRSPACE

MCB has delegated the airspace identified as Class D and E to MCAS as described in the reference. The MCAS Camp Pendleton Class D airspace is the circular airspace extending from the surface up to and including 2600 feet MSL. This airspace extends from the ARP to the following points: the 010 bearing at 4NM clockwise to the 227 bearing at 4NM direct to the 240 bearing at 2.6NM clockwise to the 355 bearing at 2.6NM direct to the 010 bearing at 4NM (See also [Appendix C](#)).

REFERENCE-

FAAO JO 7400.11, Airspace Designations and Reporting Points

2-1-9. CLASS E AIRSPACE

The MCAS Camp Pendleton Class E airspace extends from the surface up to and including 2600 feet MSL. This airspace is a continuation of Class D extending from the 010 bearing at 4 NM direct to the 018 bearing at 6.9 NM direct to the 041 bearing at 6.9 NM direct to the 050 bearing at 4 NM (See also [Appendix C](#)).

2-1-10. GCA AIRSPACE

Lima and Golf airspace is delegated to MCAS Camp Pendleton by SCT and authorized for use as described in the references below.

REFERENCES-

Coordination Procedures Letter of Agreement between MCB Camp Pendleton and MCAS Camp Pendleton

2-1-11. SPECIAL USE AIRSPACE

Hazards to aircraft in flight mandate that the airspace overlying Camp Pendleton be designated as Restricted. The airspace has been identified as R-2503A, B, C and D.

a. The boundaries of R-2503A, B, C and D encompass all airspace above MCB and MCAS. R-2503A extends from the surface to 2,000 feet MSL. R-2503B borders R-2503A and extends from the surface to 15,000 feet MSL. R-2503C, when activated through Los Angeles ARTCC, is located above the northwestern portion of R-2503B. It has a base of 15,000 feet MSL and an upper limit of FL270. R-2503D overlies R-2503A and extends from 2,000 feet MSL to 11,000 feet MSL. The controlling agency for R-2503A, B, C and D is Los Angeles ARTCC.

b. The using agency for R-2503A, B, C and D is MCB Camp Pendleton Longrifle. See [Chapter 8](#) for the assigned frequencies for Longrifle.

2-1-12. SOUTHERN CALIFORNIA TRACON (SCT)

The Federal Aviation Administration's Southern California Terminal Radar Approach Control Facility has the responsibility for IFR operations in the terminal area. SCT has delegated airspace identified as Lima and Golf to MCAS Camp Pendleton for the purpose of providing air traffic control service. SCT is located in San Diego adjacent to MCAS Miramar.

2-1-13. ALTERNATE AIRFIELDS

- a. Primary helicopter divert airport:
Carlsbad / McClellan – Palomar (CRQ)
NFG R-135 @ 9.9 NM
Rwy 6/24, 4897 x 150
Tower frequency 118.6, 276.4
- b. Primary fixed-wing divert airport:
MCAS Miramar (NKX)
NFG R-138 @ 26.5 NM
Rwy 24L/6R, 8001 x 200
Rwy 24R/6L, 12,000 x 200
South (jet) tower frequency 135.2, 298.925
North (helo) tower frequency 327.025

2-1-14. ADJACENT AIRFIELDS

- a. Oceanside Municipal (OKB)
NFG R-136 @ 3.5 NM
Rwy 6/24, 2712 x 75, asphalt
Unicom frequency 122.725
- b. Fallbrook Airpark (L18)
NFG R-040 @ 8.5 NM
Rwy 18/36, 2160 x 60, asphalt
Unicom frequency 123.05
- c. French Valley (F70)
NFG R-020 @ 22 NM
Rwy 18/36, 6000 x 75, asphalt
Unicom frequency 122.8

2-1-15. AIRFIELD DESCRIPTION

a. RUNWAYS. Runway 21 is 6005 feet long by 200 feet wide. It is the precision approach and calm wind runway and is identified with standard runway markings. It has a threshold elevation of 77 feet MSL. Runway 3 has a threshold elevation of 61 feet MSL and is a non-precision runway. Runway 21 and runway 3 have overruns of 1000 feet by 200 feet.

REFERENCES-

AOM; IFR Supplement

b. HELO SPOTS.

1. Papa 1 - Located on Papa taxiway adjacent to Alpha taxiway.
2. Papa 2 - Located on Papa taxiway adjacent to Bravo taxiway. (*Not authorized for use*)
3. Papa 3 - Located on Papa taxiway (across from the control tower) between Bravo and Delta taxiways.
4. Papa 4 - Located on Papa taxiway between Delta and Echo 1 taxiways.
5. Echo 2 - Located on Echo 2 taxiway.
6. Foxtrot 2 - Located on Foxtrot 2 taxiway.
7. Emergency landing pad (ELP) – Located between Echo 2 and Foxtrot 2 taxiways.

c. TAXIWAYS.

1. Papa taxiway is located southeast of and parallel to the runway. It is linked to the runway by Alpha, Bravo, Delta, Echo 1, and Foxtrot 1 taxiways.

NOTE-

Due to the proximity of parking spots to Papa taxiway, aircraft with wingspans larger than the C-130 wingspan of 133 feet are prohibited from taxiing on Papa south of Alpha. Aircraft such as C-17s, C-5s, and B-747s have wingspans greater than the width of taxiway Papa and pose a hazard to parked aircraft.

2. Charlie 1 and Charlie 2 taxiways link the arming and CALA areas to the runway. (See [Airfield Diagram](#))
3. Echo 2 and Foxtrot 2 taxiways link the runway to the hot fuel pits area. (See [Airfield Diagram](#))
4. Bravo, Delta, Foxtrot 1, and Papa taxiways are 100 feet wide.
5. Alpha and Echo 1 taxiways are 75 feet wide.
6. Charlie taxiways 1 and 2 are each 40 feet wide.
7. Echo 2 and Foxtrot 2 taxiways are 150 feet wide.

(a) All aircraft require prior coordination before permission is granted for taxi to the base of the control tower (VIP parking). Flight Clearance (Beast 03) will contact Ground Control to assist aircraft in parking.

(b) Load exercises / freight areas are normally located adjacent to the MALS-39 hangar.

d. LANDING AND NAVIGATIONAL AIDS (NAVAIDS). Some of these items are depicted on the airfield diagram (See also [Appendix C](#)).

1. *Oceanside VORTAC.* Channel 100, frequency 115.3, identifier OCN

2. *ATIS*. Frequency 285.45

3. *TACAN*. Channel 55, identifier NFG

4. *Precision Approach Path Indicator (PAPI)*.

(a) Runway 21: left side, approach end.

(b) Runway 3: right side, approach end.

5. Other landing aids include lighted windsocks located at the approach and departure ends and mid-field.

e. *ORDNANCE AREAS*. Locations are depicted on the airfield diagram in Appendix D. Ordnance operations will be conducted in accordance with the AOM.

REFERENCE-
AOM

1. Combat Aircraft Loading Area (CALA): Aircraft loading and unloading area.

(a) Consists of 12 numbered pads

(b) Designated as the Explosive Cargo Staging Area (ECSA).

(c) Explosive ordnance, i.e., bombs, rockets, missiles, guns, and special weapons training devices will be loaded in the CALA only. MK76 and MK106 (inert and dummy ordnance) can be loaded on the aircraft line.

(d) Hazardous cargo loading or unloading shall be conducted in the CALA.

(e) ATC procedures for CALA operations are contained in [Chapter 5](#) of this manual.

2. Arming Area. The arming area located at the northeast end of the airfield is used to arm or de-arm ordnance on aircraft. ATC procedures for arming operations are contained in [Chapter 5](#) of this manual. There are 6 designated arming pads.

3. Flight line. Aircraft para-flares may be loaded on squadron flight lines under proper supervision. Gun ammunition, 7.62mm ammunition cans only, may be loaded aboard the aircraft on the squadron flight line.

f. *FUELING AREAS*.

1. Hot fuel pits are located at the southwest end of the airfield (See also [Appendix C](#)). The fueling area has eight points. Access to the pits is via Foxtrot 2 taxiway and exit is via Echo 2. Aircraft with live ordnance aboard will not be permitted in the hot fuel pits.

EXAMPLE-

(*Callsign*), TAXI AHEAD FOR FUEL.

2. Cold refueling will be conducted in designated aircraft parking areas. It may be conducted in the CALA.

3. At no time will aircraft be fueled or defueled and uploaded or downloaded at the same time.

2-1-16. AIRFIELD WEATHER CRITERIA

a. *BASIC VFR MINIMA*. VFR minima are established as a ceiling not less than 1000 feet AGL and/or visibility not less than 3 statute miles.

b. LESS THAN VFR MINIMA. When the reported ceiling and/or visibility are less than basic VFR minima, all flight operations shall be conducted in accordance with SVFR/IFR procedures. SVFR clearance requires a minimum of 1 statute mile visibility and clear of clouds while in the Class D surface area. Minima for IFR departures and landings are contained in appropriate radar and instrument approach procedure publications.

c. SVFR MINIMA. The minima depend on the route requested or the operation being conducted. See [Paragraph 5-1-7](#).

REFERENCE-

US Code of Federal Regulations, Part 91

d. AIRFIELD MINIMA. The minima for arrivals are the published approach ceiling and visibility for the approach being conducted. Minimums for departure are specified by NAVAIR 00-80T-114. If weather conditions are below the minima for the approach requested, the controller will ensure that the pilot has the current official reported weather. If the pilot still wants the approach, a clearance will be issued by the controlling authority. An entry will be made in the Facility Log and an Incident report generated. This report will be forwarded to Airfield Operations via the ATCFO for determination. See [Chapter 2, Section 3](#) of this manual for the handling of operational hazards.

REFERENCE-

CNAF M-3710.7 Para 5.3, Instrument Flight Rules and Positive Control Procedures

2-1-17. AIRFIELD OPS DEPARTMENT

a. TRANSIENT SERVICES.

1. VISITING AIRCRAFT LINE (VAL). This section is located in the Flight Clearance office. The services include aircraft parking utilizing Beast 11. Aircraft hangar space for transient aircraft is not available. Tie down parking spaces are extremely limited. Prior Permission is Required (PPR) from Airfield Operations for transient aircraft. Exceptions to the PPR requirement are emergency aircraft and transient aircraft being hosted by a tenant squadron. Transient civil aircraft and military charter aircraft shall possess a MCAS Camp Pendleton facility license or be a member of a recognized military flying club.

2. MAINTENANCE. The airfield does not have an aircraft maintenance section. Transient aircraft may be provided with limited services during normal working hours by tenant units.

b. FLIGHT PLANNING. Complete flight planning and clearance service facilities are available during normal airfield hours. Flight Clearance has been assigned a UHF frequency of 279.3.

c. AIRCRAFT RESCUE AND FIRE FIGHTING (ARFF). ARFF at Camp Pendleton is Category II. Categories are determined by size and type of aircraft on the line. Camp Pendleton can handle all aircraft. The basic purpose of the ARFF section is to provide crash rescue service to all aircraft, regardless of type, which may land at Camp Pendleton. ARFF is manned on a 24-hour basis and posts immediate response vehicles during field hours. ARFF has the secondary function of assisting the base fire department as necessary.

d. WEATHER. The weather section is part of the National Weather Service (NWS) system. They prepare and disseminate local area weather reports to the NWS and to pilots during pilot weather briefings. Weather services are provided locally during airfield hours and at other times are available through MCAS Miramar on METRO frequency 342.4.

e. *ATC MAINTENANCE DIVISION (ATCMD)*. ATCMD is responsible for the maintenance of NAVAIDS, radar, and communications equipment for the facility. Technicians are available during field hours.

2-1-18. SEARCH AND RESCUE (SAR)

a. SAR procedures are outlined in the AOM. SAR will be activated by the Operations Officer or the Deputy Operations Officer. The unit responsible for SAR operations is the Activities Operations Center, San Diego (Coast Guard) at (619) 295-3121 (for emergencies only).

b. Air traffic controllers are not authorized to approve or disapprove SAR requests.

2-1-19. SALVAGE

Salvage operation requirements are contained in the AOM.

Section 2. Billet Descriptions

2-2-1. ATC FACILITY OFFICER (ATCFO)

a. Responsible to the Operations Officer.

b. Responsibilities include:

1. Administration and operation of the ATC Facility.
2. Coordination with local FAA representatives, local base officials, and other official representatives on air traffic control and airspace matters.
3. Establishment of Standard Operating Procedures relating to personnel, equipment, training, and qualification.
4. Coordination with the Air Traffic Control Maintenance Division (ATCMD) for maintenance support necessary to ensure efficient operation.
5. Determination of the qualifications of all ATC personnel.

c. Additional duties and responsibilities as specified by NAVAIR requirements.

REFERENCE-

NAVAIR 00-80T-114 Para 4.1.3, Billet Descriptions, USMC

2-2-2. ASSISTANT ATC FACILITY OFFICER (AATCFO)

Assists the ATCFO in operation of the ATC Facility. Additional duties and responsibilities are specified by NAVAIR requirements.

REFERENCE-

NAVAIR 00-80T-114 Para 4.1.3, Billet Descriptions, USMC

2-2-3. TRAINING AND STANDARDIZATION OFFICER (TSO)

a. Assists the ATCFO in the training and standardization of facility ATC programs and procedures. Responsibilities and duties of the TSO include:

1. Preparing tests to evaluate the results of scheduled training. A record of the test results shall be recorded on the trainee's electronic training evaluation.

2. Submitting comprehensive reports and charts to the ATCFO as required.

b. Developing electronic forms such as position logs, total sheets, qualification sheets, training evals, etc.

c. Chairing both the CEB and PEB.

REFERENCE-

NAVAIR 00-80T-114 Para 4.1.3, Billet Descriptions, USMC

2-2-4. FACILITY WATCH OFFICER (FWO)

a. The FWO shall be designated in writing by the ATCFO.

b. The FWO shall be in the facility at all times it is in operation.

- c. The FWO is responsible to the ATCFO for the overall quality of ATC services provided by crew personnel.
- d. The FWO is responsible for identifying and coordinating ATCF requirements relating to personnel, training, equipment, and supply.
- e. The FWO is responsible for ensuring currency and proficiency on rated controllers is maintained for personnel assigned to their crew.
- f. The FWO may be assigned to a supervisory position in either branch but should not be assigned to a control position, although the FWO may receive OJTI.

REFERENCE-

NAVAIR 00-80T-114 Para 4.1.3, Billet Descriptions, USMC

2-2-5. ATC FACILITY SNCOIC

- a. Responsible to the ATC Facility Officer.
- b. Performs the duties as outlined in the reference to include:
 - 1. Supervision of all assigned enlisted personnel including counseling, guidance, and dissemination of information.
 - 2. Supervision of administrative work including reports required by higher authority.
 - 3. Liaison with station headquarters on all personnel matters.
 - 4. Advising the ATC Crew Chiefs on administrative tasks, training commitments, and personnel responsibilities.

REFERENCE-

NAVAIR 00-80T-114 Para 4.1.3, Billet Descriptions, USMC

2-2-6. ATC TRAINING CHIEF

- a. Shall have all ATCS ratings.
- b. Responsibilities will be to assist the TSO in:
 - 1. Planning, executing, and supervising the ATC facility training, certification, and standardization programs.
 - 2. Maintenance of the on-the-job training program for air traffic controllers.
 - 3. Maintaining ATC certification and qualification records including MSHARP.
 - 4. Preparing designation letters and qualification letters.
 - 5. Conducting classroom training on local area, ATC equipment and procedures, as required.
 - 6. Developing local course material, visual aids, and control problems to supplement other published material such as the FAA refresher series.
 - 7. Coordinating with the TSO in the preparation of a monthly training schedule. This schedule will be supplemented with ATC information of an immediate nature, such as ATC publication changes, OPNAV instructions, etc. The schedule will also include Instrument Ground School (IGS) schedules for each month. The training schedule shall be retained for one year.

8. Maintain the indoctrination program for newly assigned air traffic controllers. The purpose of this requirement is to acquaint the new arrival with other agencies with which the trainee will be working. This can be in the form of a checklist and should include a visit to and briefing on all other ATC facilities, flight clearance, and the weather section.

9. Ensure the effectiveness and currency of ATCF directives and technical libraries.

REFERENCE-

NAVAIR 00-80T-114 Para 7.2.1.2, Training Chief

2-2-7. TOWER CHIEF

- a. Shall be designated in writing by the ATCFO.
- b. Shall perform the duties of FAA Examiner in accordance with FAA directives.
- c. Maintain a current library of facility directives and other pertinent regulations pertaining to control tower operations.
- d. Maintain tower equipment including recording of outages and action taken to correct discrepancies.
- e. Maintain operational continuity between various watch teams.
- f. Ensuring currency of controllers.
- g. Evaluate the operational readiness of branch equipment.
- h. Provide technical assistance to the ATCFO in development of procedures.
- i. With the TSO, select each operating position's training team.

REFERENCES-

FAA Order 7220.1, Certification and Rating Procedures

NAVAIR 00-80T-114 Para 9.1.2.1, Control Tower Chief

2-2-8. RADAR CHIEF

- a. Shall be designated in writing by the ATCFO.
- b. Shall perform the duties of ATCS Examiner in accordance with the reference.
- c. Maintain a current library of facility directives and other pertinent regulations pertaining to radar operations.
- d. Maintain radar equipment including recording of outages and action taken to correct discrepancies.
- e. Maintain operational continuity between various watch teams.
- f. Ensuring the currency of controllers.
- g. Evaluate and recommend to the ATCFO the operational readiness of branch equipment.
- h. Supervise FAA/military flight checks.
- i. Provide technical assistance to the ATCFO in development of procedures.
- j. With the TSO, select each operating position's training team.

REFERENCE-

NAVAIR 00-80T-114 Para 10.1.2.1, Radar Chief

2-2-9. TOWER SUPERVISOR (TS)

- a. Coordinates and directs the control of air traffic within assigned airspace and surface vehicular traffic operating on or near runways, taxiways, and other designated areas of the airfield.
- b. Briefs the control tower watch on weather conditions, traffic, equipment status, and field conditions and special evolutions prior to assuming the watch.
- c. Assigns personnel to operating positions during the shift according to individual qualifications and training requirements as directed.
- d. Notifies appropriate SAR agencies of aircraft in distress and directs the use of the Crash Phone.
- e. Advises the Tower Chief about maintenance of the current library of facility directives and regulations in the control tower.
- f. Maintains tower facility logs to include recording of outages.
- g. Maintains operational continuity of the watch.
- h. Ensuring a tower equipment checklist is completed at the beginning of each shift.
- i. Provides technical assistance to the ATC FWO in the performance of control tower operational procedures.

REFERENCES-

NAVAIR 00-80T-114 Para 9.1.2.2, Tower Supervisor
JO 7210.3, Facility Operation and Administration

2-2-10. RADAR SUPERVISOR (RS)

- a. Coordinates and directs the control of air traffic within assigned airspace.
- b. Briefs the radar watch on weather conditions, traffic, equipment status, and field conditions prior to assuming the watch.
- c. Assigns personnel to operating positions according to individual qualifications and training requirements as directed.
- d. Ensuring a radar equipment checklist is completed at the beginning of each shift.
- e. Provides technical assistance to the Radar Chief in development of procedures.
- f. Advises the Radar Chief about maintenance of the current library of facility directives and regulations in the radar room.
- g. Maintains radar facility logs to include recording of outages.
- h. Maintains operational continuity of the watch.
- i. Provides technical assistance to the ATC FWO in the performance of radar operational procedures.

REFERENCES-

NAVAIR 00-80T-114 Para 10.1.2.2, Radar Supervisor
JO 7210.3, Facility Operation and Administration

2-2-11. DESIGNATED EXAMINER

a. Responsible to the ATCFO.

b. Responsibilities include:

1. Examining controllers to ascertain proficiency on all positions. This will be accomplished with over-the-shoulder evaluations and tape talks during each controller's birth month.
2. Issuing FAA credentials.
3. Initiating transfer/withdrawal process in the credentialing system for outgoing controllers.
4. Reviewing training material and procedures.
5. Examining and qualifying controllers on all positions designated.

REFERENCES-

NAVAIR 00-80T-114 Appendix S, Administering Credentials

FAAO 7220.1C Certification and Rating Procedures for DoD Personnel

2-2-12. CREW OFFICER

a. Responsible to the ATCFO and TSO.

b. Responsibilities include:

1. Ensuring that all required ATC training is accomplished daily, to include live and simulator training.
2. Reviewing current training material for accuracy and reporting discrepancies to the TSO.

2-2-13. CREW CHIEF

a. Responsible to the ATC SNCOIC concerning personnel issues, and FWO concerning daily operations.

b. Responsibilities include:

1. Addressing all military matters that concern the personnel assigned to their crew.
2. Assisting the Tower and Radar Watch Supervisors in the overall operation of the facility.
3. Ensuring that if any controller's physical or mental health appears questionable, they are relieved from ATC duties and referred to a flight surgeon.

2-2-14. TRAINING TEAM LEADER (TTL)

a. The TTL is directly responsible to the training department.

b. Shall be an operationally current supervisor in the branch involved.

c. Review the trainee's training history prior to that individual starting OJT. All training team members shall observe the trainee perform tasks in a simulated environment prior to starting OJT.

d. Ensure all phases of training are conducted in accordance with the training plan.

e. Obtain assessment from other training team members to ensure a continuous, objective evaluation of progress during training.

f. Provide input on trainee progress to the Crew Officer or Crew Chief for training meetings.

g. Identify the need to improve performance and recommend to the ATCFO through the CEB process, as needed:

1. Continuation of OJT, or
2. Suspension or termination of OJT

h. In concert with the PI, recommend the trainee for qualification when that trainee demonstrates they meet the competencies required to operate that position under general supervision.

REFERENCE-

NAVAIR 00-80T-114 Para 7.2.5.1, Training Team Leader

2-2-15. PRIMARY INSTRUCTOR (PI)

a. Shall have MOS 7257 and be designated as a Basic Instructor (BI) prior to appointment. Appointment will be in writing and signed by the ATCFO or designated representative.

b. Directly responsible to the TTL.

c. Conduct the majority of OJT for the trainee.

d. Track the trainee's current level of proficiency throughout the training process.

d. Ensure trainees are prepared for and complete scheduled examinations in accordance with exam milestones.

e. Complete detailed daily and weekly evaluations in accordance with this manual.

f. Provide detailed input regarding progress, deficiencies, or corrective action to the CEB on behalf of the trainee.

g. In concert with the TTL, recommend the trainee for qualification when that trainee demonstrates that he or she meets the competencies required to operate that position under general supervision.

REFERENCE-

NAVAIR 00-80T-114 Para 7.2.5.1, Training Team Leader

2-2-16. TRAINEE

a. Directly responsible to the PI.

b. Attain qualifications within prescribed training timelines. Take the initiative to engage in the learning process through self-study to ensure satisfactory training progression and qualification.

c. Trainees are required to get a minimum of 2 hours position time daily.

d. Provide feedback to the TTL and PI throughout the training cycle which may assist in meeting timeline goals.

REFERENCE-

NAVAIR 00-80T-114 Para 7.2.5.3, Trainee

2-2-17. FACILITY CLERK

a. Responsible to the ATC SNCOIC.

b. Responsibilities include clerical support as directed.

Section 3. Supervisory Information

2-3-1. SUPERVISOR RATINGS

Fully qualified controllers are considered for this position as the needs of the facility dictate. These designations are granted only after demonstrated skills indicate that it is justified. The following shall be completed prior to designation.

- a. Successful completion of the appropriate supervisor syllabus.
- b. For Tower Supervisor designations, completion of at least 10 hours day and 10 hours night position training and at least 5 hours productive familiarization training in the radar facility.
- c. For Radar Supervisor designations, completion of 20 hours of position training and at least 5 hours productive familiarization training in the tower. Radar Supervisor candidates should first be qualified on Range Air.
- d. Completion of all requirements on the MTF.
- e. Candidates are required to demonstrate a comprehensive knowledge of branch and facility management by passing a board that challenges their understanding of Federal, NAVAIR, and local rules and regulations.
- f. Candidates will have demonstrated a high level of maturity and professionalism.

2-3-2. RELIEF BRIEF

Controllers and supervisory personnel shall use position and watch relief checklists as part of all relief briefings. The checklists are contained in [Appendix A](#) and shall be recorded using the ETVS Brief button.

2-3-3. ESCAT

The Emergency Security Control of Air Traffic (ESCAT) is an emergency preparedness plan that prescribes the joint action to be taken by appropriate elements of the DoD, the DOT, and the DHS in the interests of national security to control air traffic under emergency conditions.

REFERENCE-

DODINST 5030.36, Emergency Security Control of Air Traffic

2-3-4. LETTERS OF AGREEMENT

Letters of agreement are used to establish control and coordination procedures between facilities. There are several inter- and intra-facility LOAs in use at MCAS Camp Pendleton. They are updated as procedures change.

2-3-5. EMERGENCIES

Because of the infinite variety of possible emergency situations, specific procedures cannot be prescribed. However, when you believe an emergency exists or is imminent, select and pursue a course of action which appears to be the most appropriate under the circumstances and which most nearly conforms to the instructions in this section. The following is a list of some of the emergencies common to tenant aircraft. This information is provided to help controllers better assist the pilot. Controllers normally only declare an emergency on the pilot's behalf if the aircraft is NORDO and an obvious emergency exists.

a. Hydraulic emergency:

1. UH-1/AH-1. Flight control hydraulic system failures will result in degradation of aircraft control. In addition, the pooling of hydraulic fluid in the aircraft poses a chance of fire. A single system failure will result in some degradation of aircraft control, but the aircraft will remain controllable in most normal flight regimes and conditions. Aircrew will normally request a straight-in approach to the runway and landing on the runway. A dual or two system flight control hydraulic systems failure is an acute problem for the aircrew because the condition significantly reduces the aircrew's capacity to control the aircraft. Aircrew will normally request a long straight-in approach and perform a sliding landing to the runway. Aircrew will normally keep the aircraft descent rate low, between 200 – 300 feet per minute and between 40 and 60 knots. Aircraft will require tow.

2. Fixed wing. Hydraulic system failures affect several systems for the aircrew; there will be some degradation in aircraft handling (like losing power steering in a car). More time is required for aircraft to lower landing gear and go through landing checklist. Pooling of hydraulic fluid in the aircraft poses a chance of fire. Aircrew will normally request a visual check of the aircraft's landing gear by the controller. Aircraft will require a tow.

b. Chip warning light. This indicates the presence of foreign metal matter in either the engine, transmission, or other gear boxes. The aircrew may request to land as soon as possible due to possible aircraft engine, transmission, or gear box failure. It should be noted that in some cases this type of indication is not to be considered an actual emergency. Confirm with the pilot if they desire to declare an emergency.

c. Fire / fire warning light / smoke in cockpit. This is an acute emergency that requires the aircrew's full attention to perform multiple tasks in a short period of time. Obtain the aircrew's intentions and ETA when able. Most fires are electrically related so communications may be lost momentarily while the pilot secures non-essential electrical systems. Aircrew will normally request to land as soon as possible.

d. Single engine failure. (Dual engine equipped aircraft) or loss of engine power. Loss of one engine or engine power may affect the aircrew's ability to maintain level flight and may limit the capacity to maneuver the aircraft. Aircrew will adjust the approach and landing speeds as necessary or required to make a safe landing. Aircrew may request a straight-in approach to the runway for a sliding or running landing. Aircrew may declare an emergency. Aircraft may need to be towed. Be prepared to assist as necessary.

e. Hot brakes. Brakes applied too hard or too often may result in "hot brakes". Brake drums may seize, or tires may be worn flat on one side resulting in a blown tire. This may result in loss of control or damage from flying debris. ARFF has fans capable of cooling the drums. Controllers should be observant for smoke from tires indicating potential brake lock-up or blown tires. An area for cooling of hot brakes will be determined by the ARFF section leader.

f. Stuck gear. MV-22/CH-53 Aircrew that experiences a bad gear indication in the cockpit will notify ATC and may declare an emergency if unable to troubleshoot. Aircraft that experience landing gear malfunctions will proceed inbound to the ELP. The aircraft will hover until ARFF transports equipment to the ELP for the aircraft to land upon. The aircraft may request to enter the delta pattern while coordination is made to execute an emergency landing. It is not uncommon for

squadron personnel responsible for the aircraft to proceed underneath while hovering and manually pull the stuck gear down and pin it into place. Additional ELP information is available in the AOM.

2-3-6. HUNG / UNEXPENDED ORDNANCE

Hung ordnance is considered to be any practice or live ordnance carried externally that would not separate from the aircraft when release was attempted. Unexpended ordnance (no attempt was made to release) may be any practice or live forward firing ordnance carried externally that is returned to the ordnance area (CALA). Hung ordnance will be treated as an emergency. See [Paragraph 5-5-11](#) for procedures.

2-3-7. HERO CONDITIONS

In the event of an explosives mishap involving an ordnance carrier along an explosives route, the appropriate HERO UNSAFE / UNRELIABLE ORDNANCE Condition will be set by the Airfield Operations Officer (during duty hours) or the Station Duty Officer (during non-working hours). Broadcast the HERO Condition and location on all frequencies and enter the information on the ATIS. Normal separation distance from the accident site is 150 feet.

2-3-8. COURSE RULES BRIEFS

Normally, Branch Chiefs will conduct the course rules presentations. Tower or radar supervisors, may also be required to make presentations. Personnel assigned to this task will attend two such presentations prior to representing this facility at either the Instrument Ground School or classroom instruction. Presentations are normally held on the first Friday of each month. Briefing dates for the Instrument Ground School (IGS) are scheduled by HMT-303 DOSS.

2-3-9. OPERATIONAL HAZARDS

Any hazard or suspected error observed including but not limited to a loss of applicable separation, non-compliance with an ATC instruction or clearance, or aircraft / vehicle encroachment upon a movement area or within airspace under the control of ATC.

a. The following minimum documentation will be submitted to the ATCFO:

1. Operational Hazard Report

2. If the Hazard concerns an intruder aircraft, a diagram of the observed flight path entering and/or exiting Class D surface area.

b. Controller Initials - All controllers will give their operating initials to any pilot upon request over the frequency or phone. The individual will not give their name under any circumstance. If such a request is made, the ATC office will be notified as soon as possible with the reasons for such a request.

2-3-10. CONTROLLER MISHAP STATEMENTS

Controller mishap statements are submitted whenever requested by the FWO, TS or RS. Controller mishap statements are part of an accident / incident package and shall be as accurate as possible. Guidance for completing these statements is contained in NAVAIR 00-80T-114.

REFERENCE-

NAVAIR 00-80T-114 Para 4.10.2.1, ATC Hazard Reports

2-3-11. MISHAPS AND HAZARDS

In the event of a mishap or hazard the FWO shall be notified. The FWO is responsible for completing actions required by NAVAIR 00-80T-114. Mishap and hazard binders to complete required actions are located in each branch.

REFERENCE-

NAVAIR 00-80T-114 Para 4.10.2.1, ATC Hazard Reports

2-3-12. FACILITY LOGS

The tower and radar supervisors are responsible for their respective facility log. The log shall always be kept up to date and will reflect accurate and detailed information. NAVAIR 00-80T-114 contains a list of required entries.

REFERENCE-

NAVAIR 00-80T-114 Para 4.3.1, Daily Operations Log

2-3-13. TRAFFIC COUNT PROCEDURES

A tabulation of the day's traffic shall be accomplished in accordance with established guidelines when the airport closes or at midnight. Tower and GCA traffic count will be recorded in accordance with NAVAIR 00-80T-114.

REFERENCE-

NAVAIR 00-80T-114 Appendix I, Air Traffic Activity Report

2-3-14. AIRFIELD HOURS

Airfield hours are published on the Daily Brief Sheet. Extensions of hours that exceed 30 minutes and / or early closings require approval from the Airfield Operations Officer or their designated representative. FWOs may authorize an extension of no more than 30 minutes. Tower and Radar supervisor responsibilities are contained in [Paragraphs 5-6-1](#) and [6-5-1](#). Occasionally, the operations department will publish airfield hours outside those normally scheduled. If all operations are complete an hour or more earlier than the scheduled closing time, supervisors / FWOs should coordinate with flight clearance to close the airfield early.

Chapter 3. Safety

Section 1. General

3-1-1. GENERAL

The risk of electrocution and the dangers of a darkly lit radar room require a heightened sense of safety consciousness while working in ATC. Upon check-in, and annually thereafter, personnel are briefed on safety procedures, firefighting, evacuation routes, and CPR.

3-1-2. ELECTROCUTION

The potential shock hazard while working with radar equipment is significant. Should a victim be receiving a “holding” shock, secure the high voltage immediately via the main circuit breaker. If the high voltage cannot be shut off, remove the victim observing the following precautions:

- a. Protect yourself with dry insulating material.
- b. Use a rope or cane from an authorized safety board, a buckle-less belt, or clothing such as shirt or trousers to free the victim from the live wire.

WARNING-

Do not touch the victim until completely clear of the shock hazard.

- c. After securing power or freeing the victim, call for emergency assistance and administer CPR as necessary.

3-1-3. FACILITY CONTINGENCY PLAN

- a. Upon recognition of an event requiring facility evacuation, the ATCFO or FWO shall:
 - 1. Contact the tower and / or radar supervisor(s) as applicable, stating the triggering event and to execute evacuation procedures.
 - 2. Direct and ensure the prompt evacuation of all appropriate areas within the facility.
 - 3. Identify and establish a mobile / remote communication position for post-evacuation resumed control.

EXAMPLE-

“We have received a credible threat of an underway aerial drone attack directed towards the tower, execute evacuation procedures. Direct personnel and mobile comm assets to the H&HS parking lot for resumed control. (Initials).”

- 4. Forward the appropriate information pertaining to the event and actions taken to the MCAS Operations Officer, SDO, and emergency response units as applicable (ex, ARFF, hospital, PMO).
- b. Upon evacuation procedures being initiated,
 - 1. Tower supervisor (if applicable) shall,
 - (a). Evacuate all personnel within their area of responsibility.
 - (b). To the maximum extent possible, attempt to resolve all safety conditions relating to aircraft prior to evacuation, dependent on the nature of the event.

- (c). Ensure aircraft under their control are informed that ATC services are terminated.

EXAMPLE-

“Attention all aircraft, ATC tower being evacuated. ATC services are terminated, transmit intentions on 340.2.”

- (d). Ensure all departure clearances are canceled.

(e). Ensure the local and ground standalone URC-200 radios and (2) light guns are transported with evacuated personnel. Upon reaching the remote communications position, attempt to contact any remaining aircraft or vehicles.

- (f). If able, broadcast over the ATIS that ATC services are not being provided.

EXAMPLE-

“Air traffic control services are not available within Camp Pendleton airspace. Transmit intentions on three four zero point two.”

(g). Contact the ATCMD duty technician and request (2) mobile AN/GRC-258s for primary and secondary transmissions, along with any other requested assets.

2. Radar supervisor (if applicable) shall,

- (a). Evacuate all personnel within their area of responsibility.

(b). Notify SCT of evacuation and all pertinent details. Ensure the handoff of all IFR aircraft to SCT.

- (c). Transfer Lima / Golf airspace to SCT.

(d). Instruct VFR aircraft and aircraft conducting approaches under VMC that radar service is terminated. Depending on the situation, direct aircraft to either contact Camp Pendleton Tower or depart the area VFR.

EXAMPLES-

“Radar service terminated. Squawk 1200 / VFR, contact Camp Pendleton tower on 340.2 / 128.775. (Give reason if time permits)”

“Radar service terminated. Squawk 1200 / VFR, exit / remain outside Class Delta, frequency change approved. (Give reason if time permits)”

(e). Contact the ATCMD duty technician and request (2) mobile AN/GRC-258s for primary and secondary transmissions, along with any other requested assets.

3. ATCMD Maintenance Officer or duty technician (as applicable) shall,

(a). Provide and set up (2) mobile AN/GRC-258s at the position described by the ATC supervisor.

(b). Assist to the maximum extent possible with additional equipment requests from the ATC supervisor.

c. After completing a remote communications site set up at the designated position, attempt to regain communications with any aircraft remaining in Camp Pendleton’s airspace. Issue appropriate instructions and advisories as necessary, informing aircraft of the limited capabilities.

d. The facility shall remain secured until the ATCFO authorizes the site to be manned.

e. Supervisors may execute evacuation procedures without direction from the ATCFO or FWO if, in their opinion, they become aware of a situation indicating a major system failure, natural disaster, or other facility evacuation event that directly incapacitates the facility's ability to provide air traffic services and requires evacuation. If a supervisor initiates a facility evacuation, they shall inform the ATCFO, FWO, and other branch supervisor with pertinent details as soon as able.

3-1-4. FIRE BILL

Fire bills are posted in the radar room, control tower, break room, classrooms, and office spaces. Personnel utilizing these spaces shall be familiar with each. Evacuation of facilities will be accomplished in accordance with [Paragraph 3-1-3](#) d and e.

REFERENCE-
MCO P11000.11

3-1-5. CLEANLINESS

- a. Personal belongings shall be stored neatly in appropriate areas to prevent safety / trip hazards. Belongings shall not block passageways or impair ventilation.
- b. Daily cleanup will be conducted by each crew prior to relief.
- c. Facility field day is conducted weekly and / or at the discretion of the SNCOIC.

Chapter 4. Training Plan

Section 1. Training and Certification

4-1-1. TRAINING PROGRAM

The goal of the ATC training program is to produce the highest-quality, most proficient controllers while reducing qualification times and trainee downtime. The program is built on a crawl / walk / run design and should be regularly measured, analyzed, and improved. Critical elements of the program include academic instruction, simulation training, familiarization training, and on-the-job training (OJT). The program contains the minimum requirements to produce either a Local Controller (7252) or Arrival / Departure Controller (7253). These qualifications also prepare the Marine for service in the Fleet Marine Force. The Training and Readiness (T&R) Manual requires all Marines to earn two qualifications (GC & FD or FC & RD) within the first 12 months of arriving at their first ATC Facility; Marines are also required to earn NMOS 7252 or 7253 within 3 years of arriving at their first facility. Only higher headquarters can approve waivers to these standards. Position training is accomplished in three phases. The training team shall record the beginning and end of each training phase on the trainee's eval.

a. ACADEMIC AND SIMULATION PHASE. Trainees will complete all position tests and receive simulations to reinforce the material. Required simulations must be completed for each test prior to taking the next test. Trainees shall not proceed to the next phase without completing this training phase. During this training phase, students shall take at least one test and complete all associated simulations each week. Three hours of simulation (or 50 simulated approaches) are required during each week of this training phase.

b. ON-THE-JOB FAM (OJF) PHASE. This phase of training should be no longer than one week in duration. Trainees shall receive at least ten hours of familiarization training. RFC trainees should get OJF on as many approaches as possible during the week. Training received during this phase will not count against a student's OJT timeline.

c. ON-THE-JOB TRAINING (OJT) PHASE. Trainees shall receive a minimum of two hours per day on position and at least three simulator sessions per week. RFC trainees shall receive 10 simulations per day. PIs and trainees share the responsibility to comply with this requirement.

NOTE-

To the maximum extent possible, non-ATC training should be scheduled on a non-interference basis with position training.

4-1-2. CONTROLLER TRAINING PLAN

a. Prior to the start of training, the training team shall establish a training plan for each new student. The training plan shall be reviewed at least once a month, and shall include the following:

1. Simulation requirements
2. OJF plan
3. Maximum training time
4. Skill check schedule
5. Tape talk schedule

6. Projected qualification date
7. Training team responsibilities
8. Individual training needs

4-1-3. ORIENTATION COURSE

Newly assigned controllers will usually be assigned to their crew upon checking in to the unit. They will be given an orientation syllabus and training materials. The orientation course will take about a week to complete. The goal is to have the trainee familiar with tower and radar operations prior to starting position training. After completing the orientation course, trainees will be assigned to their first position based on the needs of the facility and open training pipelines.

4-1-4. SIMULATOR TRAINING

Simulation should represent a minimum of 20% total training hours. Requirements for simulator times and approaches are listed in [Table 4-1](#). In addition, simulators shall be used for the following (see [Table 4-2](#)):

- a. Maintaining proficiency on operating positions per [Paragraph 4-1-18](#).
- b. Maintaining currency on final control during periods of insufficient live traffic.

4-1-5. PERSONAL DOCUMENTATION

All controllers shall keep their FAA credential card available at all times and shall be able to present them when directed. Medical Clearance Notice (DD Form 2992) along with the Airman's Written Test (AWT) form will be kept in individual training jackets. It is the controller's obligation to advise facility leadership of changes in medical status relating to their ability to perform ATC duties.

4-1-6. LOCAL QUALIFICATION STANDARD (LQS)

LQSs and Lesson Topic Guides (LTGs) provide the minimum knowledge factors required for position qualification. Every trainee shall bring a personal copy with them when signing on position and is expected to study the source materials in preparation for training and qualification. OJT instructors will only sign items after a trainee satisfactorily demonstrates (by testing and simulation) a thorough knowledge of the material. All testing and associated simulations shall be completed prior to trainees commencing position OJT.

4-1-7. TESTING PROGRAM

- a. Once all subject matter pertaining to the appropriate LQS / LTG has been signed off by a training team member (PI or TTL), the trainee may take the corresponding test. Each line item in the LQS must be signed off individually by an OJT/PI/TTL. After completing the test, the test proctor will review the results with the trainee and document the test date and score in the training evaluation and MACCS Training Form (MTF).
- b. A minimum score of 80% is required to successfully pass all written examinations. Should any trainee fail a written examination, a make-up examination will be given no sooner than 24 hours

afterward. A subsequent failure will result in a 48-hour lockout. Each student is allowed three attempts per test. However, failure of two consecutive examinations or three exams throughout the training program may require a CEB for unsatisfactory performance. If a student fails a test, the maximum score of 80% will be recorded on the retake and the failure shall be included on the MTF.

- c. A final test will be administered for all operating positions.
- d. At a minimum, students training on a junior position will take one test every week until test complete; students training on major positions are permitted two weeks per test. Testing time limits begin once a student receives their LQS.
- e. Trainees will complete tests within 90 days of beginning training on position. Non-prior qualified controllers will be required to retest if the time between testing and position time exceeds 90 days.

4-1-8. TRAINING EVALUATIONS

- a. The training evaluation is a record of training used toward certification while signed on an ATC position or while training on simulators or in a classroom. A training evaluation is required for each trainee for each operating position.
- b. Separate training evaluations will be completed for each position on which the trainee trained during the week (i.e., training conducted on RD and ADC during the same week). Each phase of training (testing / sim, familiarization, OJT) will be documented in the appropriate section of the evaluation.
- c. After completion of training for the week, the training team shall document all training information on the evaluation; this shall be completed no later than Friday. The Branch Chiefs and TSO shall review and comment on the evaluation by the first workday of the following week.
- d. Training evaluations are maintained in electronic format. Security of the documents will be the responsibility of the TSO, Training Chief and Branch Chief.
- e. At a minimum, the following information is required on training evaluations:
 - 1. All relevant trainee information including crew assignment and trainee priority (number 1, 2 or 3 trainee).
 - 2. Actual position time totals or number of total approaches for RFC per week. Simulated approaches or times will also be entered in the appropriate space.
 - 3. Workload and complexity.
 - 4. All performance marks. It should be noted that moderate traffic for an advanced trainee may be heavy traffic for a new trainee.
 - 5. All needs improvement and/or unsatisfactory marks shall be annotated numerically in the appropriate space with a brief explanation.
 - 6. All OJTI comments shall be strictly professional and shall emphasize constructive criticism.
 - 7. Both the OJTI and trainee shall type their operating initials immediately following their comments.

8. Trainee comments are encouraged and may be made after being debriefed on the training session.

9. The evaluation session shall be documented as soon as possible after the training session.

10. Training evaluations shall be stored in the ATC Training folder on Teams.

11. Weekly comments shall reflect only constructive remarks and should provide a reasonable narrative of the work completed by the trainee. Comments shall not include subjective remarks like, “should be recommended soon,” “waiting to see more traffic,” or similar remarks.

12. Evaluations for tower or radar supervisor trainees should be maintained. In addition, training times will be logged on the branch supervisor’s position log.

13. By the end of the training week, the PI, TTL, trainee and Crew Chief will review and initial all appropriate evaluations.

14. “NTOP” or “other military duties” notes shall be accompanied with a justification.

f. Additional guidance for completing the training evaluation is contained in NAVAIR 00-80T-114.

REFERENCE-

NAVAIR 00-80T-114 Appendix N, ATC Training Evaluation Report

4-1-9. RECOMMENDATION FOR QUALIFICATION

Recommendation for qualification may be initiated by any OJTI. However, it must have the concurrence of a branch supervisor. One of the recommending controllers must be a civilian controller. Both controllers recommending qualification shall initial the corresponding block on the general tab of the eval.

4-1-10. SKILL CHECKS AND OVER-THE-SHOULDERS

a. Annual examinations for supervisors and annual over-the-shoulder checks for all qualified controllers will be conducted during each individual’s birth month. Annual over-the-shoulder checks are required for all positions (except TS / RS) and must be conducted by a Branch Chief, the Training Chief, or designated civilian. Failure of an annual over-the-shoulder will be handled in accordance with NAVAIR 00-80T-114. For controllers not getting required position time in accordance with [Table 4-2](#), currency over-the-shoulders are required prior to working unmonitored. Currency over-the-shoulders must be conducted by the Training Chief or Branch Chief. Controllers who require two consecutive monthly currency over-the-shoulders shall retake the final exam for each position requiring a currency over-the-shoulder.

REFERENCE-

NAVAIR 00-80T-114 Para 7.4.3 Annual Skill Check

b. Failure of any qualification skill check requires appropriate remediation as directed by the Branch Chief (normally not to exceed 30 days) prior to the trainee being recommended again for certification. A CEB may also be required to determine whether a subsequent qualification check will be given. Under no circumstances will more than two qualification checks be given to any trainee.

c. If a trainee is qualified while completing a performance skill check, that check must be conducted by the Training Chief or Branch Chief.

4-1-11. MAXIMUM TOTAL TRAINING TIME

The maximum total training times in [Table 4-1](#) are the time limits used in determining the trainee's 50, 75, and 100% levels of authorized training.

4-1-12. TAPE TALKS

Tape talks shall be conducted in accordance with NAVAIR 00-80T-114. The training team shall conduct tape talks on trainees at the 50% level for initial trainees. Tape talks are also required annually for each controller.

REFERENCE-

NAVAIR 00-80T-114 Para 6.2.2.5, Tape Talk

4-1-13. PERFORMANCE SKILL CHECKS

a. When a trainee reaches 50, 75, and 100% of the maximum allotted training time, the Training Chief, Branch Chief or Assistant Branch Chief will complete a performance skill check. The evaluation should present an overall picture of the trainee's progress, deficiencies, and a recommendation for qualification, to continue training, to begin skill enhancement training (SET), or to conduct a CEB.

b. For trainees who fail to qualify within the facility's maximum time, but it is determined that they can qualify within specified additional time, the ATCFO may authorize an extension. This additional time shall not exceed 120% of the facility's maximum without higher approval.

4-1-14. CONTROLLER EVALUATION BOARD (CEB)

a. Responsible for evaluating training progress, controller performance, and providing the ATCFO with recommendations for appropriate action.

b. Members of the board will be selected by the training office, and will usually consist of the following:

1. TSO and / or Training Chief
2. Branch Chiefs
3. Training Team Leader (TTL) – non-voting member.
4. Crew leadership – non-voting member
5. Four voting members will be designated by the board chairman.

c. A CEB may be conducted when any of the following take place:

1. ATCFO directs
2. Less than satisfactory performance on a skill check
3. Academic deficiencies exist
4. Trainee progress is questionable and/or unsatisfactory at any point in training

d. The board will formulate and provide a recommendation to the ATCFO concerning the trainee's performance and potential for success. The ATCFO will review the board's recommendation and make a final decision. The trainee, training team, and board members will be notified concerning the outcome.

4-1-15. SKILL ENHANCEMENT TRAINING (SET)

The purpose of SET is to improve an individual's knowledge, skills, and abilities. The Branch Chief or Training Chief may assign SET at any time in student training, typically due to an unsatisfactory performance skill check.

- a. For a trainee, SET may be assigned to enhance specific skills, techniques, or knowledge to assist in their success with the training program.
- b. SET may also be assigned to individuals certified on a position based upon supervisor observations of their performance.
- c. SET may not be used to correct deficient performance that resulted in decertification.
- d. SET must be documented on a training evaluation.

4-1-16. OJTI REQUIREMENTS

Prior to providing instruction on an ATC position, instructors shall have earned MOS 7257, be designated as Basic Instructors (BI) and meet the following requirements:

a. *INITIAL OJTI DESIGNATION*

- 1. 30 days have passed since qualification
- 2. OJTI course completion and documentation
- 3. Recommendation by a branch supervisor
- 4. Concurrence of the Branch Chief

b. *SUBSEQUENT OJTI DESIGNATION.* Newly assigned personnel who have had a prior OJTI designation may be designated locally after completing the OJTI checklist and obtaining the necessary approvals.

c. Controllers who have been away from the facility for extended periods of time automatically regain their OJTI designation after successful completion of currency requirements, as depicted in [Paragraph 4-1-17](#).

d. The Training Chief and Branch Chiefs have the authority to waive OJTI training sessions for prior designated OJTIs.

4-1-17. CURRENCY REQUIREMENTS

Controllers are required to maintain currency on each position for which they hold a qualification. Specific currency requirements are listed in [Table 4-2](#).

a. Controllers who fail to maintain currency requirements shall receive an OTS check as outlined in [Paragraph 4-1-10](#). Failure of an OTS will be handled in accordance with NAVAIR 00-80T-114.

- b.** Controllers who do not meet monthly requirements must get their OTS as soon as possible the following month.
- c.** Final controllers shall not conduct final approaches during IFR conditions unless they have controlled a minimum of five approaches in the preceding calendar month.
- d.** For extended absences from the facility of greater than 60 days but less than 90 days, returning controllers will be required to pass the final examinations for all positions qualified on before completing a currency over-the-shoulder check. If the controller fails a final examination, they will be required to pass all tests for that position prior to recommendation for qualification.
- e.** If 90 days or more have elapsed since currency on any position, returning controllers are required to earn a recommendation and pass both the final examination and a skill check. If the controller fails the final examination, they will be required to pass all tests for that position prior to recommendation for qualification.
- f.** If more than 180 days have passed since currency on any position, complete testing and retraining is required.
- g.** Test scores are only valid for 90 days prior to starting OJTI for the associated position.

REFERENCE-

NAVAIR 00-80T-114 Para 6.6.2, Currency Requirements

4-1-18. PROFICIENCY TRAINING

Annual proficiency classes and briefings cover topics required by NAVAIR, NAATSEA and subjects of interest to the facility. Additional annual training as directed from higher. These classes are assigned and/or given by the Training Chief.

4-1-19. MONTHLY CURRENCY TRACKING

The Branch Chiefs are responsible for tracking currency in their respective branch and informing the controllers with enough time to prevent a lapse in currency. The Training Chief will be made aware of any continued lapses in currency.

4-1-20. RANGE AIR STUDENT TRACKING

All Range Air students who have been training for a period greater than two months shall be briefed weekly to the ATCFO and Training department by their training team to provide an update on their status.

Any Range Air student who has been awaiting a check ride for a period greater than five days shall notify the ATCFO and Training department.

<i>TBL 4-1</i> Maximum Total Training Time and Simulation Requirements			
Position	Initial Qualification Time	Subsequent Qualification Time	Simulation Requirement (initial qualification)
Ground Control	80 days	45 days	16 hours
Tower Flight Data / Clearance Delivery	80 days	35 days	16 hours
Local Control	180 days	60 days	36 hours
Radar Final Control	90 Approaches	35 Approaches	50 approaches
Radar Flight Data	100 days	50 days	20 hours
Arrival / Departure Control / R-2503D	260 days	100 days	52 hours

<i>TBL 4-2</i> Currency Requirements	
Position	Requirement
Ground Control	2 hours
Tower Flight Data	2 hours
Local Control	2 hours
Radar Final Control	5 approaches (5 live, 5 simulated, or combination of 5 live or simulated) simulated approaches must be conducted on IPARTS
Radar Flight Data	2 hours
Arrival / Departure Control	2 hours
Chiefs, Supervisors, and Facility-Rated FWOs	4 hours on an operating position in each branch qualified

Chapter 5. Tower

Section 1. General

5-1-1. EMERGENCIES

a. An emergency can either be a distress or an urgency condition as defined in FAAO JO 7110.65. Because of the infinite variety of possible emergency situations, specific procedures cannot be prescribed. However, when you believe an emergency exists or is imminent, select and pursue a course of action which appears to be most appropriate under the circumstances and which most nearly conforms to the instructions in FAAO JO 7110.65.

b. Some of the more common emergencies and procedures for handling aircraft emergencies associated with this airfield can be found in [Chapter 2, Section 3](#) of this manual.

c. Activate the Basilone Road traffic light anytime an emergency aircraft is inbound and will overfly the road. The light should be activated when the aircraft is within 5 flying miles of the runway.

d. If an emergency aircraft is inbound to MCAS Camp Pendleton, the Tower Supervisor will notify the Radar Supervisor and GCA site as necessary. This will be determined after evaluating all information provided concerning the emergency.

e. Between sunset and sunrise, if airfield lighting becomes inoperative, broadcast an advisory on the ATIS and ensure all aircraft are advised of the outage (NITEX excluded).

5-1-2. CRASH PHONE

a. The Tower Supervisor is responsible for activating the crash phone circuit for airfield emergencies. Flight Data normally activates the crash phone.

b. The Crash phone shall be tested daily, five minutes prior to airfield opening. The phraseology for the daily crash phone test is:

PHRASEOLOGY-

“THIS IS THE TOWER WITH THE DAILY CRASH PHONE TEST. THIS IS THE TOWER WITH THE DAILY CRASH PHONE TEST. ALL STATIONS ACKNOWLEDGE WITH INITIALS WHEN CALLED. (station sequence).”

“ALL STATIONS LOUD AND CLEAR EXCEPT (if any).”

“THIS IS (operating initials) IN THE TOWER. SECURE THE CRASH PHONE.”

NOTE-

The daily facility log will reflect this test.

c. Actual emergencies

1. Phraseology is as follows:

PHRASEOLOGY-

STANDBY, STANDBY, STANDBY. (give all known emergency information). (repeat all known emergency information). ALL STATIONS ACKNOWLEDGE WITH INITIALS WHEN CALLED. (station sequence).

2. Emergency information includes aircraft callsign, type, location, nature of emergency, landing runway, and additional information as necessary. Questions from individual stations may be received; remain on the line momentarily to answer any questions.

3. When the emergency has been secured, the phone is once again activated using the following phraseology:

PHRASEOLOGY-

THIS IS THE TOWER SECURING THE EMERGENCY. THIS IS THE TOWER SECURING THE EMERGENCY AT (time) ZULU. ALL STATIONS ACKNOWLEDGE WITH INITIALS WHEN CALLED. (station sequence).

4. If an actual crash has occurred on or near the air station, initiate the call with “*CRASH, CRASH, CRASH.*” Proceed with all known information. Questions from individual stations may be received; remain on the line momentarily to answer the questions.

5-1-3. VISUAL REPORTING POINTS

[Table 5-1](#) lists local VFR reporting points.

5-1-4. VFR

VFR operations are conducted in accordance with 14 CFR Part 91, FAAO JO 7110.65, and the AOM.

5-1-5. SHIP TO SHORE FLIGHT

Since naval vessels off the coast of Southern California are not served by a Flight Service Station, flight plans shall be processed via FACSAC, North Island.

5-1-6. SHORE TO SHIP FLIGHT

Shore-to-Ship flight plans may be filed with MCAS Flight Clearance or the tower. Aircraft movement messages for flights originating from MCAS Camp Pendleton will be passed by the tower to Flight Clearance and then to FACSAC.

a. The following restrictions apply:

1. Helicopters only
2. Both arrival and departure facilities shall be open and VFR

b. The minimum information required for this type of flight plan is:

1. Aircraft callsign, number and type
2. Estimated time enroute (ETE)
3. Fuel on board (FOB)
4. Souls on board (SOB)
5. Ship's tactical callsign

5-1-7. SPECIAL VFR FLIGHTS

CFR Part 91 authorizes SVFR operations within the boundaries of Class D. SVFR operations are further restricted by excluding portions of Class D outside of Lima airspace.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between SoCal TRACON and MCAS Camp Pendleton

a. Special VFR aircraft will not be assigned fixed altitudes. The ATC tower has the authority to conduct SVFR operations at MCAS Camp Pendleton at or below 1500 feet MSL.

b. SVFR operations are limited to one aircraft operating within the Class D surface area unless one of the following conditions exists:

1. Tower has all aircraft operating in Class D in sight and provides visual separation.

2. Succeeding aircraft have preceding aircraft in sight and is instructed by ATC to maintain visual separation from that aircraft.

3. With reference to IFR aircraft, coordination may be required with the RATCF prior to resuming SVFR operations to confirm position of the IFR aircraft. For example, an IFR departure leaves 1000 feet and SVFR operations are restricted at or below 500 feet.

c. Special VFR operations are authorized for helicopters and V-22s in conversion mode.

d. SVFR/IFR operations. IFR aircraft have priority over SVFR aircraft. When IFR aircraft are cleared for an approach into Camp Pendleton airfield while SVFR operations are being conducted, SVFR operations shall be secured, (i.e., pattern aircraft have landed and / or aircraft have reported clear of Class D) prior to the IFR aircraft reaching a 5 mile final. Exception to this rule requires the application of visual separation in accordance with FAAO JO 7110.65.

e. Overflights. SVFR overflights are authorized at an altitude restriction of at or below 1500 feet MSL. CFR Part 91 requires 1 statute mile flight visibility. NAVAIR 00-80T-114 requires a 500-foot ceiling for military aircraft.

REFERENCE-

CNAF-M 3710.7 Para 5.2.4, Weather Minimums

f. Special VFR Routes. [Table 5-2](#) lists local SVFR routes. See also [Appendix C](#).

g. A Special VFR clearance may be relayed through Longrifle for those aircraft requesting clearance that are not able to establish contact with the control tower. Range Control is not authorized to issue these clearances independently and will instruct all aircraft experiencing communications difficulties to hold outside Class D airspace until a clearance can be obtained. All clearances relayed through Longrifle will be prefaced by the statement “ATC Clears...”

REFERENCE-

JO 7110.65, Para 4-2-2, Clearance Prefix

5-1-8. IFR

IFR operations are conducted in accordance with FAAO JO 7110.65 and the AOM. IFR authority is assigned to RATCF by SCT in airspace known as Lima and Golf. Airspace is described in the reference and [Appendix C](#).

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between SoCal TRACON and MCAS Camp Pendleton

5-1-9. TEC ROUTES

Tower Enroute Control routes (TEC routes) are contained within approach control airspace. These routes are designed to expedite traffic and reduce control and pilot communication requirements. None of these routes will enter ARTCC airspace. See the FAA Chart Supplement.

a. Initial altitude assignment is dependent on the associated SID with the final altitude to expect within ten minutes of departure. Final altitude assignment is based on the category of aircraft as follows:

1. Q = Non-jet, cruise speed 189 knots or less (ex: *helicopters*).
2. P = Non-jet, cruise speed 190 knots or greater (ex: *small prop*).
3. M = Turboprop / special, cruise speed 190 knots or greater (ex: *military turboprop*).
4. J = Jets

b. The heading assigned to aircraft overflying Oceanside VORTAC to join another NAVAID radial is covered in the Interfacility Coordination Procedures letter of agreement.

5-1-10. STRIP MARKING PROCEDURES

Examples of strip marking can be found in [Appendix D](#). Strip marking usage is covered in Chapter 2, Section 3 of FAAO JO 7110.65 and locally established procedures as outlined in this manual. Controllers should write clearly as strips are retained as an official record of air activity and may be used in a court of law.

- a. Strip marking for departure procedures is listed in [Table 5-3](#).
- b. Clearance limits and departure procedures for VFR-on-Top are listed in [Table 5-4](#).
- c. Special VFR route abbreviations are listed in [Table 5-5](#).

5-1-11. FLIGHT DATA INPUT / OUTPUT SYSTEM (FDIO)

This system allows for an inter- and intra-facility computer link for the transmission of generated flight strips and associated ATC information.

- a. FDIO message fields and formats are listed in [Table 5-6](#).
- b. Message types and their uses are listed in [Table 5-7](#).
- c. Local FDIO addresses:

1. NFG Tower = NFGT
2. NFG Radar = NF1A, NF1D
3. Approach = NK1A
4. NFG Flight Planning = NF3A, NF3D

5-1-12. NIGHT VISION OPERATIONS (NITEX)

a. All aircraft arriving and departing MCAS Camp Pendleton during the affected hours will use night vision devices.

b. Only aircraft participating in night vision training are permitted within Class D airspace with the exception of the station C-12. Aircraft in the pattern will not be required to land while unaided

aircraft are conducting operations. The tower and GCA are permitted to deny entry into Class D airspace to non-participating aircraft. IFR arrivals and departures are covered in the reference.

PHRASEOLOGY-

CAMP PENDLETON WILL TEMPORARILY ACTIVATE ALL AIRFIELD LIGHTING FOR STATION C-12.

c. Transient aircraft inbound to Camp Pendleton will be informed that a NOTAM is in effect for Special Operations (NVGs) and that all operations will be conducted in an unlighted environment. If the pilot cannot accept this restriction, they will be directed to depart the area. Only if an emergency is declared will this procedure be modified.

d. The maximum number of aircraft operating in the MCAS Camp Pendleton pattern during unlit periods is restricted to not more than four aircraft in the touch and go pattern. Not more than two aircraft are allowed to conduct low work.

e. All vehicles shall practice light discipline when operating on MCAS Camp Pendleton during unlit periods.

f. Broadcast the following prior to commencing NVG Operations:

PHRASEOLOGY-

CAMP PENDLETON WILL SECURE ALL AIRFIELD LIGHTING IN FIVE MINUTES FOR NVG OPERATIONS. ALL NON-PARTICIPATING AIRCRAFT REMAIN WELL CLEAR.

PHRASEOLOGY-

CAMP PENDLETON WILL SECURE ALL AIRFIELD LIGHTING IN ONE FIVE SECONDS FOR NVG OPERATIONS. ALL NON-PARTICIPATING AIRCRAFT REMAIN WELL CLEAR.

g. If TS deems NVG operations unsafe due to the weather, they may secure NITEX for safety. If the field becomes IFR, approach lights will be turned on unless pilots request otherwise.

h. Refer to the AOM for further information concerning NITEX Operations.

5-1-13. LONGRIFLE OPERATIONS

Restricted Airspace is managed by Marine Corps Base Range Control. Coordination procedures are contained in the reference. ATC will:

a. Advise MCB Longrifle of the following:

1. Opening and closing of the airfield.
2. Radar and TACAN status.
3. If an aircraft under ATC approaches an active firing point and cannot be diverted.

b. Advise aircraft of active training facilities within Area One and Area Two.

c. Ensure aircraft under ATC control remain outside of restricted airspace excluding Area One and Area Two unless otherwise coordinated.

REFERENCE-

Control and Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and MCB Camp Pendleton

5-1-14. AREAS ONE AND TWO

The areas described in the reference were established so a safe working environment could exist between the hazardous-to-aircraft operations conducted in restricted airspace and IFR / VFR operations at MCAS Camp Pendleton.

a. Area One is that portion of R-2503B between 1000 feet MSL and 6000 feet MSL southeast of a line 3NM northwest and parallel to the final approach course for runway 21, excluding airspace defined as Class D and E.

b. Area Two is that portion of R-2503A from 1000 feet MSL to 2000 feet MSL southeast of a line 3NM northwest of and parallel to the extended runway centerline, excluding the airspace defined as Class D.

REFERENCE-

Control and Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and MCB Camp Pendleton

5-1-15. COMBAT AIRCRAFT LOADING AREA (CALA)

The CALA is located at the northwest corner of the airfield. The AOM outlines its use.

a. Aircraft may be towed or taxied to this area utilizing Taxiway Charlie.

b. 500-foot AGL overflight restriction – During CALA operations, all inbound aircraft will be informed by the tower, or ATIS verification that the restriction is in effect. Aircraft shall be instructed to remain at or above 500 feet AGL over any vehicles, aircraft or known activity in the CALA area. All landings to the right grass shall be made after passing the displaced threshold located 2288 feet SW of the runway 21 threshold.

TBL 5-1 VFR Reporting Points			
Reporting Point	Position	Aircraft Usage	Remarks
North Initial	NFG R-032 @ 9	Rwy 21 Inbound Rwy 3 Outbound	Mandatory Reporting Point
Intersection	NFG R-360 @ 5	Inbound / Outbound	Mandatory Reporting Point
LCAC	NFG R-249 @ 2.3	Inbound	Mandatory Reporting Point
MASS-3	NFG TACAN Site	Inbound	Secondary Reporting Point
Lake O'Neill	NFG R-032 @ 4.8	Rwy 21 Inbound Rwy 3 Outbound	Secondary Reporting Point
Point Canyon	NFG R-017 @ 6.7	Inbound / Outbound	Mandatory Reporting Point
Morro Hill	NFG 062 @ 6.2	Outbound	Mandatory Reporting Point
Oceanside VORTAC (South Initial)	NFG R-210 @ 2.3	Fixed-wing initial for Overhead Break	Mandatory Reporting Point

<i>TBL 5-2</i> Special VFR Routes		
Route	Weather Minimums	Restrictions
VORTAC	800 – 1	Departure only
LCAC	800 – 1	Arrival only
Intersection	800 – 1	Arrivals and Departures
Point Canyon	800 – 1	Arrivals and Departures
North Initial	500 – 1	Arrivals and Departures
Pattern	500 – 1	Local Airport Area

<i>TBL 5-3</i> Strip Marking for Departure Procedures	
Departure Procedure	Abbreviation
Quentin	QUNTN#
Corri	CORRI#
Bulldog	BLDG#
Munn	MUNN#

<i>TBL 5-4</i> VFR-on-Top Clearance Limits	
Clearance Limit	SID Abbreviation
VISTA	BLDG#
CORRI (helicopter)	CORRI#
QUENTIN	QUNTN#

<i>TBL 5-5</i> SVFR Route Abbreviations	
Route	Abbreviation
Intersection route	NW
LCAC route	SW
VORTAC route	SW
North Initial route	NE
Point Canyon route	N
Pattern work	W (circled)

5-1-16. ARMING AREA

- a. After completing uploading in the CALA, aircraft will proceed to the arming area to change the status of the ordnance from “cold” to “hot”.
- b. 500 foot AGL overflight restriction - Aircraft shall be instructed to remain at or above 500 feet AGL over any vehicles, aircraft or known activity in the arm/de-arm, upload / download areas.
- c. Aircraft in the arming area will face away from airport movement areas. Up to six aircraft are authorized to be in the arming / de-arming area at any time.

5-1-17. FLIGHT CHECK AIRCRAFT

These aircraft are used for the inspection and evaluation of NAVAIDS. They determine whether established tolerances are maintained by ATCMD / Airway Facilities. Flight Check aircraft shall be given priority over other aircraft in the Class D surface area in accordance with FAAO JO 7110.65.

5-1-18. PIREPS

Procedures for soliciting and disseminating PIREPs are covered in FAAO JO 7110.65. Flight Data shall forward all PIREPs to Weather.

5-1-19. VIP PROCEDURES

The Tower Supervisor will assist Operations with VIP procedures in the following manner:

- a. Ensure that Flight Clearance is notified as soon as possible of the estimated arrival time of aircraft with codes aboard (if possible, at least 10 minutes prior to landing.)
- b. Ensure that the VIP aircraft receive expeditious handling during all phases of movement.

5-1-20. STATE AND LOCAL CALLSIGN DESIGNATIONS

The following callsigns are FAA approved and have been assigned to aircraft conducting specific operations:

- a. *BRAVO*. Assigned to Benbow Helicopters Inc. Use of this callsign will be for VFR use only.
- b. *DUKE*. Assigned to Orange County Sheriff's Department (OCSD) Use of this callsign will be for VFR/SVFR use only.
- c. *EAGLE*. Assigned to Airborne Law Enforcement (ABLE). Use of this callsign will be for VFR/SVFR use only.
- d. *HB*. Assigned to Huntington Beach Police Dept Traffic Aero Bureau. Use of this callsign will be for VFR use only.
- e. *IDAHO*. Law enforcement agency use on bona fide air interdiction/surveillance missions.
- f. *SUPPLY*. Assigned to Orange County Regional Narcotics Suspension Program (RNSP) Use of this callsign will be for VFR/SVFR use only.
- g. *TV-7*. Assigned to West Coast Helicopters. Use of this callsign will be for VFR/SVFR use only.
- h. *EIGHTEEN AERO*. Assigned to Orange County Sheriff - Coroner Department (OCSD) Use of this callsign will be for VFR/SVFR use only.

i. *POLICE 40*. Assigned to Huntington Park Police Department. Use of this callsign will be for VFR / SVFR use only.

j. *ASTRIA*. Assigned to San Diego County Sheriff Department.

5-1-21. HIGH POWER RUN-UPS

Jets and prop aircraft with more than two engines shall conduct high power run-ups on the runway. Exceptions are made only with prior approval by operations. C-12 / C-35 type aircraft may conduct run-ups at the hold short line serving the duty runway with the exception of Delta taxiway.

a. Helicopters

1. Test area. The North Grass is normally used for hover checks. In the north grass, aircraft will remain centered between the runway and parallel taxiway. Hover checks will not exceed 50 feet AGL without approval from the tower.

2. In-flight test area. The Punch Bowl area located 2NM southeast of the field.

b. Fixed wing tests

1. Ground Test area. The south end of the runway may be used to allow a fixed wing aircraft to conduct high power (above 80%) ground test. Low power (below 80%) ground tests may be conducted on the flight line and taxiways.

2. In-flight test area. Fixed wing aircraft may conduct their flight checks in R-2503B after contacting the appropriate facility for clearance.

REFERENCE-

AOM, Paragraph 3003, Traffic Patterns and Authorized Operations

5-1-22. CLOSED TOWER OPS

Requests for closed tower operations shall be submitted to the Air Station Operations Officer via Flight Clearance. MAG-39 squadrons shall coordinate closed tower requests through the MAG-39 Operations Officer. See also the AOM.

5-1-23. CONSOLIDATED WAKE TURBULENCE - TOWER

MCAS Camp Pendleton operates under the Consolidated Wake Turbulence (CWT) separation standards contained in the FAAO JO 3760.1, 7110.126, and 7110.65. Controllers shall hand write the appropriate CWT category on all strips until system updates enable CWT categories to be entered via the FDIO.

<i>TBL 5-6</i> FDIO Message Fields and Formats				
Field Number	Description	Abbreviation	Format	Example
01	Message Type		2 letters	FP, AM, FR, GI
02	Aircraft ID	AID	2 – 7 characters	ATLAS06
03	Aircraft Type	TYP	2 – 9 characters	H1/P, C130/I, BE20/F
04	Beacon Code	BCN	4 digits	4540
05	Speed	SPD	2 – 4 digits	120
06	Coordination Fix	FIX	2 – 12 characters	BONDO, VISTA
07	Coordination Time	TIM	‘P’ and 4 characters	P1730
08	Assigned Altitude	ALT	2 – 20 characters	30, 110, 250
09	Requested Altitude	RAL	2 – 7 characters	150, 210, OTP/40
10	Route	RTE	First and last item shall be a fix or a special condition indicator	NFG.BLDG2.VISTA. V458.BONDO.NKX
11	Remarks	RMK	‘o’ followed by plain language	oPAR OTS

<i>TBL 5-7</i> FDIO Message Types			
Code	Message Type	Description	Context
FP	Flight Plan	Enter a flight plan into the system	FP ATLAS06 H1/P 120 NFG P2130 030
AM	Amendment Message	Modify previously filed data	AM 315 TYP H1/X
GI	General Information	Plain language entry of operational information	GI NK1A oPAR RTS
SR	Strip Request	Obtain a printed flight strip	SR 032 NFG NFGT
RS	Remove Strip	Remove a flight plan from the system	RS 127
WR	Weather Request	Obtain the most recent weather observation for an airport	WR NKX
DM	Departure Message	Manually enter an aircraft’s departure time	DM 445 1732
FR	Full Route	Obtain a printout of an aircraft’s flight plan	FR VV7M624
Other messages are possible; this table lists the most common.			

Section 2. Ground Control

5-2-1. DUTIES & RESPONSIBILITIES

The ground controller is responsible for exercising surveillance and general control of the airport movement area, excluding the runway and grass landing areas. Primary duties are listed in NAVAIR 00-80T-114 and FAAO JO 7110.65. Additional duties may include:

- a. Operating the Airfield Lighting Control System (AFLCS).
- b. Generating Shore to Ship, IFR and SVFR flight strips as necessary.

5-2-2. AIRCRAFT MOVEMENT

- a. Taxiing aircraft have priority over movement of vehicles except those vehicles responding to an emergency.
- b. Do not use the words "cleared" or "takeoff" in conjunction with authorization for aircraft to taxi. Use the words "*TAXI / HOLD / PROCEED*" as appropriate. Caution should be exercised in using the phrase "go ahead" in conjunction with aircraft movement. Additionally, the word "continue" is used in conjunction with aircraft forward movement and shall not be used in conjunction with hold short instructions.
- c. All transient aircraft parking at the MALS-39 line or VIP parking area will be parked by VAL personnel or Aircraft Rescue and Fire Fighting when VAL personnel are not available.
- d. It may be necessary to issue progressive taxi instructions to transient aircraft. Progressive taxi instructions involve calling each turn far enough in advance to prevent abrupt stops or delays.

5-2-3. VEHICLE MOVEMENT

- a. Always use full call-signs and never the word "cleared" in conjunction with vehicle movement instructions. Use "*CROSS, PROCEED, or HOLD SHORT*" as appropriate. Caution should be exercised in using the phrase "go ahead" in conjunction with vehicle movement. Additionally, the word "continue" is used in conjunction with vehicle forward movement and shall not be used in conjunction with hold short instructions.
- b. The tower has no vehicle control in the fuel pits, parking areas, or ramp areas.
- c. Vehicular traffic on runways and taxiways is strictly controlled at all times. Reports of personnel and vehicle operators not conforming to the following procedures should be forwarded to the Airfield Operations Office. Communication requirements are as follows:
 - 1. *Emergency Vehicles*. Emergency vehicles or ground support vehicles equipped with direct two-way radio communications are allowed access to any of the movement areas after receiving clearance from the tower.
 - 2. *Other Vehicles*. All other vehicle operators shall report to Flight Clearance (Bldg 2399) prior to operating on the airport. They will receive:
 - (a) A two-way radio or escort for communication with tower.
 - (b) Orange and white checkered flag.

5-2-4. EMERGENCY PROCEDURES

a. When the crash phone is activated, ARFF will get into position as soon as possible. The emergency standby positions are southeast of the runway at the hold short lines of Alpha, Bravo, Delta, Echo 1 and Foxtrot 1. In addition, ground shall stop all airfield traffic.

Phraseology-

“Attention all aircraft and vehicles, this is Camp Pendleton ground. Emergency in progress. Hold your position until further advised. Emergency in progress.”

b. The ground controller will only speak to the section leader, via the Crash Net, to reduce frequency congestion. Relay all updated information to the command vehicle as soon as possible. See [Appendix C](#).

c. Once crash vehicles have responded, ground may allow movement on the airfield only after obtaining approval from the ARFF section leader.

d. Once the emergency aircraft has landed, local may authorize rescue vehicle access onto the runway. They are required by OPNAVINST to escort the aircraft to its line. Neither pilot nor controller may override this order. If the pilot states that no further assistance is required, it shall be relayed to ARFF. The emergency aircraft shall hold its position after landing until released by the ARFF section leader. Inform the tower supervisor when the section leader advises the emergency has been secured. An entry will be made in the facility log to this effect.

e. Off-station emergency procedures are as follows:

1. ARFF dispatcher will notify the tower supervisor upon change to airfield response status.

(a) The tower supervisor will in turn restrict all flight operations, other than aircraft involved in support of an emergency. Tower will also ensure that Flight Clearance has been informed.

(b) Flight Clearance is responsible for informing the Operations Officer, Operations Officer, or Operations SNCOIC (in the order shown) of any change in the status of the airfield.

(c) This status will remain until either the Operations Officer or Airfield Operations Officer reopens or closes the airfield.

2. The ARFF section leader will contact the tower supervisor when support for operations can be met.

5-2-5. DEPARTURE PROCEDURES

All departures from MCAS Camp Pendleton shall be planned, filed, cleared, and conducted in accordance with IFR procedures, with the exception of:

a. MEDEVAC / SAR operations

b. Contingency operations as directed by higher authority.

c. Local and VFR flights.

d. Flights departing for entry into the tower pattern.

5-2-6. COORDINATION AND PHRASEOLOGY WITH LOCAL CONTROL

Ground control shall coordinate all helo spot or taxiway departures, runway crossings, north grass operations, and aircraft being taxied to an intersection with local via ETVS communication. Ground shall also coordinate when the use of the runway is complete by stating “clear deck” via ETVS communications. When coordinating, include type of aircraft and location. Abbreviated ETVS communications are authorized between Ground and Local and the need to identify the caller and receiver is not necessary.

EXAMPLE-

“H1 Papa 4”, “H1 North grass”, “Cross Runway 21 at Echo, H1.”

5-2-7. AREA OF HIGH VIGILANCE

The area of the airfield with the greatest potential for an operational hazard is Echo 2 taxiway; aircraft may easily taxi past the hold short lines onto Echo 2. It is crucial for controllers to ensure aircraft (especially transient aircrews) hold short prior to entering Echo 2.

Section 3. Clearance Delivery

5-3-1. DUTIES & RESPONSIBILITIES

The clearance delivery controller is responsible for obtaining, posting, and relaying ATC clearances and advisories. Normally this position is combined at the Flight Data position. Primary duties are listed in NAVAIR 00-80T-114. Additional duties may include:

- a. Exercising general surveillance of the airport.
- b. Issue clearance items as appropriate.
- c. ATC clearances received from SCT shall be relayed to the pilot verbatim. Exceptions for obtaining / issuing a clearance are covered by LOA.
- d. Assign a VFR-on-Top altitude. See [Table 5-9](#).
- e. Aircraft departing on an IFR flight plan shall be issued an Obstacle Departure Procedures and initial altitude in accordance with the routes in the SCT LOA. See [Table 5-8](#).

EXAMPLE-

“CLEARED TO NAVY NORTH ISLAND AIRPORT VIA THE QUNTN ONE OBSTACLE DEPARTURE PROCEDURE, QUNTN, SKATE, THEN DIRECT. ON DEPARTURE, CLIMB AND MAINTAIN 3000, DEPARTURE FREQUENCY IS (127.3 OR 323.0), SQUAWK 4534”

REFERENCE-

*Interfacility Coordination Procedures Letter of Agreement between SoCal TRACON and MCAS Camp Pendleton
FAA JO 7110.65 4-3-2 Departure Clearances*

TBL 5-8 Initial Altitudes for Departure Procedures	
Departure Procedure	Assigned Altitude
Quntn	3000
Bulldog	4000
Munn	4000
Corri	Maintain as coordinated

TBL 5-9 VFR-On-Top Altitude Assignment	
Clearance Limit	Assigned Altitude
VISTA	4000
QUNTN	3000
CORRI (helicopter)	Runway 21 – 3800

5-3-2. GCA BOX REQUESTS

Aircraft requesting the CORRI ODP to remain in the GCA box pattern shall have a clearance limit of KNFG. Aircraft will expect vectors to Camp Pendleton Airport after CORRI. In the remarks sections of the flight plan, the words “Requesting GCA Box” shall be added to alleviate any confusion with Southern California TRACON.

Section 4. Tower Flight Data

5-4-1. DUTIES & RESPONSIBILITIES

The flight data controller assists both the ground controller and the local controller. Primary duties are listed in NAVAIR 00-80T-114. Additional duties may include:

- a.** Obtaining a release for all departing IFR aircraft
- b.** Activating the crash phone as directed by the tower supervisor.
- c.** Monitoring local control frequencies.
- d.** Monitoring flight strips for updating of INREQ / ALNOT status.
- e.** Clearance Delivery when combined.

5-4-2. WEATHER

The flight data controller is responsible for documenting, posting, and forwarding weather information as follows:

- a.** Maintain and update the ATIS broadcast.
- b.** Pass hourly and special weather observations to RATCF and SCT.
- c.** Post PIREPs and relay them to the weather office and to RATCF.
- d.** Pass tower visibility observations whenever the visibility is less than four miles.
- e.** Request a special observation as directed by the tower supervisor.

5-4-3. FLIGHT CLEARANCE COORDINATION

Maintain liaison with Flight Clearance as follows:

- a.** Relay departure and arrival times to Flight Clearance.
- b.** Relay range status information.
- c.** Relay airfield status (i.e., IFR to VFR, taxiway closures, NAVAID outages).
- d.** Relay Flight Plan Not Received (FPNO) information.
- e.** Other information as directed (i.e., VIP information, pilot requests).
- f.** Request VAL assistance in parking VIP or transient aircraft as soon as the tower is notified the aircraft is inbound.
- g.** Forward all requests for cold refuel. The minimum information required from the pilot for relay to Flight Clearance is:
 - 1.** Callsign
 - 2.** Type
 - 3.** Side numbers / tail numbers
 - 4.** Location
 - 5.** Amount of fuel in gallons or pounds

5-4-4. SCT COORDINATION

Flight data will coordinate the following information with SCT:

- a. IFR clearances when the FDIO is not operational.
- b. Releases for IFR aircraft whenever RATCF is not operational.
- c. ATIS code and all hourly and special weather sequence reports via GI message over the FDIO.
- d. Airfield and NAVAID status.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between SoCal TRACON and MCAS Camp Pendleton

5-4-5. RATCF COORDINATION

Tower Flight Data will coordinate the following information with RATCF:

- a. All IFR releases (Arrival / Departure Controller).
- b. Emergencies in progress (Radar Flight Data).
- c. Traffic calls as directed by the Local Controller (Appropriate control position, i.e., P1 / P2 / A2 / A3)
- d. Field status and conditions (Radar Flight Data).

5-4-6. AUTOMATIC TERMINAL INFORMATION SERVICE (ATIS)

The ATIS is used to continuously broadcast pertinent airfield non-control information, weather and NOTAMS to pilots. This system is designed to reduce frequency congestion and controller workload of repeating basic airfield information to individual pilots. The ATIS is broadcast over UHF frequency 285.45. Operation of the ATIS is covered in [Chapter 8](#) of this manual.

Significant changes in field conditions or weather will require a new ATIS broadcast. In this case the ATIS code will change but not the time of the weather. Only information that directly affects arriving and departing aircraft as outlined in FAAO JO 7110.65 will be transmitted over the ATIS.

Section 5. Local Control

5-5-1. DUTIES & RESPONSIBILITIES

The local controller's primary duties are listed in NAVAIR 00-80T-114.

5-5-2. LOST COMMUNICATIONS

Aircraft experiencing lost communications while VFR will remain VFR and squawk 7600. In accordance with the AOM, the aircraft should enter Class D via normal course rules and orbit northwest of the airport.

a. Between sunrise and sunset, fixed wing aircraft will move ailerons or rudders while on the ground and will rock their wings while in flight. Rotary wing aircraft will flash landing lights while facing the tower or rock the tip path plane.

b. Between sunset and sunrise, aircraft should flash navigation or landing lights.

5-5-3. COORDINATION BETWEEN POSITIONS

Local and ground shall exchange information as necessary for the safe and efficient operation of the airfield.

a. Local Control shall coordinate with Ground via ETVS communication whenever they plan to use a portion of the airfield that normally belongs to Ground (i.e., helo spots and taxiways). Abbreviated ETVS communications are authorized between Ground and Local and the need to identify the caller and receiver is not necessary. Helo spot usage will be limited to departures and approaches to a full stop (Touch-and-gos and low approaches are not authorized). All helo classifications are authorized for use on P-3. To ensure minimum FOD and possible damage to parked aircraft on the ramp, P-1 and P-4 are restricted to SUCO and UH1Y aircraft.

EXAMPLE-

"Request Papa 1."

"Request Bravo."

"Request Papa 3."

b. Local shall coordinate arrival and departure information with flight data and direct flight data to accomplish coordination with other facilities.

c. The local controller is responsible for providing traffic advisories to the arrival and final controllers. During periods that RATCF/GCA has no traffic, the tower will not be required to coordinate VFR tower departures/arrivals/traffic.

EXAMPLE-

"Traffic outbound Morrow Hill, V22"

d. Landing clearances when the VISCOM is OTS (Appropriate control position, i.e., P1 / P2 / A2 / A3)

5-5-4. PREFERRED RUNWAY USE

Runway 21 is the calm wind and Precision Approach runway. Whenever sustained winds of 5 knots or more exist, the runway most nearly aligned with the wind shall be used, unless an operational advantage exists.

5-5-5. VFR DEPARTURE PROCEDURES

All VFR helicopter departures from MCAS Camp Pendleton will be in accordance with the AOM. These routes have been established by the operations department and MAG-39 as the primary routes departing Class D airspace. Approval may be made on a case-by-case basis for aircraft requesting departure routes other than those published.

- a.** Unless operationally advantageous, all departures will be from the runway. (Helicopters cleared for takeoff from the North Grass may be instructed to proceed to the runway prior to commencing the departure).
- b.** Consider aircraft ready for takeoff when they call for departure. Pilots are required to advise tower of any delay they may need on the runway. Aircraft may require two to five minutes on the runway for run-ups.
- c.** VFR departures shall report departing the Class D surface area after takeoff unless otherwise specified. Aircraft desiring to stay within Class D will be monitored by the control tower. Aircraft departing via Morro Hill should be instructed to squawk VFR due to potential conflicts with other VFR & IFR traffic.
- d.** All aircraft departing Class D intending to enter R-2503 shall contact Longrifle for clearance through the restricted area (310.3 or 123.2).
- e.** All aircraft departing MCAS Camp Pendleton shall avoid overflying the Ranch House and the old Naval Hospital. A 1,500-foot restriction exists over Mainside, as well as a 500-foot restriction over the base stables and Naval Weapons Station.
- f.** When runway 3 is in use and the aircraft requests multiple VFR practice approaches, unless otherwise coordinated, the aircraft will complete the circling approach, enter a right downwind, and depart via the Oceanside VORTAC or depart via Morro Hill and contact RATCF when instructed by tower.
- g.** VFR aircraft within class D airspace requesting local GCA pattern will be coordinated from the tower to RATCF for a frequency and beacon code. Beacon codes will range from 4534 to 4543.

5-5-6. IFR DEPARTURE PROCEDURES

All IFR departures that enter SCT airspace shall be in accordance with the standard instrument departure procedures published for MCAS Camp Pendleton unless otherwise directed.

- a.** Flight Data will request a release from Arrival Control for all IFR departures, no sooner than 5 minutes after obtaining an IFR clearance via FDIO or 5 minutes via phone from SCT.
- b.** A release is valid for three minutes. If the aircraft has not started takeoff roll by that time, the takeoff clearance will be canceled and coordination for another release will be required.
- c.** IFR release authority shall be obtained from Arrival Control. Arrival may authorize an IFR departure, on an individual aircraft basis, if visual separation can be applied by the tower in

accordance with FAAO JO 7110.65. If the tower controller cannot provide visual separation, they will inform the arrival controller. Arrival will then become responsible for providing a release time for the departure. Arrival is not obligated to solicit releases requiring the application of visual separation.

d. Departure frequencies will be assigned as follows:

1. Aircraft departing runway 21 on an IFR flight plan will be directed to contact SCT.
2. Aircraft departing runway 3 on an IFR flight plan will be switched to RATCF on departure frequency 236.3 or 128.45.
3. Aircraft departing on a Corri departure shall be instructed to contact RATCF on assigned frequency.

REFERENCE-

*Interfacility Coordination Procedures Letter of Agreement between SoCal TRACON and MCAS Camp Pendleton
Para 6-3-2, Departure Coordination*

5-5-7. OPPOSITE DIRECTION OPERATIONS (ODO)

VFR ODO are authorized as follows:

- a. Tower must ensure departing aircraft are issued a turn to avoid conflict with opposing GCA traffic.
- b. Local shall coordinate all North Initial departures with Arrival.
- c. Local shall issue traffic to final controllers for all departures when GCA aircraft are within 5 miles of the runway. The phrase “OPPOSITE DIRECTION” must be stated.

EXAMPLE-

“Traffic, H-1 departing runway 3, opposite direction, Point Canyon.”

- d. Tower shall use the VISCOM and CTRD as memory aids during ODO.
- e. For radar responsibilities during VFR opposite direction operations, see [Paragraph 6-4-14](#).

5-5-8. SECTION DEPARTURES AND ARRIVALS

In accordance with NAVAIR 00-80T-114, section touch and go approaches are not authorized. Flight aircraft shall break up unless the pilot requests a full stop, stop & go, or low approach. More than one C-130 on the runway for takeoff is not authorized.

REFERENCE-

CNAF M-3710.7, General Flight and Operating Instructions

5-5-9. VFR ARRIVAL PROCEDURES – HELICOPTER

All VFR helicopter arrivals to MCAS Camp Pendleton will be in accordance with [Appendix C](#). These routes have been established by the operations department and MAG-39 as the primary routes for entry into Class D airspace. Approval may be made on a case-by-case basis for aircraft requesting arrival routes other than those published.

- a. Helicopters inbound to MCAS Camp Pendleton shall be in contact with tower prior to entering the Class D surface area or at a designated VFR reporting point, to allow for sequencing.
- b. All helicopters shall avoid:

1. Overflying congested areas below 1000 feet AGL.
 2. Overflying the Ranch House or the old Naval Hospital.
 3. Flights over Mainside at an altitude below 1500 feet AGL.
- c. Aircraft requesting the overhead maneuver will report the North or South initial.
- d. Breaks southeast of the airport are not authorized below pattern altitude or at night.

5-5-10. VFR ARRIVAL PROCEDURES – FIXED-WING

VFR fixed-wing aircraft inbound to MCAS Camp Pendleton will report the North or South Initial for entry into the Class D surface area. Approval may be made on a case-by-case basis for aircraft requesting arrival routes from other points.

- a. Section arrivals by fixed-wing aircraft C-130 or larger in physical size are prohibited.
- b. Overhead Break. The normal break pattern for fixed-wing aircraft is northwest of the runway at an altitude of 1600 feet MSL.

5-5-11. HUNG ORDNANCE ARRIVAL PROCEDURES

This situation is considered an emergency. Aircraft inbound with hung ordnance shall advise the tower, make a straight in approach and after landing taxi to de-arming. Aircraft with forward firing hung ordnance shall have their aircraft face 350° in the arming area.

a. Pilots will:

1. Advise Camp Pendleton Tower on initial contact, stating the type of hung ordnance the aircraft has and request a straight in approach to a full stop landing.
2. After landing, hold in de-arming until the aircraft is de-armed and released by ARFF.
3. Aircraft with hung ordnance and having lost radio contact should be escorted to a final landing by a wingman. In the event that a situation arises in which an unaccompanied aircraft with hung ordnance shall recover after experiencing radio failure, the pilot will, to the maximum extent possible, avoid flying overpopulated areas while complying with lost communication procedures contained in the AOM.

b. Controllers will:

1. Activate the Crash Phone and pass the standard emergency information as well as type ordnance, if known.
2. If *Time Share* is being conducted, advise GCA of the type aircraft, position, type ordnance and estimated arrival time. See [Paragraph 6-1-13](#).

5-5-12. IFR ARRIVALS

Aircraft operating on IFR clearances shall follow published instrument procedures and comply with instructions from SCT until handed off to Camp Pendleton Arrival. Aircraft inbound to Camp Pendleton will be handled as follows:

- a. *Visual approaches*. Visual approaches shall be restricted to the airspace south and east of the runway extended centerline. Coordination for a Visual approach will only come from RATCF

when Lima Airspace is activated. Whenever Lima Airspace is returned to SCT, coordination for the Visual Approach will come from the SCT controller.

b. Radar Approaches. Precision Approach Radar (PAR) and Surveillance Approaches (SA) are conducted into Camp Pendleton. These approaches shall be handled as follows:

1. Radar flight data will forward the following minimum inbound information to Tower Flight Data no later than 10 flying miles from the airfield:

- (a) Aircraft position
- (b) Aircraft callsign, number and type aircraft
- (c) Intentions

2. Radar Final controller coordination will be handled in accordance with [Paragraph 6-2-7](#) for VISCOM.

NOTE-

The tower will not clear an aircraft on the VISCOM prior to the aircraft reaching 3/4 miles.

3. The following phraseology will be used when the VISCOM is out of service. Coordination is covered in [Paragraph 6-2-8](#) of this manual:

- (a) 6/7-mile check; tower will instruct GCA to check 3/4.

PHRASEOLOGY-

SIX / SEVEN MILES (callsign, type aircraft, intentions)

CHECK THREE / FOUR

EXAMPLE-

GCA: "Six miles, Atlas22, H-1, full stop."

Tower: "Check three."

- (b) 3/4-mile check; tower will give clearance or other instructions.

PHRASEOLOGY-

THREE / FOUR MILES (type aircraft, intentions)

EXAMPLE-

GCA: "Three miles, H-1, full stop."

Tower: "Cleared to land"

"Cleared low approach at or above 600"

"Continue"

4. In the event a wave-off is initiated by RATCF after the tower has issued a clearance, the RATCF controller will activate the wave-off light (aircraft can be expected to climb to 2200, fly runway heading).

c. Runway 3 procedures. Aircraft on surveillance approaches circling to runway 3 will be handed off from the final controller at 2 miles. IFR aircraft that have been cleared to circle shall not be instructed to extend downwind. See also [Paragraph 6-4-13](#).

d. Climb out instructions are contained in [Paragraph 6-4-11](#).

e. Restricted low approaches may be used when men, equipment, or aircraft are on the runway. Tower will issue to RATCF *"CLEARED LOW APPROACH AT OR ABOVE 600."* Unless otherwise coordinated. The low approach will be conducted at or above 600 feet MSL.

f. *RATCF Go-Around / Wave Off procedures.* Go-Around / Wave Off instructions indicate the aircraft will fly runway heading and maintain 2200 feet. This procedure will be executed via VISCOM or verbally under the following circumstances:

1. The final controller does not receive a clearance from the control tower by two miles.
2. The tower has an aircraft holding in position.
3. Any other condition that requires RATCF aircraft to discontinue the approach.

5-5-13. VFR TRANSITIONS

VFR transitions are to be conducted on a non-interference basis with pattern and departure / arrival operations. Approval for overflights is at the discretion of the local controller.

EXAMPLE-

TRANSITION APPROVED, (altitude, if needed). REPORT, (as needed).

A flight strip may be written and posted at the local control position for each aircraft operating in Class D airspace, (i.e.: photo mission, TERF route, Punch Bowl, etc.)

5-5-14. LZ Phoenix

LZ Phoenix is located in the Ysidora Basin south of the airport (see [Appendix C](#)). This basin area is locally referred to as the *"Punchbowl."* Because it is located within Class D airspace the following rules apply:

- a. All operations will be conducted during VFR conditions.
- b. All operations will be conducted at or below 500 feet AGL. Up to 700 feet may be authorized by the tower. Also, safety of flight will be considered regarding other traffic.
- c. All aircraft will monitor UHF Guard (243.0).
- d. All pattern aircraft will use 254.95, a non-interfering discrete UHF frequency.
- e. Tower is not responsible for the separation of LZ Phoenix pattern aircraft.
- f. Tower shall advise aircraft inbound to LZ Phoenix of other known aircraft using the LZ.
- g. All aircraft recovering from the LZ Phoenix to the airfield will do so only after establishing communications with the tower and receiving specific instructions for re-entry to the tower pattern. Aircraft that are unable to establish communications with the tower shall utilize Lost Communication procedures.

REFERENCE-

Control Responsibilities and Procedures for TALA Memorandum of Understanding

5-5-15. LANDING ZONE PROCEDURES

All aircraft within Class D airspace shall advise the tower prior to departing from, arriving at, or transiting to and from landing zones.

- a. The following LZs are located in Class D airspace and may conflict with pattern aircraft:

- | | | |
|----------|------------------------|----------------|
| 1. LZ 1 | 6. LZ 17 | 11. LZ Pelican |
| 2. LZ 11 | 7. LZ 22 | 12. LZ 32 |
| 3. LZ 12 | 8. LZ 27 | 13. LZ 33 |
| 4. LZ 14 | 9. 25 Area Combat Town | 14. Cal Site 1 |
| 5. LZ 16 | 10. LZ Toucan | 15. Cal Site 2 |

5-5-16. DZ BASILONE

This drop zone is in Class D airspace. When notification of drop activity is received, the following guidelines apply:

- a. Class D surface area shall be VFR.
- b. Maintain contact with the drop aircraft while in Class D unless otherwise coordinated.
- c. Receive a 5-minute prior to live drop report from pilot.
- d. Close the Intersection and Point Canyon routes as appropriate.
- e. Close the northwest pattern to fixed-wing aircraft.
- f. Do not authorize circling approaches.
- g. Include notification of DZ activity on the ATIS.

5-5-17. PIPELINE PATROL

Civilian aircraft routinely transition through Class D airspace to inspect the gas pipeline. The usual route flown is depicted in [Appendix C](#). These aircraft will be assigned the squawk 4035. After contact and proper code assignment, tower will coordinate with Longrifle.

5-5-18. MIKE TERF ROUTE

This is a helicopter training route located in the Mike Training area, southwest of the airfield. MAG-39 aircraft using this area will normally have one aircraft flying the route and a safety aircraft directly above, with a ceiling of 700 MSL, acting as an escort. See [Appendix C](#). The following requirements apply to use of this area:

- a. Weather shall be VFR.
- b. Tower shall advise aircraft inbound to the TERF route of the runway in use, weather, and field conditions.
- c. Tower shall advise aircraft on a left turnout of traffic in the area.
- d. TERF aircraft shall monitor tower frequency while in the Class D surface area.
- e. Aircraft exiting the area will establish communications prior to climbing above 500 feet.

5-5-19. MEDEVAC FLIGHTS

Emergency response aircraft may transit the Class D surface area utilizing the callsign *MEDEVAC* or *Mercy Air*.

5-5-20. TOWER TRAFFIC PATTERN ALTITUDES

[Table 5-10](#) lists standard tower traffic pattern altitudes. All altitudes are MSL.

5-5-21. TRAFFIC PATTERN CAPACITIES

There are no restrictions on the number of aircraft permitted in the pattern except for NVG operations. It is the responsibility of each controller to know their own limitations with respect to the number of aircraft they can handle in the pattern. Each controller has the authority to limit the number of aircraft in the pattern based on their best judgment. Supervisors should be cognizant of each controller's abilities, limitations and projected traffic when assigning positions.

5-5-22. NORTH GRASS PROCEDURES

Helicopters conducting low work in the North Grass will be restricted at or below 50 feet AGL unless approval for higher altitude from local control is granted. This will ensure any aircraft on final that may require a go-around have ample room to avoid this traffic.

- a.** No more than 2 aircraft are allowed in the North Grass at the same time. Exit from the North Grass will be direct to the parallel taxiway after establishing communication with Ground Control.
- b.** Helicopters may operate in the north grass below 50 feet AGL during all runway operations except those involving heavy jets. Ensure traffic is issued whenever fixed-wing aircraft are involved.

5-5-23. HELO SPOT OPERATIONS

The following procedures shall apply when aircraft are taxied to helo spots:

a. Duty runway 21 operations:

- 1.** Papa Spots 1 and 3 may be used for both arrivals and departures with no restriction to the size and type of helicopter. Papa 1 and 4 are to be used only for H-1 departures unless otherwise directed by the Airfield Operations Officer.
- 2.** For Papa 1 or 3 departures, coordination with ground control is necessary if the departure route will overfly any portion of the taxiway.
- 3.** Papa 4 departures will overfly the parallel and avoid overflight of the fuel farm. It is understood that aircraft calling the local controller from Papa 4 will overfly the remainder of Papa taxiway on departure.

b. Duty runway 3 operations:

- 1.** Papa Spots 1 and 3 will primarily be used for departures with no restriction to the size and type of helicopter.
- 2.** Papa 4 is to be used only for arrivals. Aircraft must avoid overflying the fuel farm.
- 3.** For Papa 3 departures, coordination with ground control is necessary if the departure route will overfly any portion of the taxiway.

c. Echo 2 and Foxtrot 2 operations:

- 1.** Aircraft may be instructed to land directly on Foxtrot 2.
- 2.** Aircraft may be issued takeoff clearance directly from Echo 2.

5-5-24. USE OF TOWER RADAR DISPLAY

a. The following procedures are set forth for the training, specific conditions, and limitations for authorized use of the Tower Display Workstation (TDW).

1. To determine an aircraft's identification, exact location, or spatial relationship to other aircraft.
2. To provide aircraft with radar traffic advisories.
3. To provide a direction or suggested heading to VFR aircraft as a method for radar identification or as an advisory aid to navigation.
4. To provide information and instructions to aircraft operating within the surface area for which the tower has responsibility.

b. Where STARS data from a system common to the TRACON and the tower is presented on the Tower Display Workstation (TDW), and an operational advantage will be gained by using the QUICK LOOK function, the following shall be adhered to:

1. Handoffs and point outs between the tower and radar may be accomplished via the CTRD.
2. Coordination to enter the class D and E airspace shall be accomplished via CTRD no later than 10 flying miles from the airport.
3. When the RATCF is closed, these functions are authorized for use by SoCal TRACON.
4. The use of the QUICK LOOK functions are authorized for use during all periods of equipment use.
5. The Local Controller has the responsibility to determine whether to use the QUICK LOOK function or another form of communication with radar (i.e., verbal, VISCOM, computer handoff). Factors to be considered are those found in the references.

REFERENCES-

NAVAIR 00-80T-114 Para 9.2.7.4, STARS Tower Display Workstation Quick Look Functions, Para 9.2.6, Tower Radar Display

5-5-25. V-22 MAINTENANCE PATTERN

Procedures for use of the MV-22 maintenance pattern:

- a.** MV-22 aircraft desiring to enter the maintenance pattern shall contact Camp Pendleton Arrival on their primary frequency and make the specific request. Controllers should approve the operation, taking weather and traffic conditions into consideration. ATC will assign a block altitude in Lima airspace area 'B' for the aircraft to operate. Using left turns, aircraft will enter a racetrack pattern parallel to the final approach course and operate under their own navigation while in VMC.
- b.** Aircraft are expected to remain within the lateral and vertical limits of Lima airspace area 'B'. ATC may issue vectors or initiate a point out from SCT in the event of a potential spill out. If additional airspace is needed, coordination shall be made with SCT.
- c.** Controllers shall use their best judgment in any circumstances not covered under these procedures.

5-5-26. ROZ JULIET

Restricted Operating Zone (ROZ) Juliet is located at the eastern edge of Class D and R-2503B (see [Appendix C](#)). This ROZ is scheduled to be active weekly, Monday through Friday. The overflight restriction is 1500 feet MSL.

TBL 5-10 Tower Traffic Pattern Altitudes	
Patterns Northwest of the Runway	
Helicopter	600
Fixed-Wing	1100
Patterns Southeast of the Runway	
Helicopter	1000
Fixed-Wing	1500
Break Altitudes	
Helicopter	1100
Fixed-Wing	1600
Fixed-Wing (southeast)	Not below pattern altitude and not after sunset

Section 6. Tower Supervisor

5-6-1. DUTIES & RESPONSIBILITIES

The Tower Supervisor (TS) is directly responsible to the FWO for all matters pertaining to air traffic control service. Primary duties are listed in NAVAIR 00-80T-114. The TS is also responsible for opening and closing procedures contained in [Appendix A](#). Additional duties and responsibilities are contained in [Chapter 2](#).

NOTE-

TS may be combined with FWO.

5-6-2. WATCH RELIEF

The possibility of an operational error during the first few minutes after assuming the watch is significant. The TS shall ensure watch checklists and position relief checklists are used during relief briefings. The watch checklist and position checklists are contained in [Appendix A](#) and are located at each operating position.

5-6-3. MANNING

Ensuring sufficient personnel are on hand to staff the facility is a joint responsibility of the TS and FWO. Tower manning is considered one qualified controller for each operating position (TS, LC, FD, and GC). During limited operational periods such as weekends or holidays only, reduced manning is authorized in accordance with [Paragraph 5-6-4](#).

5-6-4. COMBINING POSITIONS

FWOs may authorize the tower supervisor to combine operating positions during limited operational periods. Before combining operating positions, tower supervisors shall evaluate training schedules, projected traffic, weather conditions and controller abilities. In no case will safety of flight be compromised. The following positions may be combined:

- a. TS and LC
- b. FD / CD and GC

NOTE-

Circumstances may dictate that only one of the above positions is combined. Special consideration shall be given before combining the supervisor position; only under very limited operations should the supervisor position be combined.

5-6-5. TRAINING

The TS shall train trainees at all positions to the maximum extent practical. The TS is responsible to ensure ATC training is accomplished and is annotated on performance evaluation sheets. The TS's training responsibilities are as follows:

- a. Ensure that training (particularly pertinent control information) is standardized. If not, advise the tower chief or TSO and make recommendations.
- b. Ensure both OJT and classroom instruction are provided as trainees go through the ATC training syllabi.
- c. Ensure instructors provide debriefs after trainees receive time on position.

5-6-6. EQUIPMENT

The TS is responsible for coordinating equipment outages with appropriate maintenance personnel and shall assure that an equipment checkout is performed at the beginning of each shift. They shall keep the FWO advised of operational capabilities and report malfunctions and limitations as appropriate. Complete equipment descriptions and maintenance responsibilities are covered in [Chapter 8](#) of this manual.

5-6-7. BASILONE BRIDGE TRAFFIC LIGHT

Due to the height of the Basilone Bridge, a traffic signal has been installed to automatically restrict any vehicle over 9.9 feet from traversing that area. When the red light is activated, an alarm will sound; reset to green only after all IFR air traffic (including VFR practice instrument and radar approaches and all fixed wing aircraft on final approach to Runway 21 and departing Runway 3) is clear of the Approach / Departure corridor. This traffic signal may be overridden by the remote-control box located at the Local Control console. The control box shall be set to the 'auto' position during times the field is open and set to green at all other times.

NOTES-

1. *Reset procedure: from 'red', switch selection button to 'green' then back to 'auto.'*
2. *Maintenance of this light shall be conducted by Public Works (725-9803).*
3. *A mute button is available for times when there are multiple GCAs.*

5-6-8. PROCEDURES AND POLICY ENFORCEMENT

Supervisory personnel are responsible for enforcing facility directives and policies. The supervisor's responsibility in dealing with outdated or operationally ineffective directives is to bring it to the attention of the chain of command. The Procedures Evaluation Board (PEB) is the primary method for addressing improvements to the ATC system. See [Paragraph 1-1-4](#). Whenever safety is at stake, take appropriate action and advise the FWO and / or TSO to reevaluate the procedure.

5-6-9. COORDINATION

The TS will be required to coordinate with outside agencies from time to time. The supervisors shall always present themselves as professionals during all communications. Any problems that cannot be resolved between controllers shall be brought to the attention of the FWO. Additional coordination may be required in the following areas:

- a. With pilots and other controllers immediately following an incident. The TS will cooperate fully when support is required for SAR or when aircraft in distress require assistance.
- b. Information gathering is a part of fact finding. Any internal request for information regarding an incident shall be directed to the ATCFO. No information shall be released to personnel outside the chain of command without prior coordination. Any outside request for information shall be directed to the ATCFO or SNCOIC.

5-6-10. INTRUDER ALERT

Every effort will be made to identify and track unauthorized civil aircraft that enter Restricted Airspace and/or Class D airspace. The checklist in [Appendix A](#) will be used.

5-6-11. EMERGENCY LOCATOR TRANSMITTER

Notify the SCT supervisor immediately upon receipt of an Emergency Locator Transmitter (ELT) signal. Attempt to obtain fixes or bearings with the help of ATCMD personnel. Solicit the assistance of other aircraft known to be operating in the area and coordinate all additional information with the SCT supervisor to include the termination time.

REFERENCE-

FAAO JO 7110.65, Para 10-2-10, Emergency Locator Transmitter (ELT) Signals

5-6-12. EXTENDED FIELD HOURS

Changes to airfield opening or closing time shall be approved by the FWO after normal working hours and the operations department / Airfield Operation Officer during normal business hours. The FWO may only allow the field to remain open for aircraft established on final approach to the runway or within close proximity to the field at the scheduled time of closing.

a. Requests for extended airfield hours over 30 minutes require operations department approval. Extensions of 30 minutes or less may be approved by the FWO.

b. TS, as directed by the FWO, shall notify the following:

1. ARFF
2. Flight Planning
3. RS
4. Weather

5-6-13. BIRD AND AIRCRAFT STRIKE HAZARD PROGRAM (BASH)

a. All information regarding birds or animals that may be a hazard to flight safety will be reported to Flight Clearance immediately and entered in the BASH section of the facility database or emailed to the tower chief.

b. If it appears that the birds or animals will remain in the area for an extended period, add the advisory to the ATIS broadcast.

5-6-14. DEN REPORTING

a. The following items must be reported to the Domestic Events Network (DEN). This list is not all-inclusive; supervisors shall use their best judgment in any situation not covered.

1. Unauthorized laser illumination of aircraft.
2. Unusual aircraft activities in the vicinity of special interest sites.
3. Suspected human trafficking.
4. Hijackings.

b. Primary telephone number for the DEN is (844) 432-2962. Alternate numbers are (844) 663-9273 and (405) 225-2444.

Chapter 6. Radar

Section 1. General

6-1-1. EMERGENCIES

Due to mountainous terrain in the vicinity of Camp Pendleton, aircraft may require altitude changes to improve radio communication or radar reception.

- a. Chapter 10 of FAAO JO 7110.65 contains information requirements for handling emergencies.
- b. Common emergencies are outlined in [Paragraph 2-3-5](#) of this manual.
- c. NAVAIR 00-80T-114 Emergency Procedures Manual contains complete information pertaining to emergencies.

6-1-2. RADAR HANDOFF / COORDINATION FIXES

The following fixes are used for reference points when coordinating with San Diego TRACON and intra-facility coordination for handoffs, point-outs, and initial approach/holding fixes. See Low Altitude FLIPs for other fixes.

- a. **Oceanside VORTAC (OCN)**. NFG R-210 @ 2.3 DME. The Oceanside VORTAC is most often utilized by San Diego TRACON for holding aircraft inbound to Oceanside and Palomar Airports. Aircraft inbound from San Diego TRACON, using this point, are usually vectored Northeast at 4000 feet. Caution should be used around this area due to airway V23 crossing the OCN VORTAC NW-SE. The VORTAC is not located within Lima airspace.
- b. **BONDO**. NFG R-092 @ 18.7 DME. BONDO is often utilized by San Diego TRACON for coordinating handoffs on fixed-wing aircraft from the east. BONDO is not located within Lima Airspace.
- c. **CASPO**. NFG R-031 @ 19 DME. ONTOS is the IAF/IF for the TACAN Z Runway 21 Approach. Aircraft handed off for the TACAN Z are normally vectored to intercept the NFG 19 DME arc prior to CASPO or for a straight-in approach. Minimum crossing altitude at CASPO is 4900 feet MSL.
- d. **FISAM**. NFG R-031 @ 10 DME. FISAM is the Final Approach Fix (FAF) for the TACAN Z Rwy 21 Approach.
- e. **TEMEC**. OCN R-026 @ 20 DME. TEMEC is the IAF for the VOR/DME or TACAN Y Runway 21 Approach. This fix is situated approximately one-tenth of a mile outside the northeastern edge of Lima airspace. Aircraft inbound for the VOR/DME or TACAN Y Runway 21 Approach will normally be vectored to intercept the OCN 20 DME arc.
- f. **CUPIS**. OCN R-026 @ 16 DME. CUPIS is the Missed Approach holding fix for the TACAN Rwy 3 Approach. CUPIS is also the coordinated non-radar holding fix with SCT. Min / max holding altitude at CUPIS is 5200 feet MSL. It is located within Lima airspace; however, the holding pattern extends into San Diego TRACON airspace. Coordination is required with San Diego TRACON for holding.
- g. **GOZER**. NFG R-083 @ 9 DME. GOZER is the Holding and IAF for the Copter TACAN RWY 21 Approach. The airspace around GOZER should be monitored for aircraft holding or departing

GOZER for the Copter TACAN Rwy 21 Approach. Minimum crossing altitude at GOZER is 4200 feet MSL. Maximum holding altitude at GOZER is 6000 feet MSL; minimum is 4200 feet MSL.

h. CORRI. NFG R-047 @ 9 DME. CORRI is an initial approach fix for the Copter TACAN Rwy 21 Approach and the clearance limit for the CORRI 4 departure. It is within Lima airspace, less than one mile north of Fallbrook Airpark. RATCF can clear an aircraft to CORRI for the Copter TACAN Rwy 21 Approach without leaving Lima airspace or coordinating with San Diego TRACON.

NOTE-

Clearance from the Alpha portion of Lima airspace requires a point out with TRACON.

i. AMFIB. NFG R-211 @ 7.9 DME. AMFIB is the missed approach holding fix for the TACAN Z, Copter TACAN, and GPS Rwy 21 Approaches. AMFIB is outside Lima airspace.

j. HILKU. OCN R-026 @ 11 DME. HILKU is the FAF for the VOR/DME or TACAN Y Runway 21 Approach.

k. HOPIL. NFG R-031 @ 6 DME. HOPIL is the FAF for the Copter TACAN Rwy 21 Approach.

l. JEMOD. Bearing 031° @ 6.5 NM from Runway 21. JEMOD is the FAF for the RNAV (GPS) Rwy 21 Approach.

m. VISTA. OCN R-083 @ 9 DME. VISTA is utilized as a crossing fix for the BULLDOG Departure. VISTA lies outside Lima Airspace. It is the missed approach Holding Fix for the VOR-A Approach at Oceanside Municipal Airport.

n. Fallbrook Airpark (L18). NFG R-040 @ 8.5 DME. Fallbrook is a civilian airpark located within Lima airspace. It is also used as a visual reference point to aid pilots unfamiliar with the Camp Pendleton area. Fallbrook is situated on the RATCF downwind between 5 and 6 miles. Use caution due to civil air traffic in the vicinity of the airpark.

o. WIPAM. OCN R-206 @ 9 DME. WIPAM is the Holding and IAF for the TACAN RWY 3 Approach. WIPAM is also the holding fix for the published missed approach procedure for aircraft conducting surveillance and precision approaches.

p. AECIA. Bearing 027° from JEMIS, AECIA is the Initial Approach Waypoint IAWP for the RNAV (GPS) RWY 21 Approach.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-1-3. OBSTRUCTIONS

[Table 6-1](#) lists prominent obstructions within 10 NM of Camp Pendleton.

TBL 6-1 Obstructions		
Obstruction	Location	Height (MSL)
ASR-9 Antenna	113° @ 1.5	682
Water Tank	180° @ 4	449
Fallbrook Water Tower	033° @ 6.5	970

6-1-4. LANDING ZONES AND CONFINED AREA LANDING SITES

All information concerning the location, restrictions, and utilization of all out-lying landing zones is contained in the AOM and MCB's Range and Training Regulations.

6-1-5. GCA CRITICAL DATA

[Table 6-2](#) lists critical data for MCAS Camp Pendleton.

6-1-6. MODE 3/A CODE ASSIGNMENTS

a. The beacon codes assigned to MCAS Camp Pendleton for use within Lima airspace are 4534 to 4543. Codes 4001 to 4037 are assigned to MCB Camp Pendleton for use within the restricted area.

NOTE-

Code 4037 may be used by the control tower.

b. Code 1255 has been assigned to active firefighting aircraft.

REFERENCE-

FAAO JO 7110.65, Para 5-2-10, VFR Code Assignments

6-1-7. OBSTACLE DEPARTURE PROCEDURES

There are four Obstacle Departures published for MCAS Camp Pendleton: Quntn, Bulldog, Corri, and Munn departures. These departures are found in the FLIP Approach Plates.

6-1-8. INSTRUMENT APPROACH PROCEDURES

Instrument approaches are published in the FLIP Approach Plates. The approaches depict procedures for runways 21 and 3.

6-1-9. FLIGHT DATA INPUT / OUTPUT SYSTEM (FDIO)

The system allows for a computer link between facilities for the transmission of generated flight strips and associated ATC information. Procedures and terminology are contained in [Paragraph 5-1-11](#) of this manual.

6-1-10. FLIGHT PROGRESS STRIPS

Flight strips shall be used to post current data on air traffic movement. Use standard hand printed characters to prevent misinterpretation. Enter the estimated times, clearance information, position reports, and any other IFR flight data received over any communication channel. Maintain current data on the strip and file it in the appropriate place when no longer required for control purposes. The blocks of a flight strip are numbered as shown in [Figure 6-1](#).

TBL 6-2 GCA Critical Data	
Item	Description
Field Elevation	78 MSL
Threshold Elevation	Rwy 21 = 77 MSL RWY 3 = 61 MSL
Glideslope Angle	3.5°
Decision Altitude	393 MSL 315 AGL
ASR MDA	940 MSL
PAR w/o GS MDA	760 MSL
ASR MAP	0.85 NM from threshold
Final Approach Course	Heading 211°
PAR Final Approach Fix	5.03 NM

Examples of flight strips are contained in [Appendix D](#). Enter handwritten information on flight strips as follows:

a. *Standard Strip Marking*

1. Block 1: Aircraft callsign
2. Block 2: blank
3. Block 3: Number and type aircraft with equipment suffix and CWT category
4. Block 4: blank
5. Block 5: Beacon code

b. *Practice Approaches*

1. Blocks 1-5: As listed above
2. Block 6: blank
3. Block 7: Coordination fix, or position of radar contact
4. Block 8: Time of radar contact and check mark for inbound information passed to tower. An 'I' will be placed in block 8 of the flight strip to identify approaches conducted in IMC.
5. Block 9:
 - (a) Altitude in hundreds of feet
 - (b) SFA button number. 'C' around the SFA once communications have been established.
 - (c) Radar contact status
 - (d) Additional remarks
6. Block 10: Type of approach (use blocks 13 and 16 for subsequent approaches)
7. Block 11: Landing intentions (use blocks 14 and 17 for subsequent approaches)
8. Block 12: Bay assignment (use blocks 15 and 18 for subsequent approaches)

c. *IFR Arrivals*

Radar arrival strips will be marked in the same manner as the practice approach strips.

1. Block 6: Previous fix (or inbound airway if handwritten)
2. Blocks 1-5, and 7-12 are the same for an instrument approach.

Tower arrival strips:

1. Block 15: Aircraft arrival time
2. Block 18: Flight planning initials

NOTE-

IFR arrivals and VFR practice approaches will not be combined on one strip.

d. *IFR Departures*

Blocks 1, 3 and 5 remain the same. Blocks 2, 4, and 6-18 are marked as follows:

1. Block 2: Revision number

2. Block 4: Computer Identification Number (CID)
3. Block 6: Proposed departure time
4. Block 7: Requested altitude
5. Block 8: Departure airport, if other than NFG
6. Block 9:
 - (a) Clearance limit, route, altitude, remarks
 - (b) Radar contact/handoff symbol, if applicable
7. Block 12: A 'B' once the clearance has been broadcast to the pilot
8. Block 13: Time release requested
9. Block 14: Actual release time
10. Block 15: Departure time

Tower Departure strips:

1. Block 15: Aircraft departure time
2. Block 18: Flight planning initials

e. Fallbrook (IFR) Departures

Blocks 1-8 remain the same. Blocks 10-14 and 16-18 remain blank. Blocks 9 and 15 are marked as follows:

1. Block 9: Clearance void time annotated by a 'V<' and the void time.
2. Block 14: Time aircraft is airborne.
3. Block 15: Time aircraft departs Lima airspace.

f. Overflights/Point Outs

A flight strip will be written any time an aircraft is approved to transition through Lima airspace. Blocks 1-8 remain the same as an arrival strip. Blocks 9-18 are marked as follows:

1. Block 9:
 - (a) Altitude and remarks
 - (b) A 'V' will be placed in the middle of the block if the aircraft is VFR. No symbol indicates the strip is IFR.
 - (c) Routing of the aircraft
 - (d) Radar contact/handoff symbol (if applicable)
2. Block 17: Handoff facility or Point out symbol with facility identifier (i.e., PSCT or RIV)
3. Block 18: Time of handoff or exiting Lima airspace

g. Nonradar

The following information will be entered on a Nonradar flight strip:

1. Block 9:

(a) Clearance limit, 2-letter identifier for fixes on the approach procedure or holding fix. See [Table 6-3](#).

(b) Time aircraft reaches the fix

(c) Altitude

(d) Time aircraft transferred to SCT or landed

FIG 6-1
Flight Progress Strip

1	5	8	9	10	11	12
2	6			13	14	15
3				16	17	18
4						

6-1-11. ABBREVIATIONS

In addition to the contractions and abbreviations contained in FAAO JO 7110.65, abbreviations contained in [Table 6-3](#) shall be used in strip marking.

6-1-12. CIVILIAN PRACTICE GCAs

Civilian aircraft require prior approval from Airfield Operations to conduct practice GCAs.

6-1-13. TIME SHARE

MACS-1 Company K is the air traffic control company at Camp Pendleton. The company utilizes the ATNAVICS (Air Traffic Navigation Integration and Coordination System) to provide ATC services. Procedures for time share are covered by letter of agreement. See also [Paragraph 6-5-9](#).

REFERENCE-

Timeshare Letter of Agreement between MCAS Camp Pendleton and MACS-1

6-1-14. RADAR ACCEPTABILITY

Controllers shall, in accordance with FAAO JO 7110.65 and AN/FPN-68A User's Manual, check for display acceptability. PAR control interface information can be found in [Chapter 8](#).

6-1-15. DISABLING OF STARS MSAW AND CA ALARMS

a. Only the Radar Chief is authorized to temporarily inhibit the MSAW, Approach Path Monitor portion of MSAW, and/or CA functions when their continued use would adversely impact operational priorities.

b. In the unlikely event this takes place, the Facility Officer and Facility Watch Officer shall be notified as soon as possible of the circumstances requiring this action. An entry in the Daily Record of Facility Operation shall be made whenever these functions are inhibited.

c. Only during times of heavy traffic may the “inhibit” or “suppress” function for individual aircraft be used.

6-1-16. OPERATIONAL MODE TRANSITION PROCEDURES

When degradation of equipment makes it necessary to operate in other than Full-Service Level (FSL), appropriate action must be taken to notify the FWO and return Lima and / or Golf airspace to SCT. Following the action, the FWO will notify the ATCFO.

6-1-17. CONSOLIDATED WAKE TURBULENCE - RADAR

MCAS Camp Pendleton operates under the Consolidated Wake Turbulence (CWT) radar separation standards contained in the FAA JO 3760.1, 7110.126, and 7110.65. Controllers shall write the appropriate CWT category on all strips until system updates enable CWT categories to be entered via the FDIO.

<i>TBL 6-3</i> Abbreviations			
Meaning	Abbreviation	Meaning	Abbreviation
Surveillance Approach	SA	Missed Approach	MA
Precision Approach	PA	Radar Contact	R
TACAN Z Approach	TZ	Radar Service Terminated	R
TACAN Y Approach	TY	Radar Contact Lost	R
TACAN Rwy 3 Approach	T3	Radar Handoff	Ⓡ
Copter TACAN Approach	CT21	Point out to a Facility	P
Visual Approach	VA	SoCal TRACON	SCT
Full Stop	FS	Clearance Void Time	V<
Low Approach	LA	RNAV	RNAV
Touch and Go	TG	Depart VFR	V
Stop and Go	SG	Depart IFR	I
Option	OPT	ONTOS	ON
Circling	③	SCUPI	SC
Quentin Departure	QUNTN	AMFIB	AM
Bulldog Departure	BLDG	GOZER	GO
Corri Departure	CORRI	CORRI	CO
Munn Departure	MUNN		

Section 2. Radar Final Control

6-2-1. DUTIES & RESPONSIBILITIES

The final controller's duties and responsibilities include, but are not limited to:

- a. Obtaining landing clearance from the tower and relaying to the pilot on final approach.
- b. Relaying traffic information, clearances, restrictions, and other instructions from the tower or Radar Supervisor verbatim.
- c. Additional duties are outlined in NAVAIR 00-80T-114.

6-2-2. INTRAFACILITY HANDOFFS

The final controller should initiate the handoff from the arrival controller.

- a. The final controller, when the aircraft is on dogleg, will advise the arrival controller of radar contact.

PHRASEOLOGY-

(callsign), (position), RADAR CONTACT.

- b. Arrival controller will transfer control of the aircraft to the final controller.

PHRASEOLOGY-

YOUR CONTROL

(the final controller will then initiate communications with the aircraft)

PHRASEOLOGY-

YOUR CONTROL ON STANDBY

(the final controller will initiate communications with the aircraft only after the arrival controller advises the aircraft to STANDBY FOR FINAL CONTROLLER)

PHRASEOLOGY-

YOUR CONTROL ON CONTACT

(the arrival controller will instruct the aircraft to contact the final controller on a GCA frequency)

- c. Any additional instructions or restrictions will be coordinated as necessary.

6-2-3. FINAL APPROACH ABNORMALITIES

In addition to the requirements in FAAO JO 7110.65, the following procedures apply for final approach abnormalities:

- a. Safe limit exceeded. Once the aircraft is established on glide path and has been instructed to begin descent:

- 1. If, at any time, the Final Controller observes a radical target deviation, "LOW ALTITUDE ALERT. CHECK YOUR ALTITUDE IMMEDIATELY" shall be transmitted to the pilot.

PHRASEOLOGY-

LOW ALTITUDE ALERT. CHECK YOUR ALTITUDE IMMEDIATELY.

2. If, after final descent, the aircraft target touches the lower safe limit cursor, issue optional wave off. Additionally, if at a distance between one mile and where decision height intersects the glide path cursor, the aircraft is greater than 25 feet below glide path, issue optional wave off as follows:

PHRASEOLOGY-

AT LOWER SAFE LIMIT. TOO LOW FOR SAFE APPROACH. IF RUNWAY / APPROACH LIGHTS NOT IN SIGHT, (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

3. If, at a distance between one mile and where decision height intersects the glide path cursor, the aircraft target is greater than 50 feet above the glide path cursor, issue optional wave off as follows:

PHRASEOLOGY-

TOO HIGH FOR SAFE APPROACH. IF RUNWAY / APPROACH LIGHTS NOT IN SIGHT, (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

4. The safe limit has been exceeded on a surveillance approach if the target is not touching the azimuth cursor at one mile.

5. The safe limit has been exceeded on a surveillance or PAR approach if, at a distance between three miles and greater than one mile / where decision height intersects glideslope, the aircraft target is more than one target's width right / left of the centerline cursor or touches the azimuth safe limit cursor.

Note: the azimuth safe limit cursor lines are defined as parallel lines 100 feet left and right of the centerline starting at runway threshold and extending to one mile from the runway point of intercept (RPI). Then from one mile at 100 feet left and right of centerline to a point three miles from RPI and 350 feet left and right of centerline.

6. In either case, if the safe limit has been exceeded, an optional wave off shall be issued.

PHRASEOLOGY-

TOO FAR RIGHT / LEFT FOR SAFE APPROACH. IF RUNWAY NOT IN SIGHT, (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

7. If, at a distance between one mile and decision height, the aircraft target is not touching and above the glide path cursor, issue an optional wave off as follows:

PHRASEOLOGY-

TOO HIGH FOR SAFE APPROACH. IF RUNWAY NOT IN SIGHT, (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

b. Radar contact lost. The final controller shall issue an optional waveoff:

PHRASEOLOGY-

RADAR CONTACT LOST. IF RUNWAY / APPROACH LIGHTS NOT IN SIGHT, (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

1. When initiating a missed approach, notify tower by using the *Wave Off* button on the VISCOM. If time permits, coordinate with tower flight data on the reason for missed approach.

c. Tower directs/clearance not received / cancelled:

PHRASEOLOGY-

TOWER DIRECTS/ CLEARANCE NOT RECEIVED / CANCELLED. (Appropriate missed approach or wave off instructions). ACKNOWLEDGE.

6-2-4. MISSED APPROACH / WAVE OFF INSTRUCTIONS

In the event of a missed approach or wave off, the final controller shall issue appropriate instructions to the pilot in accordance with [Paragraph 6-4-8](#).

6-2-5. APPROACH COMPLETION INSTRUCTIONS

After completion or termination of each radar approach, the final controller shall issue appropriate completion instructions to the pilot in accordance with [Paragraph 6-4-9](#).

6-2-6. APPROACH COMPLETION COORDINATION

After completing the approach, the final controller shall coordinate with the tower or arrival controller, as appropriate.

a. For aircraft completing a touch and go / low approach to remain with GCA:

PHRASEOLOGY-

BUTTON (number) ON THE GO.

b. For aircraft making a full stop:

PHRASEOLOGY-

BUTTON (number) CLEAR.

c. For a change of landing intentions

PHRASEOLOGY-

(callsign) MADE A FULL STOP. BUTTON (number) CLEAR

PHRASEOLOGY-

(callsign) ENTERED TOWER PATTERN. BUTTON (number) CLEAR

PHRASEOLOGY-

(callsign) ON THE GO (WITH or NO, as appropriate) CLIMBOUT INSTRUCTIONS.

6-2-7. VISCOM PROCEDURES

The VISCOM will normally be used to exchange information between the final controller and the tower.

a. VISCOM button descriptions are contained in [Chapter 8](#) of this manual.

b. The following procedures shall be used:

1. At seven miles for jets, or six miles for helicopters and propeller aircraft, the final controller will activate buttons for type aircraft, duty runway, position, and intentions. Tower will respond by acknowledging the flashing plane figure.

2. At four miles for jets, or three miles for helicopters and propeller aircraft, the final controller will activate the 3/4 button. Tower will respond by acknowledging the flashing plane figure which relays the intentions.

3. The final controller will respond by acknowledging the flashing plane figure.

4. If *CONT* is activated and no clearance is received by 2 miles, issue a go-around (for VFR aircraft) or missed approach instructions (for IFR aircraft) unless verbal coordination is completed with the tower.

5. Any changes of aircraft intentions prior to the issuance of a clearance will be coordinated with tower flight data via the interphone. After the clearance is issued, changes shall be coordinated with the local controller.

6. Missed approaches (whether pilot- or controller-initiated) and tower clearance not received shall be coordinated via the VISCOM using the WAVE OFF button.

7. At the completion of each approach, the final controller will reset the VISCOM console by pressing the *RESET* button. The tower will respond by acknowledging the flashing plane figure.

8. During runway 3 operations, the final controller will initiate “circle to rwy 3” on the VIDS to signify the circling intentions to tower.

6-2-8. GCA – TOWER COORDINATION WITHOUT VISCOM

When the VISCOM is out of service or unavailable, coordination for instrument approaches shall be accomplished as follows:

a. At seven miles for jets, or six miles for helicopters and propeller aircraft, the final controller will coordinate with tower flight data via the interphone and pass the following information:

1. Aircraft position
2. Aircraft callsign and type
3. Intentions

PHRASEOLOGY-

SIX MILES, (callsign, type aircraft, intentions)

b. At four miles for jets, or three miles for helicopters and propeller aircraft, the final controller will coordinate with the local controller for a clearance and pass the following information:

1. Aircraft position
2. Type aircraft
3. Intentions

PHRASEOLOGY-

THREE MILES, (type aircraft, intentions)

4. The local controller will respond by issuing one of the following:

- (a) Clearance
- (b) Continue
- (c) Wave off

c. If “continue” is relayed to the final controller and no clearance is received by 2 miles, issue a go-around (for VFR aircraft) or missed approach instructions (for IFR aircraft).

d. Missed approaches (whether pilot- or controller-initiated) and tower clearance not received shall be coordinated as soon as practical. Notify the local and arrival controllers.

EXAMPLE-

LOCAL, P-1, (callsign) MISSED APPROACH.

EXAMPLE-

ARRIVAL, P-1, (callsign) MISSED APPROACH.

e. Runway 3 coordination requirements are contained in [Paragraph 6-4-13](#).

6-2-9. GCA TRAFFIC INFORMATION

The tower will pass traffic information to the appropriate final controller via the interphone. The final controller will forward the traffic information verbatim. Radar traffic advisories will be issued as deemed necessary.

6-2-10. TOWER CLEARANCES

Clearances will normally be received at the 3/4-mile check.

PHRASEOLOGY-

RWY 21 WIND (direction and speed), CLEARED TO/FOR (type landing), (other instructions as given by tower).

a. Tower may issue a low approach with an altitude restriction over personnel or other aircraft. A 600-foot MSL restriction is standard.

PHRASEOLOGY-

CLEARED LOW APPROACH AT OR ABOVE SIX HUNDRED

b. The final controller will issue the clearance verbatim.

PHRASEOLOGY-

CLEARED LOW APPROACH AT OR ABOVE SIX HUNDRED

6-2-11. PRECISION APPROACH RADAR (PAR) PHRASEOLOGY

Criteria for issuing PAR approach phraseology is contained in FAAO JO 7110.65.

Elevation Position Information:

a. The Final Controller shall issue precise glidepath information for the pilot to establish and maintain a proper rate of descent. The controller shall use the following criteria when providing information from the designated target on the elevation:

1. On glidepath - the target shall be within $\pm$ ten (10) feet of the glidepath cursor.
2. Slightly above/below glidepath - the target shall be between 11 and 25 feet of the glidepath cursor.
3. Above/below glidepath - the target shall be between 26 and 50 feet of the glidepath cursor.
4. Well above/below glidepath - the target is greater than 50 feet from the glidepath cursor.

Azimuth Position Information:

b. The Final Controller shall issue precise headings along with azimuth information for the pilot to establish and maintain proper course alignment. The controller shall use the following criteria when providing information from the designated target on the azimuth.

1. On course - the target shall be within ± 50 feet of the azimuth cursor.
2. Left/right of course - the target shall be between 51 and 100 feet of the azimuth cursor.
3. Well left/right of course - the target is greater than 100 feet from the azimuth cursor.

Critical data for PAR is located in [Table 6-2](#).

6-2-12. SURVEILLANCE APPROACH (ASR) PHRASEOLOGY

Criteria for issuing surveillance approach phraseology is contained in FAAO JO 7110.65. The arrival controller will normally issue required surveillance approach information. If the information has not been passed to the aircraft, the final controller will issue the information in [Paragraph 6-4-7](#) prior to final descent.

a. Descent instructions straight-in runway 21:

1. Using ASR:

PHRASEOLOGY-

9 MILES FROM RUNWAY, DESCEND AND MAINTAIN 2600

7 MILES FROM RUNWAY, DESCEND AND MAINTAIN 2000

6 MILES FROM RUNWAY, PREPARE TO DESCEND IN ONE MILE

5 MILES FROM RUNWAY, DESCEND AND MAINTAIN 1120

4 MILES FROM RUNWAY, ALTITUDE SHOULD BE 1620

3 MILES FROM RUNWAY, DESCEND TO YOUR MINIMUM DESCENT ALTITUDE

2 MILES FROM RUNWAY

OVER MISSED APPROACH POINT, PROCEED VISUALLY,

or,

IF RUNWAY NOT IN SIGHT, EXECUTE MISSED APPROACH

2. Using PAR azimuth:

PHRASEOLOGY-

7 MILES FROM TOUCHDOWN, DESCEND AND MAINTAIN 2000

6 MILES FROM TOUCHDOWN, PREPARE TO DESCEND IN ONE MILE

5 MILES FROM TOUCHDOWN, DESCEND AND MAINTAIN 1180

4 MILES FROM TOUCHDOWN, ALTITUDE SHOULD BE 1580

3 MILES FROM TOUCHDOWN, DESCEND TO YOUR MINIMUM DESCENT ALTITUDE

2 MILES FROM TOUCHDOWN ALTITUDE SHOULD BE 820

OVER MISSED APPROACH POINT, PROCEED VISUALLY,

or,

IF RUNWAY NOT IN SIGHT, EXECUTE MISSED APPROACH

b. Descent instructions, circling to runway 3:

1. Using PAR azimuth**PHRASEOLOGY-**

7 MILES FROM TOUCHDOWN, DESCEND AND MAINTAIN 2000

6 MILES FROM TOUCHDOWN, PREPARE TO DESCEND IN ONE MILE

5 MILES FROM TOUCHDOWN, DESCEND AND MAINTAIN 1180

4 MILES FROM TOUCHDOWN, ALTITUDE SHOULD BE 1580

3 MILES FROM TOUCHDOWN, DESCEND TO YOUR MINIMUM DESCENT ALTITUDE

2 MILES FROM TOUCHDOWN

(a) If the airport is VFR,

PHRASEOLOGY-

TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY 3, (Observe aircraft turn) CONTACT TOWER.

(b) If the airport is IFR, continue the approach to one mile before issuing circling instructions

PHRASEOLOGY-

OVER MISSED APPROACH POINT. TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY 3, (Observe aircraft turn) CONTACT TOWER.

or,

IF RUNWAY NOT IN SIGHT, EXECUTE MISSED APPROACH.

2. Using ASR:**PHRASEOLOGY-**

9 MILES FROM RUNWAY, DESCEND AND MAINTAIN 2600

7 MILES FROM RUNWAY, DESCEND AND MAINTAIN 2000

6 MILES FROM RUNWAY, PREPARE TO DESCEND IN ONE MILE

5 MILES FROM RUNWAY, DESCEND AND MAINTAIN 1120

4 MILES FROM RUNWAY, ALTITUDE SHOULD BE 1620

3 MILES FROM RUNWAY, DESCEND TO YOUR MINIMUM DESCENT ALTITUDE

2 MILES FROM RUNWAY

(a) If the airport is VFR,

PHRASEOLOGY-

TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY 3, (Observe aircraft turn) CONTACT TOWER.

(b) If the airport is IFR, continue the approach to one mile before issuing circling instructions

PHRASEOLOGY-

OVER MISSED APPROACH POINT. TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY 3, (Observe aircraft turn) CONTACT TOWER.

or,

IF RUNWAY NOT IN SIGHT, EXECUTE MISSED APPROACH.

6-2-13. PAR DISPLAY CONTROL MENU

a. The AN/FPN-68A Precision Approach Radar (PAR) display control menu shall have the following, at a minimum, active when conducting approaches:

- 1.** WHI
- 2.** WX
- 3.** MAP
- 4.** HIST
- 5.** Radar Cover

b. Range/AZ functions will be utilized to provide the most accurate course/glidepath calls during each approach.

Section 3. Radar Flight Data

6-3-1. DUTIES & RESPONSIBILITIES

The primary function of radar flight data is to coordinate arrival information with the tower and final controllers. The data controller shall receive a complete brief from the arrival controller on duty limitations. The flight data controller shall not take action to initiate or accept handoffs, point outs, or releases without approval from the arrival controller or radar supervisor. Additional duties include, but are not limited to:

- a. Receiving and posting current weather and runway in use.
- b. Updating the Equipment Status Board
- c. Other duties as set forth in NAVAIR 00-80T-114.

6-3-2. DEPARTURE COORDINATION

Radar data will assist the arrival controller at their discretion in obtaining IFR releases. Obtaining and forwarding these releases shall be handled in accordance with [Paragraph 6-4-4](#).

6-3-3. WEATHER INFORMATION

The tower is responsible for forwarding weather information to TRACON and GCA. The radar data controller will update the weather on the control display each hour or as necessary. Radar data shall ensure the supervisor, arrival, and final controllers are aware of the new weather information.

6-3-4. POSTING FLIGHT STRIPS

- a. The FDIO shall be used for processing information on aircraft departing Camp Pendleton. Verbal coordination is required with TRACON flight data when the FDIO is out of service.
- b. Radar data shall ensure all clearances for aircraft departing the GCA pattern on an IFR flight plan are handled in accordance with the letter of agreement.
- c. Radar data is responsible for preparing and placing all pending departure and arrival strips on the inactive board.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-3-5. FORWARDING FLIGHT PROGRESS INFORMATION

Radar flight data shall forward the departure time to tower flight data on all aircraft that depart Camp Pendleton on an IFR flight plan from the GCA pattern.

6-3-6. FINAL CONTROLLER COORDINATION

Radar flight data is responsible for the initial inbound coordination to the final controller. Radar data shall pass the following information:

- a. Position
- b. Callsign (if target is primary-only, say so)
- c. Type aircraft
- d. Type approach and landing intentions

e. Frequency or button number

EXAMPLE-

P-1, DATA, INBOUND. CORRI, 3 SOUTH, ATLAS 22, H-1, PAR, LOW APPROACH TO REMAIN ON BUTTON 3.

f. Changes to previously passed information

EXAMPLE-

(callsign) CHANGED INTENTIONS TO (new intentions)

6-3-7. TOWER COORDINATION

a. Radar flight data is responsible for the initial inbound coordination to the tower. This coordination may be accomplished via automation or landline. Radar data will forward inbound information to the tower as it is available and usually no later than 10 flying miles from the airport. Verbal coordination is required for all TACAN approaches to runway 3 and any non-standard procedures.

NOTE-

Non-standard intentions may include, but are not limited to stop and go, the option, or short approach.

1. Automation procedures. Once the aircraft is tagged with the appropriate scratchpad information, radar data will initiate an automated point out with the tower. Once the tower accepts the point out, the coordination is complete. Scratchpad information is listed in [Table 6-4](#).

2. Verbal procedures. Radar data will forward inbound information to tower data. At a minimum, this information should include callsign and type aircraft.

EXAMPLE-

DATA, DATA, INBOUND. 8 MILES SOUTH OF CAMP PENDLETON, GUNFIGHTER 22, H-1, TACAN RUNWAY 3.

b. Subsequent approach coordination will be conducted by the final controllers in accordance with [Paragraphs 6-2-7](#) and [6-2-8](#).

TBL 6-4					
Scratchpad Information					
Frequency		Type Approach		Intentions	
1	236.3	S	ASR	B	GCA BOX
2	277.325	G	GPS / RNAV	F	FULL STOP
3	339.8	P	PAR	I	DEPART IFR
4	319.45	T	TACAN	Z	TOWER PATTERN
5	314.075	V	VISUAL	C	CIRCLE TO RWY 3
6	306.175	O	OVERHEAD	T	TOUCH & GO
7	128.45			V	DEPART VFR
8	310.725				
9	234.325				
10	134.1				

Section 4. Arrival / Departure Control

6-4-1. DUTIES & RESPONSIBILITIES

The Arrival Control position shall combine arrival and departure functions for MCAS Camp Pendleton. The Arrival Controller's duties and responsibilities include, but are not limited to:

- a. Controlling and coordinating the movement of all instrument controlled traffic within Camp Pendleton Arrival Control area of responsibility by determining the interval to be used between successive approaches / departures; taking into consideration all aspects of the ATC situation including weather, runway in use, adequacy of NAVAIDS, reliability of communications, types of aircraft, and any other factors which may affect the overall operation.
- b. Issuing clearances to aircraft operating under arrival control jurisdiction.
- c. Initiating and accepting radar handoffs / point outs with Approach Control.
- d. Providing assistance and priority services to emergency aircraft.
- e. Maintaining radar surveillance of the MCAS Camp Pendleton Arrival Control area of responsibility and providing additional radar services, as workload permits.
- f. Incorporating the assistance of other operating positions within the RATCF, as necessary, to assist in the control of air traffic.

NOTE-

There are no restrictions on the number of aircraft permitted in the GCA pattern. It is the responsibility of each controller to know their own limitations with respect to the number of aircraft they can handle in the pattern. Each controller has the authority to limit the number of aircraft in the pattern based on their best judgment. Supervisors should be cognizant of each controller's abilities, limitations and projected traffic when assigning positions.

6-4-2. STRIP BOARD MANAGEMENT

Arrival controllers shall maintain an active and inactive strip board organized from bottom to top in order of priority.

- a. The active board shall contain flight strips on all active aircraft, to include arrivals, departures, and overflights. Strips shall be organized in the bay in the following manner from bottom to top:
 1. Departures in the order released.
 2. Arrivals in sequence from first to last.
 3. Overflights / point outs.
- b. The inactive board will contain flight strips on aircraft proposed to depart Camp Pendleton or Fallbrook, aircraft with arrival estimates to Camp Pendleton or Fallbrook and aircraft with estimates to overfly the airspace.

6-4-3. CONTROL TRANSFER

The arrival controller is responsible for coordination necessary for the transfer of control on IFR aircraft. The Radar Flight Data controller shall complete handoffs / point outs only at the direction of the arrival controller.

6-4-4. DEPARTURES

Arrival Control is responsible for the separation of all Camp Pendleton and Fallbrook IFR departures.

a. Due to the close proximity of the airport to the edge of airspace boundary with SCT, all IFR aircraft departing MCAS Camp Pendleton require prior authorization from SCT. These departures will be handled as follows:

1. Tower Flight Data will call Radar Arrival/Departure Controller and request release for Camp Pendleton departures.

2. The arrival controller is responsible for initiating the release request with SCT. The request may be made by the flight data controller when directed by the arrival controller.

3. When requesting a release from SCT, the controller shall state callsign, CID, the proposed SID and amendment number with the release request.

EXAMPLE-

CAMP PENDLETON, REQUEST RELEASE (Callsign, CID and amendment number, SID).

4. Any read back of flight plan routing by SCT shall be considered a full route or amended clearance whether or not it is stated by the controller as a full route or amended clearance. The flight plan information shall be copied verbatim by controller accepting the release and forwarded to the tower as a full route or amended clearance.

5. Once the release is obtained from SCT and all conflicts have been resolved by the arrival controller, a release along with any necessary restrictions will be forwarded to the tower.

EXAMPLE-

(Callsign, CID, restrictions, if any) RELEASED.

6. The arrival controller is responsible for coordinating with Range Air on all departures over Oceanside VORTAC.

7. When R-2503D is active with fixed-wing aircraft and the departing aircraft will overfly the Oceanside VORTAC, coordinate all IFR releases with the R-2503D controller.

b. Corri Departures. IFR Departures remaining within Camp Pendleton airspace will be handled as standard IFR departures in accordance with this paragraph.

c. Fallbrook departures shall be handled in accordance with the letter of agreement. SCT will call Arrival and request release for Fallbrook departures. Arrival control is the releasing authority for departures and responsible for separation between these departures and aircraft operating within Lima airspace.

d. All aircraft departing runway 21 on an IFR flight plan will be directed to contact SCT on 323.0 or 127.3 when communications with the aircraft are no longer required.

e. Aircraft departing runway 3 on an IFR flight plan shall be switched to Camp Pendleton Arrival (236.3 or 128.45) for sequencing and traffic advisories. A radar hand-off will be affected with SCT before the aircraft leaves Lima airspace and all potential conflicts have been resolved.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-4-5. PATTERN DESCRIPTION

- a. Upwind Leg. Outbound runway heading climbing to 3000.
- b. Crosswind Leg. At approximately 1 NM from the end of runway 21, turn left heading 050 climbing to 3000. The crosswind vector shall not be issued until the aircraft has passed 1200 feet for runway 21 departures.
- c. Downwind Leg. Heading 030, parallel to the runway and final approach course. Altitude assignment will depend on MVA requirements.
- d. Base Leg. Heading 300-320 (100-120 for approaches from the north) at pattern altitude.
- e. Dogleg. Heading 210-230 (190-210 for approaches from the north) for aircraft intercepting the final approach course less than 2 miles from the approach gate. Additional intercept information is contained in FAAO JO 7110.65.
- f. Final. Aircraft vectored for a precision or surveillance approach will be instructed to maintain an altitude at or above the minimum vectoring altitude until established within the lateral confines of the intermediate segment as depicted on the Lima video map. Once established within the confines of the intermediate segment, an aircraft may be vectored below the minimum vectoring altitude until commencing the approach as follows:
 - 1. Between 11 and 7 miles on final: an altitude at or above 2600.
 - 2. Between 7 and 5 miles on final: an altitude at or above 2000.

REFERENCES-

*FAAO JO 7110.65, Para 4-8-1, Approach Clearance, and
FAAO JO 7110.65, Para 5-9-1, Vectors to Final Approach Course*

6-4-6. ARRIVALS

SCT shall provide arrival information to GCA in accordance with the letter of agreement. SCT is responsible for issuing approach clearances for aircraft conducting TACAN, VOR/DME, or GPS approaches.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-4-7. RADAR APPROACHES

Aircraft requesting radar approaches shall be vectored in accordance with FAAO JO 7110.65. In addition, Arrival control is responsible for providing the following to aircraft conducting radar approaches:

- a. Runway / weather information, if different from that reported to SCT.
- b. Lost communications, missed approach and climb out instructions as required by FAAO JO 7110.65.
- c. Position and landing check information as required by FAAO JO 7110.65.

6-4-8. MISSED APPROACH / WAVE OFF INSTRUCTIONS

a. Fixed-wing instrument approaches. The missed approach procedures for fixed-wing aircraft conducting instrument approaches are published on the instrument approach procedure. The clearing authority (SCT) is responsible for protecting the published missed approach on the first approach after handoff (unless alternate missed approach instructions are coordinated). Arrival Control shall issue the missed approach procedure to fixed-wing aircraft on radar (ASR/PAR) approaches in accordance with the letter of agreement.

b. Fixed-wing radar approaches. The missed approach procedures for fixed-wing aircraft conducting radar approaches are outlined in the letter of agreement with SCT.

NOTE-

In the event fixed-wing aircraft are conducting multiple approaches, coordination shall be accomplished to protect for the aircraft's missed approach prior to the aircraft reaching a 3-mile final for a full stop.

c. Helicopters. Arrival control shall vector helicopters executing missed approach from instrument approaches to remain within Lima airspace. Arrival shall issue the radar missed approach procedure to all helicopters conducting radar approaches. The procedure is climb and maintain 2200 via heading 211 and expect radar vectors.

PHRASEOLOGY-

YOUR MISSED APPROACH PROCEDURE IS CLIMB AND MAINTAIN 2200 VIA HEADING 211 AND EXPECT RADAR VECTORS.

d. Wave Offs. Aircraft that are waved off by the tower or have not received a clearance by 2 miles will be issued wave off instructions. The procedure is climb and maintain 2200 via heading 211 and expect radar vectors.

PHRASEOLOGY-

CLIMB AND MAINTAIN 2200 VIA HEADING 211 AND EXPECT RADAR VECTORS.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-4-9. APPROACH COMPLETION INSTRUCTIONS

Arrival control shall issue approach completion instructions to each aircraft making approaches into Camp Pendleton.

a. Full stops.

PHRASEOLOGY-

AFTER LANDING, STOP SQUAWK, CONTACT TOWER (frequency for transient aircraft)

b. Low approach to tower pattern.

PHRASEOLOGY-

AFTER COMPLETING LOW APPROACH, MAINTAIN VFR, STOP SQUAWK, CONTACT TOWER (frequency for transient aircraft)

c. Low approach to depart VFR.

PHRASEOLOGY-

AFTER COMPLETING LOW APPROACH, SQUAWK AND MAINTAIN VFR, CONTACT TOWER (frequency for transient aircraft)

d. Low approach to remain in the GCA pattern or to depart IFR.

PHRASEOLOGY-

REMAIN THIS FREQUENCY FOR DEPARTURE CONTROL

6-4-10. LOST COMMUNICATIONS PROCEDURES

Lost communications procedures shall be issued in accordance with FAAO JO 7110.65. The procedures for Camp Pendleton are as follows:

PHRASEOLOGY-

IF NO TRANSMISSIONS ARE RECEIVED FOR ONE MINUTE IN THE PATTERN OR FIVE / FIFTEEN SECONDS ON FINAL APPROACH, ATTEMPT CONTACT CAMP PENDLETON TOWER ON 340.2 / 128.775 AND PROCEED VFR. IF UNABLE...

a. TACAN-equipped aircraft runway 21.

1. Helicopters:

PHRASEOLOGY-

CLIMB AND MAINTAIN 4200, PROCEED DIRECT CORRI, EXECUTE COPTER TACAN RUNWAY 21 APPROACH.

2. Fixed-wing / V-22:

PHRASEOLOGY-

CLIMB AND MAINTAIN 3800, PROCEED DIRECT ONTOS, EXECUTE TACAN Z RUNWAY 21 APPROACH

b. TACAN-equipped aircraft runway 03.

CLIMB AND MAINTAIN 3800, PROCEED DIRECT ONTOS, EXECUTE TACAN Z RUNWAY 21 APPROACH, CIRCLE TO RUNWAY 3.

c. If the TACAN is out of service, ask pilot to state intentions.

REFERENCE-

JO 7110.65, Para 5-10-4, Lost Communications

6-4-11. CLIMBOUT INSTRUCTIONS

Before an aircraft which plans to execute a low approach or touch-and-go begins final descent, Arrival Control shall issue climbout instructions. Limited airspace south of Camp Pendleton prevents the routine issuance of IFR climbout instructions. If weather conditions permit, issue VFR climbout instructions followed by an IFR clearance when the aircraft is established at an altitude at or above the MVA, within Camp Pendleton assigned airspace, and appropriately separated from other traffic. When weather reports indicate the aircraft is likely to encounter IFR conditions on climbout, issue the IFR climbout instructions before the aircraft reaches a 3-mile final and obtain a point out from SCT, if necessary. If SCT denies the point out, instruct the arriving aircraft to commence the climbout early enough to avoid entering SCT airspace. At the controller's

discretion, aircraft requesting practice approaches may be instructed to maintain VFR throughout the approach.

a. Runway 21 VFR PAR

PHRASEOLOGY-

~~AFTER COMPLETING LOW APPROACH / TOUCH AND GO, MAINTAIN VFR, FLY RUNWAY HEADING. CLIMB TO ONE THOUSAND TWO HUNDRED THEN TURN LEFT HEADING 050 CLIMB AND MAINTAIN THREE THOUSAND, EXPECT RADAR VECTORS.~~

b. Runway 21 IFR PAR

PHRASEOLOGY-

AFTER COMPLETING LOW APPROACH / TOUCH AND GO, FLY RUNWAY HEADING. CLIMB TO ONE THOUSAND TWO HUNDRED THEN TURN LEFT HEADING 050 CLIMB AND MAINTAIN THREE THOUSAND, EXPECT RADAR VECTORS.

c. Runway 21 IFR ASR

PHRASEOLOGY-

AFTER COMPLETING LOW APPROACH / TOUCH AND GO, FLY RUNWAY HEADING. CLIMB TO ONE THOUSAND SIX HUNDRED THEN TURN LEFT HEADING 050 CLIMB AND MAINTAIN THREE THOUSAND, EXPECT RADAR VECTORS.

d. Runway 3 VFR procedures shall be in accordance with [Paragraph 5-5-5](#) f.

e. Multiple IFR operations to runway 3 are not authorized.

f. Runway 21 VFR ASR

PHRASEOLOGY-

AFTER COMPLETING LOW APPROACH / TOUCH AND GO, MAINTAIN VFR, FLY RUNWAY HEADING. AFTER THE UPWIND NUMBERS AND LEAVING ONE THOUSAND SIX HUNDRED, TURN LEFT HEADING 050 CLIMB AND MAINTAIN THREE THOUSAND, EXPECT RADAR VECTORS.

REFERENCE-

JO 7110.65, Para 5-10-4, Lost Communications

6-4-12. RADAR CONTACT LOST PROCEDURES

If radar services cannot be provided due to equipment failure, the following procedures will take place:

a. All previously radar controlled aircraft will be issued amended control instructions in compliance with non-radar procedures, as outlined in FAAO JO 7110.65 and the Radar Supervisor's Handbook.

1. If radar contact is lost in the radar pattern and the aircraft has not started final approach, clear the aircraft to the appropriate fix and issue holding instructions in accordance with the SCT letter of agreement.

2. Once non-radar separation has been ensured and all conflicts resolved, coordinate to return Lima airspace to SCT.

b. SCT becomes the IFR authority into MCAS Camp Pendleton. Aircraft requesting an IFR approach will be cleared for the TACAN approach and communications transferred to the tower prior to the Final Approach Fix. Tower will coordinate directly with SCT to obtain releases.

REFERENCE-

Interfacility Coordination Procedures Letter of Agreement between MCAS Camp Pendleton and SoCal TRACON

6-4-13. RUNWAY 3 OPERATIONS

The following procedures apply when the duty runway is 3.

a. Instrument / radar approaches to runway 21 are authorized providing the following conditions are met:

1. The procedure shall initially be approved by the tower supervisor.
2. Arrival shall, as part of the handoff, inform the final controller on each approach that the approach is *straight-in runway 21*.
3. North Initial opposite direction operations are not authorized (i.e., VFR aircraft outbound North Initial while GCA aircraft is on final)
4. Arrival will inform aircraft when runway 21 operations are terminated. Aircraft inside 6 miles shall normally be allowed to complete the approach in progress. Alternate missed approach instructions may be issued as necessary.

EXAMPLE-

RUNWAY 21 OPERATIONS TERMINATED. EXPECT (type of approach), CIRCLE TO RUNWAY 3.

b. Circling approaches shall be conducted as follows:

1. VFR

- (a) Arrival shall advise pilots requesting circling approaches to expect to circle at 2 miles.
- (b) If the weather is such that a maneuver at two miles is impractical or if the pilot is unable to accept the conditions, coordination shall be affected for the aircraft to conduct a TACAN runway 3 approach.
- (c) Final controller shall issue circling instructions at 2 miles or as directed by the tower. All other instructions received from the tower shall be forwarded to the aircraft verbatim. The final controller shall ensure the aircraft commences the maneuver and will perform the communications transfer to the tower.

PHRASEOLOGY-

TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY THREE. CONTACT TOWER.

- (d) For tower responsibilities when multiple approaches are requested, see [Paragraph 5-5-5](#).

2. IFR

- (a) Final controller shall issue circling instructions at 1 mile or as directed by the tower. When the aircraft commences the maneuver, the final controller shall transfer control of the aircraft to the tower after pilot acknowledges the instructions.

PHRASEOLOGY-

TURN RIGHT, CIRCLE NORTHWEST FOR LEFT DOWNWIND RUNWAY THREE. CONTACT TOWER.

(b) The aircraft must report the runway in sight prior to commencing the circling maneuver. If the pilot does not report the runway in sight by one mile, issue missed approach instructions.

PHRASEOLOGY-

REPORT FIELD / RUNWAY IN SIGHT.

3. The missed approach and tower clearance not received instructions remain the same per [Paragraph 6-4-8](#).

6-4-14. OPPOSITE DIRECTION OPERATIONS

IFR ODO are not authorized (i.e., an aircraft is inbound on a surveillance approach to runway 21, circle to runway 3, while aircraft are departing runway 3 IFR or are established on the TACAN Runway 3 approach).

REFERENCE-

FAAO JO 7210.3, Paragraph 2-1-38, Opposite Direction Operations

- a. Tower must ensure departing VFR aircraft are issued a turn to avoid conflict with opposing GCA traffic.
- b. Radar must ensure separation between runway 3 IFR departures / TACAN 3 arrivals and GCA pattern aircraft (i.e., pattern aircraft shall be extended or placed in holding until the arriving aircraft has landed or the departing aircraft has achieved required separation).
- c. VISCOM and verbal coordination shall be used as memory aids during ODO.
- d. For tower responsibilities during VFR opposite direction operations, see [Paragraph 5-5-7](#).

6-4-15. VIDEO MAP REQUIREMENTS

a. At a minimum, arrival controllers shall have the following maps displayed while operating the position:

- 1. MVA Chart
- 2. Extended runway centerline
- 3. Lima airspace boundaries

Section 5. Radar Supervisor

6-5-1. DUTIES & RESPONSIBILITIES

The Radar Supervisor (RS) is directly responsible to the FWO for all matters pertaining to air traffic control service. Primary duties are listed in NAVAIR 00-80T-114. The RS is also responsible for opening and closing procedures contained in [Appendix A](#). Additional duties and responsibilities are contained in [Chapter 2](#) of this manual.

6-5-2. WATCH RELIEF

The possibility of an operational error during the first few minutes after assuming the watch is significant. The RS shall ensure watch checklists and position relief checklists are used during relief briefings. The watch checklist and position checklists are contained in [Appendix A](#) and are located at each operating position.

6-5-3. MANNING

Ensuring sufficient personnel are on hand to staff the facility is a joint responsibility of the RS and FWO. Radar manning is one qualified controller for each operating position (RS, ADC, RD, and 2 final controllers). During limited operational periods such as weekends or holidays, reduced manning is authorized in accordance with [Paragraph 6-5-4](#).

6-5-4. COMBINING POSITIONS

FWOs may authorize the radar supervisor to combine operating positions during limited operational periods. Radar supervisors shall evaluate training schedules, projected traffic, weather conditions and controller abilities before combining operating positions. In no case will safety of flight be compromised. The following positions may be combined:

- a. RS with ADC
- b. RS with RD
- c. ADC with RD
- d. FC shall not be combined with any other position.

NOTE-

Circumstances may dictate that only one of the above positions is combined. Special consideration shall be given before combining the supervisor position; only under very limited operations should the supervisor position be combined. Training is not authorized during periods in which a control position is combined with any other position. Range Air may not be combined; staffing Range Air is required by letter of agreement. The Range Air controller may be relieved only by MCB range supervisory personnel. RFC may not be combined; however, staffing may be reduced to one final controller.

6-5-5. TRAINING

The RS shall train trainees at all positions to the maximum extent practical. The RS is responsible to ensure ATC training is accomplished and is annotated on performance evaluation sheets. The RS's training responsibilities are as follows:

- a. Ensure that training (particularly pertinent control information) is standardized. If not, advise the radar chief or TSO and make recommendations.

b. Ensure both OJT and classroom instruction are provided as trainees go through the ATC training syllabi.

c. Ensure instructors provide debriefs after trainees receive time on position.

d. *Weather Minimums for PAR Trainees*

1. Except in certain situations, PAR trainees shall not work when weather is less than 1000 feet ceiling and/or 3 miles visibility.

2. With written approval from the Radar Chief, trainees nearing qualification or who have achieved a prior RFC rating may work under supervision when weather is not less than 500 feet ceiling and/or 1 mile visibility.

6-5-6. EQUIPMENT

The RS is responsible for coordinating equipment outages with appropriate maintenance personnel and shall assure that an equipment checkout is performed at the beginning of each shift. They shall keep the FWO advised of operational capabilities and report malfunctions and limitations as appropriate. Complete equipment descriptions and maintenance responsibilities are covered in [Chapter 8](#) of this manual.

6-5-7. PROCEDURES AND POLICY ENFORCEMENT

Supervisory personnel are responsible for enforcing facility directives and policies. The supervisor's responsibility in dealing with outdated or operationally ineffective directives is to bring it to the attention of the chain of command. The Procedures Evaluation Board (PEB) is the primary method for addressing improvements to ATC procedures. See [Paragraph 1-1-4](#). Whenever safety is at stake, take appropriate action deemed necessary and advise the FWO and/or TSO to reevaluate the procedure.

6-5-8. COORDINATION

The RS will be required to coordinate with outside agencies from time to time. The supervisors shall always present themselves as professionals during all communications. Any problems that cannot be resolved between controllers shall be brought to the attention of the FWO. Additional coordination may be required in the following areas:

a. With pilots and other controllers immediately following an accident or incident. The RS will cooperate fully when support is required for SAR or when aircraft in distress require assistance.

b. Information gathering is a part of fact finding. Any internal request for information regarding an incident or accident shall be directed to the ATCFO. No information shall be released to personnel outside the chain of command without prior coordination. Any outside request for information shall be directed to the ATCFO or SNCOIC.

6-5-9. TIME SHARE PROCEDURES

When necessary to conduct Time Share operations from the AN/TPN-31A (ATNAVICS), the following will apply:

a. The RS and FWO will be responsible for ensuring the midfield site is properly manned. Minimum manning will be a Controller in Charge (CIC) and 2 final controllers. The CIC will at a

minimum be qualified and current on FC and RD as well as be designated in writing by the ATCFO.

b. The RS shall ensure the hourly weather sequence, ATIS code, and information on known airport conditions are forwarded to the midfield site.

c. The CIC will be responsible for the quality of service provided by controllers on duty at the ATNAVICS. Additionally, the CIC will be responsible to:

1. Evaluate the operational readiness of the equipment using the attached checklist.
2. Notify the RS of the operational status of the ATNAVICS.
3. Report equipment outages and limitations to the company technicians.
4. Work as the coordinator between the RATCF and the final controllers.
5. Assign personnel to operating positions according to individual qualifications and training requirements and assign trainees to qualified controllers for supervision.
6. Maintain a Daily Record of Facility Operations (DRFO) to include, but not limited to, facility opening and closing, supervisor on duty, watch checklist completion, and equipment outages.
7. Ensure controllers use the RFC Run Log in accordance with normal RATCF procedures.

d. Checklist for the AN/TPN-31A:

1. Ensure all displays are operational.
2. Ensure all the frequencies are displayed and operational on the PROCOMM 2000.
3. Complete a landline check with the RATCF and tower.
4. Ensure that all required paperwork, to include RFC Run Logs and DRFO, is inside the AN/TPN-31A.

NOTE-

The AN/TPN-31A's digital display performs constant self-checks and is considered operational without the standard PAR alignment procedures as defined in NAVAIR 00-80T-114 as long as the controllers on duty and the ATNAVICS Supervisor are satisfied.

e. Coordination

1. When Time Share operations are in progress, coordination between the RATCF, tower, and ATNAVICS will be completed as follows:

2. RATCF responsibilities:

(a) RATCF is responsible for coordinating inbound aircraft with the tower via the STARS *Point Out* function.

(b) Verbal coordination shall be completed with the ATNAVICS when inbound aircraft are in the downwind of a standard traffic pattern, or as soon as practical for a non-standard pattern. The minimum amount of information that will be passed is *position, aircraft callsign and type, type approach, intentions, and frequency*.

(c) In the event a communication breakdown occurs between the ATNAVICS and tower, the RATCF will become responsible for coordination of any current operations. After completion of

the final approach, Time Share operations will cease until two-way communication can be established between the ATNAVICS and tower.

3. ATNAVICS responsibilities:

(a) The ATNAVICS will coordinate with the tower and RATCF via dedicated and recorded ring lines. In the event of an emergency, a shared frequency may be used to coordinate with either the RATCF or the tower for one approach. In the ring lines cannot be fixed in a timely manner, Time Share operations will cease, and approaches will be conducted in the RATCF.

(b) Verbal coordination shall be completed with the tower following the same verbal coordination procedures set forth in [Paragraph 6-2-8](#).

(c) When an aircraft completes an approach to return to the pattern, the CIC will coordinate with the Arrival controller as specified in [Paragraph 6-2-6](#).

(d) In the event an aircraft executes a missed approach, the CIC will advise RATCF of the instructions given.

PHRASEOLOGY-

(callsign) EXECUTING MISSED APPROACH.

4. Tower responsibilities:

(a) All coordination will be initiated by the ATNAVICS or RATCF unless there is a safety of flight issue that shall be passed from the tower. The tower is responsible for responding to the requests from both radar facilities in a timely manner.

(b) In the event a restricted low approach or a tower-directed waveoff is issued from the tower, the phraseology will follow [Paragraph 6-2-10](#).

6-5-10. EXTENDED FIELD HOURS

Changes to airfield opening or closing time shall be approved by the FWO after normal working hours and the Operations/Airfield Operation Officer during normal business hours. The FWO may only allow the field to remain open for aircraft established on final approach to the runway or within close proximity to the field at the scheduled time of closing.

a. Requests for extended airfield hours in excess of 30 minutes require Operations department approval. Extensions of less than 30 minutes may be approved by the FWO.

b. RS, as directed by the FWO, shall notify the following:

1. ATCMD
2. Longrifle
3. SCT

Chapter 7. Facility Watch Officer

Section 1. General

7-1-1. DUTIES & RESPONSIBILITIES

FWO duties are outlined in NAVAIR 00-80T-114. Additional information can be found in the FWO binder in the ATC office. The FWO may be assigned as a branch supervisor but should not be assigned to a control position.

- a. The FWO shall always be available when the facility is in operation. They will be responsible for the military performance and overall quality of ATC services provided by crew personnel.
- b. The FWO may be combined with a branch supervisor, but should not be assigned to a control position, but may be a trainee on any position. An FWO must be present at the ATC facility whenever the airfield is open and must be available to attend to any applicable matter within the facility.
- c. All complaints directed to the facility involving pilots, adjacent facilities, and general public regarding ATC services or procedures shall be forwarded to the ATCFO. The FWO is responsible for the accumulation and forwarding of initial documentation, along with complaints regarding services and procedures provided by the facility.
- d. Minimum qualifications
 - 1. Officer FWOs shall be qualified on FC and GC.
 - 2. Enlisted FWOs shall have earned a 7257 designation.
- e. FWOs shall be designated in writing by the ATCFO.

7-1-2. QUIET HOURS PROCEDURES

- a. If the airfield or Ranch House is in full quiet hours as defined in the AOM, no aircraft shall enter or depart Class Delta until approval has been granted by the ATCFO, Deputy Operations Officer, Operations Officer, Station Executive Officer, or Station Commanding Officer.
- b. The FWO is responsible for contacting the appropriate personnel listed above to determine the status of quiet hours.
- c. Notification received via Flight Planning that quiet hours have ended is sufficient to resume normal operations.

7-1-3. STATION DUTY OFFICER NOTIFICATION

- a. In the event of any incident occurring outside normal duty hours and falling under the Commanding Officer's CCIRs, the FWO will notify the Station Duty Officer as soon as time permits.
- b. Additionally, a call will be made to an ATC staff member (ATCFO / TSO / SNCOIC / Branch Chief) before the end of the watch.

7-1-4. MANNING EXCEPTIONS

Facility minimum manning levels are described in the branch supervisor sections of this manual. Any further reduction in manning levels must be approved by the ATCFO.

7-1-5. DAILY POSITION ASSIGNMENTS

Facility Watch Officers will prepare operating position assignments in accordance with NAVAIR 00-80T-114.

Chapter 8. Equipment

Section 1. Common Equipment

8-1-1. GENERAL

The ATC Facility contains all the necessary equipment to provide air traffic services. Equipment common to both the tower and radar is contained in this section.

8-1-2. VIDS

a. Each workstation has a touch screen monitor, keyboard and trackball that are used to enter data into the VIDS (Visual Information Display System). Data may be entered from workstations and maintenance stations.

b. The VIDS uses trackballs, keyboards, and a touch screen to provide primary physical interface with the system. To select operations, the operator may use any of these interfaces.

c. The VIDS replaces the Digital Altimeter Setting Indicator (DASI), VISCOM, time indicator, and wind indicators.

d. The following information can be derived from this system:

1. Digital altimeter
2. Time indicator
3. Wind speed and direction
4. Flight Data Input / Output (FDIO)
5. Automated Surface Observation System (ASOS)
6. Automatic Terminal Information System (ATIS)
7. Video camera

8-1-3. ETVS

The ETVS (Enhanced Terminal Voice Switch) allows the controller to select preset frequencies, land lines or intercom lines for transmitting and receiving or receiving only. The supervisor's display allows controllers to build custom maps for individual displays and allows controllers to record individual positions for tape talks. The Touch Entry Display (TED) is a touch-sensitive screen that displays all frequencies and land-line communication links. The equipment is operated as follows:

a. *Frequency selection.* Touching the screen over the frequency you wish to select will activate that frequency. A green light will appear. You may select to either monitor a frequency (indicated by 'RX' on the screen) or transmit (indicated by 'TX' on the screen). Additional options allow for headset or speaker selection.

b. *Land line selection.* Touching the screen over the land line you wish to use will activate that line. For those lines that require a dial code, a keypad will automatically display.

c. Emergency Radio Push-to-Talk. The ERAD PTT button, when held, transmits over the UHF and VHF emergency frequencies.

8-1-4. 1939 / 4977 / 4779 LINES

a. When using these lines, it is important to first ensure that no other controllers are using the line. For the dial lines, it is standard procedure and professional etiquette to ask if the line is clear prior to dialing a number.

b. The 1939 line uses a dial line for communications with stations outside this facility. Incoming calls are indicated by a flashing green light and a tone is sounded. All communications are recorded, and information shall be limited to ATC information. See [Table 8-1](#) for a listing of stations and numbers on the 1939 line.

c. The 4977 line is a voice activated line for communications with SCT. Incoming calls are indicated by a flashing green light and the controller's voice. All communications are recorded, and information shall be limited to ATC information.

d. The 4779 line uses a dial code for communications with stations outside the facility. Incoming calls are indicated by a flashing green light and a tone. When using a headset, the 4779 line is active once selected. Hand-held receivers shall be keyed to transmit any message. All communications are recorded, and information shall be limited to ATC information. See [Table 8-2](#) for a listing of stations and numbers on the 4779 line.

<i>TBL 8-1</i>	
1939 Line Stations and Numbers	
Number	Station
20	Palomar Tower
27	Miramar Tower
29	Miramar Ops
44	Camp Pendleton GCA
60	TRACON Area Manager
62	Palomar Sector
70	Supervisor
75	Del Mar Sector
77	Flight Data

<i>TBL 8-2</i>	
4779 Line Stations and Numbers	
Number	Station
23	Camp Pendleton Ops
24	North Island Ops
25	Miramar Ops
28	DCG (San Diego CG)
32	San Diego FSS
40	Camp Pendleton Tower

8-1-5. DALR

- a. The DALR (Digital Audio Legal Recorder) is in the terminal equipment room of building 2399.
- b. The DALR provides the legal recording capability between air traffic controllers, pilots and ground based TRACONs and towers.
- c. The DALR is capable of 24-hour recording of 64 channels of audio and GPS time code. The recorder is also capable of inserting incident tags onto tapes so that incidents may be referenced later.

8-1-6. VISCOM

- a. Controllers can access the VISCOM on the upper right side of the VIDS monitor.

- b. Each of the active radar displays is identified on the VIDS as follows:

- 1. FC1 = 01
- 2. FC2 = 02
- 3. FC3 = 03

- c. This facility is configured as shown in [Table 8-3](#).

<i>TBL 8-3</i> VISCOM Buttons and Meanings	
Button	Meaning
HELO	Helicopter
PROP	Propeller-driven aircraft
JET	Jet aircraft
OTHER	Other type aircraft
V22	V22 aircraft
FULL STOP	Full stop landing
T/G	Touch and go
LOW PASS	Low approach
TGZ	Touch and go to the tower
WAVE OFF	Initiating a waveoff
EMER...	Emergency
RESET	Reset the VISCOM

8-1-7. SUPERVISOR'S CONFIGURATION TERMINAL

The Supervisor's Configuration Terminal allows the radar supervisor to record 2 positions in radar and 2 positions in the tower simultaneously. Each facility has 2 small cassette recorders available for tape talks. Use the following steps to configure the recorders.

- a. Step 1: Using the ETVS computer, log into the ETVS network.

- 1. Select NETWORK, then scroll down to LOG IN.

- 2. Scroll down on USER ID to TWRSUP or RDRSUP.

- 3. Enter password for selected position ('rs' for radar supervisor or 'ts' for tower supervisor). Click OK.

- b. Step 2: Select home position.

- 1. Select NETWORK. Scroll down to SET HOME POSITION.

- 2. Enter password for selected position ('rs' for radar supervisor or 'ts' for tower supervisor).

- 3. Click OK.

- 4. Select position; RDRSUP or TWRSUP.

- 5. Click OK.

- c. Step 3: Assign recorders to individual positions.**
- 1.** Select ONLINE. Scroll down to SUPERVISORY RECORD.
 - 2.** Enter password for selected position ('rs' for radar supervisor or 'ts' for tower supervisor).
 - 3.** Click OK.
 - 4.** Click twice on the position(s) you want to record under the POSITIONS SELECTIONS list (it will appear under the supervisor's selection list).
 - 5.** Click on the box to the left of the recorder position in the supervisor's selection's list (an 'X' will appear in the box).
 - 6.** Click SEND.
 - 7.** Click CLOSE.
- d. Step 4: Log out. Select NETWORK, scroll down to LOGOUT.**

Section 2. Radar Equipment

8-2-1. ASR-9

The ASR-9 Airport Surveillance Radar is located at latitude 33° 17' 14.45" N and longitude 117° 19' 50.63" W, 1.5 miles from the Airport Reference Point. The FAA-owned antenna provides the same data to both this facility and TRACON.

8-2-2. GCA FREQUENCIES

GCA frequencies are listed in [Table 8-4](#).

NOTE-

128.775 may be used by RATCF only with tower's concurrence.

8-2-3. STARS INDICATOR

a. Radar systems' performance checks shall be made daily by a qualified controller through observation and use of the system. These checks should be made on a continuous basis, but shall be accomplished at least once each watch, available traffic permitting. Both primary and secondary radar systems shall be checked. MSAW / CA alarms shall not be inhibited. Acceptability of the radar is a controller / ATC supervisor determination that cannot be usurped by non-controller personnel.

b. In addition to using the systems' test map, verification of the accuracy of maps shall be accomplished in the following manner:

1. Using 'targets of opportunity' flying over displayed fixes, navigation aids, etc., verify runway centerline by observing landing and / or departing aircraft.

2. Useable target return strength of 2 or 3 will be maintained along the radar pattern.

c. Video map assignments:

1. Map 82 = Class D & E
2. Map 105 = Shell of Range airspace
3. Map 111 = Fixes
4. Map 120 = Lima airspace
5. Map 143 = Low altitude airways

d. Brightness controls: set to give the best presentation of target information.

TBL 8-4			
GCA Frequencies			
Button	Function	UHF	VHF
1	Arrival	236.3	
2	GCA	277.325	
3	GCA	339.8	
4	GCA	319.45	
5	GCA	314.075	
6	GCA	306.175	
7	GCA		128.45
8	R-2503D	310.725	
9	R-2503D	234.325	
10	R-2503D		134.1
	GCA		*128.775
	Guard	243.0	121.5
	Base Ops	279.3	
	Range	310.3	123.2

8-2-4. FPN-68 PAR

The AN/FPN-68A Precision Approach Radar (PAR) is an all-digital, color, three-dimensional azimuth, and elevation precision approach display with a 20-mile range. It incorporates:

- a. Quantitative alphanumeric presentations of deviation from glideslope and course.
- b. Deviation is displayed to the nearest foot.
- c. Distance to touchdown displayed to nearest tenth of a nautical mile.
- d. AN/FPN-68 passwords will be changed annually and disseminated for use by:
 - 1. ATCFO/SNCOIC
 - 2. Training/Branch Chiefs

NOTE:

Radar Final Controllers will have the option, based on their personal preference, whether they assign a call sign to a designated target. Controllers shall ensure that positive identification of aircraft is maintained throughout the approach.

8-2-5. TACAN MONITOR

The TACAN monitor is in the radar room between ASR-2 and PAR-1. It indicates the status on the TACAN. It has an audio and visual alarm to indicate any malfunction. All alarms shall be reported to ATCMD.

Section 3. Tower Equipment

8-3-1. TOWER FREQUENCIES

Tower frequencies are listed in [Table 8-5](#).

NOTE-

128.775 may be used by RATCF only with tower's concurrence.

8-3-2. CRASH PHONE

The crash phone is located between the flight data and local control positions. It provides direct telephone access to several stations. This system is used to inform all key personnel of any emergency and all pertinent information relating to an emergency. The opening tower crash phone test will include all stations. Proper phraseology is required to prevent any unnecessary reaction from ARFF personnel. Stations on the crash phone circuit are:

- a. ARFF
- b. PMO
- c. Emergency Dispatch Center (Mainside)
- d. Flight Clearance
- e. Weather

8-3-3. FIRE DEPARTMENT LINE

This line is located on the ETVS (FIRE RNG). It gives the tower direct contact with the base Fire Department.

8-3-4. LONGRIFLE LINE

The Longrifle positions have intercom lines to all ATC positions via the ETVS. These lines are a direct link between ATC and Range Control personnel. They are used to determine the status of ranges, artillery firing areas, mortar positions and drop zones.

8-3-5. LIGHT GUNS

There are two light guns in the tower. They are located between flight data and local and between ground control and flight data. The light guns should be kept in the charging cradles until required for use. At least quarterly, the spare battery should be put in use and the original battery put in the spare battery socket on the charger base. The light gun has a rubber eyecup which is used to aim the light gun at the aircraft. Press the appropriate color-coded button located on the handle. The authorized color signals are specified in FAAO JO 7110.65. During daytime use the light gun can be seen up to 3.6NM. Nighttime range is further.

8-3-6. AIRFIELD LIGHTING

- a. Runway 3/21 is equipped with High Intensity Runway Lights (HIRLs). All runway lights are spaced 200 feet apart. Runway lights are white. Green threshold lights mark the approach ends of the runways. Red lights indicate the departure ends of the runways.

TBL 8-5 Tower Frequencies		
Function	UHF	VHF
Tower Primary	340.2	128.775
Clearance Delivery	271.6	134.675
Ground Control	360.2	134.675
Base Ops	279.3	
Guard	243.0	121.5

b. All taxiways are outlined by blue lights.

c. A Short Approach Lighting System (SALS), type A-2, is installed at the approach end of runway 21. The approach lights are white. This system is used to aid the pilot in identifying the landing runway. A roll guidance system is available. It is operated at night, when requested by the pilot, or as deemed necessary by the controller if not contrary to the pilot's request to leave it off. The proper settings are listed in FAAO JO 7110.65.

NOTE-

Use the MALS setting table from FAAO JO 7110.65.

d. Sequenced Flashing Lights (SFLs or Strokes) are associated with the SALS. They are controlled by the Airfield Lighting Control System (AFLCS). The strokes are used only when the approach lights are activated. Reducing the intensity of the approach lights will not influence the intensity of the strokes. The intensity can be set at Low, Medium, or High. They are operated in accordance with FAAO JO 7110.65.

e. Before any taxiway or runway lights are turned off, step their intensity down to the OFF position.

f. The rotating beacon is a standard military green and split white light. The beacon is located northeast of the tower and is controlled by the AFLCS. The rotating beacon will be turned on during airfield hours in accordance with FAAO JO 7110.65.

g. The Precision Approach Path Indicator (PAPI) is a visual glide-slope indicator arranged in a single row of four lights. The effective range of the PAPI is 5 miles during the daytime and 20 miles at night. They are located near the approach end of each runway on the southeast side.

8-3-7. AFLCS

The airfield lighting control panel is utilized to control the various lighting on the airfield: taxiway lights, runway lights, approach lights and strokes, rotating beacon and PAPI. This equipment is maintained by personnel of the Facilities Department. The computer control system for operating the airfield lighting is part of the VIDS display. The AFLCS control screen is activated by pressing the 'Display AFLCS' button on the VIDS.

a. To turn on the runway, taxiway, PAPI, approach or strobe lights, click on the desired setting with the mouse or trackball.

b. Light intensity settings can be found in FAAO JO 7110.65. The strokes will only activate when the approach lights are set at step 3 or higher.

c. Pilot-Controlled Lighting (PCL) is a system that allows the pilot to activate airfield lighting when the tower is closed. All airfield lighting except the rotating beacon, compass rose and bird bath are activated by clicking the radio transmitter on 340.2 (3 times for low, 5 times for medium, 7 times for high). PCLs will be checked for operability and set to 'Enable' before closing the tower for the day.

8-3-8. TOWER DISPLAY WORKSTATION

The tower display workstation (TDW) is located between the flight data and local control positions.

8-3-9. ATIS

The ATIS is used to continuously broadcast pertinent non-control airfield information, weather, and NOTAMs to pilots. This system is designed to reduce frequency congestion and controller workload of repeating basic airfield information to individual pilots. The ATIS is broadcast on 285.45. The ATIS frequency may be used for emergency communications to aircraft. The procedure for making an ATIS broadcast is as follows:

- a.** Click the ATIS code button to open the ATIS dialog box.
- b.** To manually change the ATIS code, click the code drop-down arrow and select the appropriate code. If the next code in sequence is to be used, clicking 'OK' will automatically select it.
- c.** New weather information is automatically displayed in the 'New METAR Message' field upon receipt when the ATIS dialog box is opened.
- d.** If no information is received from the ASOS, the operator can enter weather information manually.
- e.** Enter the weather information on the ATIS screen. Enter airfield information as necessary under the weather observation text field.
- f.** Click 'Test' to listen and verify that the new message is correct. This will start the text to voice process.
- g.** Click 'OK' to replace the current ATIS broadcast with the new one.
- h.** Click 'Cancel' to deactivate the ATIS window without saving any changes.

Section 4. Backup Equipment

8-4-1. EMERGENCY POWER

- a. Backup power for the FPN-68 radar van is provided by an electrical generator operated by an automatic switch. The radar set is also connected to an Uninterrupted Power Supply (UPS) that will provide about 30 minutes of service in the event of a total loss of power.
- b. The RATCF site also has an automatic generator switch-on capability when a power fluctuation or failure occurs. In addition to generator power, the RATCF is equipped with a UPS that can support critical systems for about 60 minutes.
- c. The tower is supplied emergency power from the same generator as the RATCF. There are also UPSs located in the tower to support the tower equipment for a short time in the event of a total loss of power.

8-4-2. BACKUP COMMUNICATIONS

- a. *Secondary Transmitter or Receiver.* In case of tower communication failure, there is a selector within the ETVS console which allows engagement of a secondary transmitter or receiver for five preset frequencies. To activate the secondary system, press the desired button (Main or Standby). Inform supervisory personnel and ATCMD whenever this is done.
- b. *UHF and VHF Tunable.* The GRC-171 is a UHF transceiver with selectable frequencies, and the GRC-211 is its VHF counterpart. One of each is located at the supervisor's console in the RATCF and on the local control position. To operate the tunable radios, the power switch shall be on, the desired frequency shall be selected, and the respective UHF or VHF transmitter or receiver button shall be selected in the ETVS.
- c. *Portable Radios.* The GRC-258 Radio Set consists of two URC-200s. The URC-200 (V2) VHF/UHF Transceiver is a lightweight, tactical, Line-Of-Sight transceiver that provides AM and FM communication of non-secure voice or data. It is suitable for ManPack, vehicular, or fixed station applications.
- d. *Emergency Communications System (ECS).* The ECS provides transmit and receive capability on a system that is independently powered. This system also has a battery backup system that provides 30 minutes of operation. The system provides transmit and receive capability on UHF & VHF Guard, Tower UHF & VHF primary, and RATCF UHF & VHF primary frequencies. Use of the ECS in one facility affects the backup radios in the other facility.
- e. *Antenna Locations.* The tower's radio transmitter antennas are located on the roof of building 2360. The tower's radio receiver and ECS antennas are located on the roof of building 23100.

Section 5. Maintenance

8-5-1. MAINTENANCE RESPONSIBILITY

Maintenance on equipment used by the air traffic control facility is performed by personnel from Air Traffic Control Maintenance Division, Telephone Office, and Public Works. Maintenance responsibility is divided as follows.

8-5-2. AIR TRAFFIC CONTROL MAINTENANCE DIVISION (ATCMD)

ATCMD is responsible for the maintenance of the landlines, intercoms, radio equipment, TACAN, and radar systems. This includes the ATIS recorders and the ETVS.

8-5-3. PUBLIC WORKS (PWD)

Marine Corps Base Facilities Maintenance Department conducts weekly checks of the generators. MCAS PWD is responsible for the maintenance of items not relating to air traffic control (i.e., air conditioning, heating, water services).

8-5-4. AIRFIELD LIGHTING CONTROL

Any AFLCS control problems should be brought to the attention of the Tower Supervisor, who will call ATCMD to advise the technicians of the problem. Any AFLCS mechanical problems (i.e., burned out lights or the rotating beacon not turning) should be brought to the attention of PWD.

8-5-5. BASE TELEPHONE

Base telephone is responsible for the maintenance of the crash phone. Crash phone outages will be called to the Branch Chief during working hours. After working hours notify telephone trouble service at 725-4114 and brief the Branch Chief the next duty day.

Appendix A. WATCH ITEMS

- [1. Tower Opening Procedures](#)
- [2. Tower Closing Procedures](#)
- [3. Radar Opening Procedures](#)
- [4. Radar Closing Procedures](#)
- [5. Tower Relief Checklist](#)
- [6. Radar Relief Checklist](#)
- [7. Intruder Alert Checklist](#)

1. Tower Opening Procedures

OPENING PROCEDURES

The Tower Supervisor will ensure that the tower is manned, at a minimum, by one qualified Flight Data person and one supervisor fifteen minutes prior to field opening. The radar supervisor will ensure the radar facility is manned, at a minimum, by one supervisor fifteen minutes prior to field opening. A full crew manning of both the tower and radar facility will take place no later than five minutes prior to scheduled opening time. Tower and radar supervisors will ensure that all facility logs are kept up to date. The following actions are required to open the ATC tower for operations:

- a. *Position panels.* Properly adjust volume, panel lighting, etc. Disable Pilot Controlled Lighting (PCL) and ensure lights are no longer responsive.
- b. *Equipment requirements.* If weather conditions dictate certain equipment be operational prior to the field opening (lighting, etc.) and an outage is discovered, forward the information to the Operations department.
- c. *Fire department line (FIRE RNG).* Obtain a line check.
- d. *Longrifle coordination.* Obtain status of pertinent ranges and impact areas.
- e. *Weather:*
 - 1. Obtain current weather information for the ATIS broadcast by calling the weather office. Advise them of opening time and airfield status.
 - 2. Send a GI message to radar flight data and advise them of the weather. When the FDIO is out of service, call flight data and advise them of the airfield status and ATIS information.
 - 3. *ATIS broadcast.* Make the opening ATIS.
- f. *Aircraft Rescue and Firefighting.* Advise them of the duty runway and opening time.
- g. *Flight Clearance.* Advise them of the duty runway, airfield and NAVAID status, hot ranges, and any other pertinent information.
- h. *Crash phone test.* Initiate the crash phone test five minutes prior to opening. Advise the tower supervisor of any stations not responding to the test.
- i. *Portable Radios.* Verify that all stored frequencies are correct each day. Check the battery power daily.
- j. *Facility log.* Make appropriate opening entries in the facility log.
- k. *Traffic light.* Reset the Basilone Road traffic light to *AUTO*.
- l. *Equipment status.* Obtain status of tower equipment and log any outages.

OPENING BROADCAST

An opening broadcast will be made on all recorded frequencies except guard. The broadcast shall include the status of Class D airspace, duty runway, and other relevant information.

EXAMPLE-

“Camp Pendleton commencing ATC operations. Duty runway 21, field VFR.”

2. Tower Closing Procedures

CLOSING PROCEDURES

The Tower Supervisor will ensure that the tower is secured upon field closure. The supervisor will ensure that the following has been completed prior to securing the tower:

- a. *Field lighting.* All field lighting is off (except tower obstruction lights, ground obstruction lights, and main power). Pilot-controlled lighting will be tested and enabled for use by pilots departing or returning to the airfield during closed field operations.
- b. *Closing calls.* The following sections are notified:
 - 1. ARFF – get closing initials.
 - 2. Flight Clearance – get closing initials.
- c. *Logs and evals.* All logs are closed; all training evaluations are completed and saved in the proper location.
- d. *Flight strips.* Bundle the flight strips and bring them to the Branch Chiefs' office.
- e. *Monitors.* Turn down the brightness of all ETVS and VIDS monitors.
- f. *Traffic light.* Basilone Road traffic light shall be set to green.

CLOSING BROADCAST

A closing broadcast will be made on all recorded frequencies except guard. The broadcast shall include the status of Class D airspace and other relevant information.

EXAMPLE-

“Camp Pendleton ceasing ATC operations. Field VFR.”

CLOSING ATIS BROADCAST

An ATIS message shall be made at field closing, including all relevant information for any closed-field operations.

EXAMPLE-

“Marine Corps Air Station Camp Pendleton ATIS information (code). Camp Pendleton airport is closed. No arrivals or departures are authorized without prior permission. All aircraft use 340.2 for UNICOM when the airport is closed. Contact Longrifle on 310.3 for current range conditions prior to taxi and entry to R-2503. Closed field departures to the Intersection are not authorized. Weather is available on Metro, 342.4. Camp Pendleton airport is scheduled to reopen at (time) on (date).”

3. Radar Opening Procedures

OPENING PROCEDURES

The radar supervisor will ensure the radar facility is manned, at a minimum, by one supervisor, a radar data, and a final controller fifteen minutes prior to field opening. A full crew manning of both the tower and radar facility will take place no later than five minutes prior to scheduled opening time. Tower and radar supervisors will ensure that all facility logs are kept up to date. The following actions are required to open the RATCF for operations and shall be repeated by the oncoming crew:

- a. *Position panels.*** Properly adjust volume, panel lighting, etc.
- b. *Equipment requirements.*** Complete Equipment Checklist (Appendix A-6) and test functions described in Chapter 8.
- c. *Aural alarm.*** Check aural alarms. (MULTIFUNC, Z, A, Enter)
- d. *Longifle coordination.*** Obtain status of pertinent ranges and impact areas.
- e. *Weather.*** Determine weather based off VIDS.
- f. *Equipment status.*** Advise tower of radar equipment status.
- g. *Passdown log.*** Review Passdown Log
- h. *Facility log.*** Make appropriate opening entries in the facility log.

SOCAL COORDINATION

No later than 5 minutes prior to scheduled opening time, call SOCAL for airspace and departure frequency.

EXAMPLE-

“SOCAL, Camp Pen, request Lima airspace at the top of the hour and departure frequency.”

Appendix A-6

2. Radar Equipment Checklist

☐ ASR1	☐ NAVAID MONITOR
☐ ASR2	☐ ERAD PTT
☐ ASR3	☐ ETVS ADC
☐ PAR1	☐ ETVS RFD
☐ PAR2	☐ ETVS ASR2
☐ VISCOM1	☐ ETVS PAR1
☐ VISCOM2	☐ ETVS PAR2
☐ VISCOM3	☐ ETVS ASR3
☐ VIDS06	☐ ETVS SUP
☐ VIDS07	☐ B1
☐ VIDS08	☐ B2
☐ VIDS09	☐ B3
☐ VIDS10	☐ B4
☐ VIDS11	☐ B5
☐ VIDS12	☐ B6
☐ ECS	☐ B7
☐ FDIO	☐ B8
☐ 1939	☐ B9
4977	B10
☐ MSAW ALARMS	

4. Radar Closing Procedures

CLOSING PROCEDURES

The radar supervisor will ensure that the radar facility is secured upon field closing. The supervisor will ensure that the following has been completed prior to securing the radar room.

- a.** Equipment status and discrepancies are reported. Turn down brightness of all ETVS, VIDS, and STARS monitors.
- b.** Ops count completed and entered into the Air Traffic Activity Log in the VIDS. Input AT Coach and IPART Simulators usage into the training simulator tracker, as appropriate.
- c.** Facility log closed and training evals completed.
- d.** Manuals returned to the library.
- e.** Appropriate passdown log entries made for the next working day.
- f.** Notify SCT of field closing time, return Lima Airspace, and log their initials.
- g.** Flight strips, and Radar Final Control Run Logs submitted to the ATC office as required.
- h.** Notify ATCMD of closing time.

5. Tower Relief Checklist

EQUIPMENT CHECKOUT

- a. NAVAIDs
- b. Radar
- c. Communications / ETVS
- d. Lighting
- e. FDIO
- f. Crash phone

AIRPORT STATUS

- a. Duty runway
- b. IFR / VFR

HOT RANGES

AIRPORT CONDITIONS / STATUS

- a. Runway
- b. Grass areas
- c. Taxiways
- d. Fuel pits

WEATHER TRENDS

TRAFFIC

- a. Proposed inbound / outbounds
- b. Arrivals / departures
- c. Pattern aircraft
- d. Special requests / instructions

SPECIAL ACTIVITIES

- a. Inbound / outbound route closures
- b. Range activities
- c. Drop zones
- d. Landing zones
- e. Punchbowl / TERF route / LZ Phoenix

OTHER POSITIONS

- a. Controller workload
- b. Controller fatigue
- c. Controller training

6. Radar Relief Checklist

Arrival/Departure Control Relief Checklist

EQUIPMENT CHECKOUT

a. Communications / ETVS

b. STARS (A1)

c. NAVAIDS

d. VIDS

e. FDIO

f. GCA Capabilities

1. PAR

2. ASR

g. Alignment / acceptability check

1. ASR MAPS

DUTY RUNWAY

AIRPORT CONDITIONS

WEATHER

RANGE STATUS

CURRENT TRAFFIC

SPECIAL INSTRUCTIONS

a. SUAs Activity

Radar Watch Supervisor Relief Checklist

EQUIPMENT CHECKOUT

a. Communications / ETVS

b. STARS (A1, A2, A3)

c. NAVAIDS

d. VIDS

e. FDIO

f. PAR (P1, P2)

g. Alignment check

h. VISCOM

DUTY RUNWAY

AIRPORT CONDITIONS

WEATHER

RANGE STATUS

CURRENT TRAFFIC

SPECIAL INSTRUCTIONS

a. SUAs Activity

OTHER POSITIONS

a. Controller workload

b. Controller fatigue

c. Controller training

Radar Final Control Relief Checklist**EQUIPMENT CHECKOUT**

a. Communications / ETVS

b. STARS (A3)

c. VIDS

d. PAR (P1, P2)

e. Alignment / acceptability check

1. PAR (DH/Glideslope)

2. ASR (Alignment/SA MAP)

f. VISCOM

DUTY RUNWAY**AIRPORT CONDITIONS****WEATHER****TRAFFIC****SPECIAL INSTRUCTIONS**

a. Range activities

b. Student Training Status

1. SIMS

c. New Facility Directives/Publications

Radar Flight Data Relief Checklist**EQUIPMENT CHECKOUT**

a. Communications / ETVS

b. STARS (A2)

c. NAVAIDS

d. VIDS

e. FDIO

g. Alignment / acceptability check

1. ASR MAPS

DUTY RUNWAY**AIRPORT CONDITIONS****WEATHER****RANGE STATUS****CURRENT TRAFFIC****SPECIAL INSTRUCTIONS**

a. Closures

b. Range activities

c. Drop zones

d. Landing zones

7. Intruder Alert Checklist

UNAUTHORIZED LANDING

- a. Activate the crash phone.

PHRASEOLOGY-

“This is the tower with an intruder alert.”

- b. Instruct the aircraft to clear the runway at the approach or departure end and hold until airfield security has concluded their investigation.
- c. For departure, taxi the aircraft via the runway. The aircraft shall not be allowed to use the parallel taxiway at any time.
- d. Make appropriate entries in the facility log to include the type aircraft, callsign, entry route, etc.
- e. Complete an operational hazard report and indicate the observed flight path of the aircraft on an intruder map. Written statements are required from all controllers who observed the aircraft.

UNAUTHORIZED OVERFLIGHT OF CLASS D AIRSPACE

- a. Attempt to establish communications with the aircraft.
- b. Issue traffic advisories as necessary.
- c. Make appropriate entries in the facility log to include the color, type aircraft, callsign, entry route, etc.
- d. Obtain as much information as possible (callsign, pilot's name, pilot's rating and license number, mailing address, color and type aircraft, etc.). Contact adjacent facilities, airfields, airport managers, or the pilot.
 - 1. During normal working hours, request the pilot to call the ATC office at (760) 725-0368 or (760) 725-0354.
 - 2. During evening hours, request the pilot to call Flight Clearance at (760) 725-8016. Flight Clearance personnel should be advised to transfer the call to the tower supervisor or FWO.
- e. Complete an operational hazard report with controller statements and indicate the observed flight path of the aircraft on an intruder map. Written statements are required from all controllers who observed the aircraft.
- f. Forward all reports and statements to the ATCFO or TSO.
- g. If requested, notify the appropriate SCT supervisor of relevant aircraft information.
 - 1. San Diego sector (858) 537-5900.
 - 2. Coast sector (858) 537-5913.
 - 3. Upon request from the FAA supervisor, fax the information to SCT at (858) 537-5893.
- h. Any infraction of the CFRs shall be forwarded to the TSO in order to complete and forward all requirements to the FSDO within 10 days.

Appendix B. EQUIPMENT

FDIO – Flight Data Input / Output System

GENERAL

The FDIO is the primary method of transmitting and receiving IFR flight plans and movement messages. It consists of a monitor, keyboard, and printer.

COMMON FDIO ADDRESSES

- a. *NFGT* = Camp Pendleton Tower.
- b. *NF1A* or *NF1D* = Camp Pendleton Radar.
- c. *NK1A* = SCT Flight Data.

MESSAGE TYPES

- a. *AM* = Amendment message.
- b. *FP* = Flight plan.
- c. *HM* = Hold message.
- d. *RS* = Remove strip.
- e. *SR* = Strip request.
- f. *FR* = Full route.
- g. *WR* = Weather request.

MESSAGE FIELDS

- a. *01* = Message type.
- b. *02* = Aircraft ID.
- c. *03* = Type aircraft.
- d. *04* = Beacon code.
- e. *05* = Speed.
- f. *06* = Coordination fix.
- g. *07* = Time.
- h. *08* = Assigned altitude.
- i. *09* = Requested altitude.
- j. *10* = Route.
- k. *11* = Remarks.

COMMON MESSAGE FORMATS

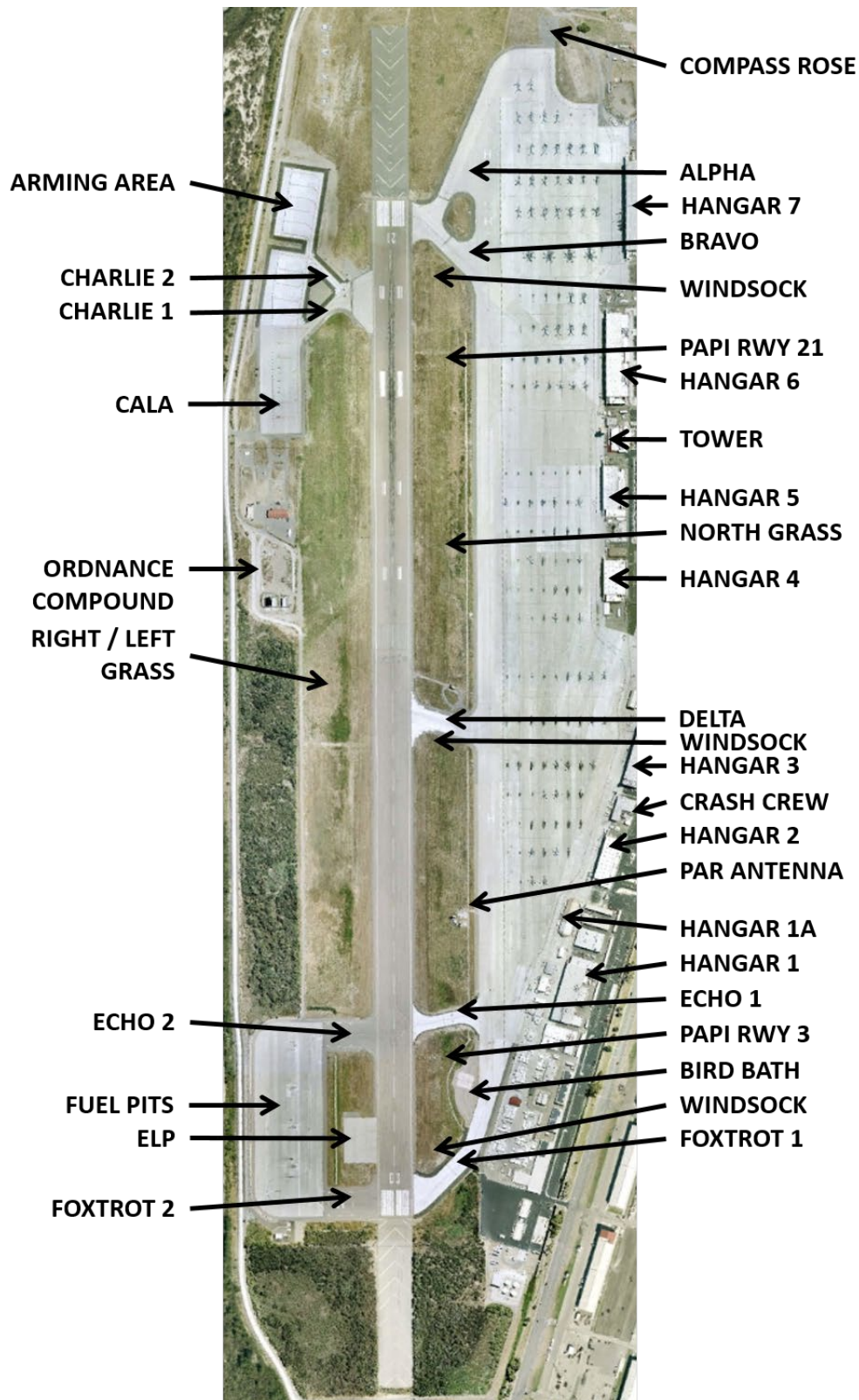
- a. *Flight Plan*. FP_Callsign_Type_Speed_Departure Fix_Proposed Time_Altitude_Route(enter)
- b. *Amendment*. AM_CID_MessageField_Amendment(enter)
- c. *Strip Request*. SR_CID_DeparturePoint_Requester(enter)
- d. *Remove strip*. RS_CID(enter)
- e. *Full route* = FR_CID(enter)
- f. *Weather request* = WR_Airport(enter)

Appendix C. AIRFIELD AND AIRSPACE

- [1. Airfield Diagram](#)
- [2. Location Identifiers](#)
- [3. Squadrons](#)
- [4. Low Altitude Airway Structure](#)
- [5. Runway 3 SVFR Arrival and Departure Routes](#)
- [6. Runway 21 SVFR Arrival and Departure Routes](#)
- [7. Special VFR Lost Communications](#)
- [8. Mike TERF Route](#)
- [9. Pipeline Patrol Route](#)
- [10. MVA Chart](#)
- [11. Lima and Golf Airspace](#)
- [12. Fallbrook and Oceanside GPS Approaches](#)
- [13. Class D and E Airspace](#)
- [14. Ranges and Impact Areas](#)
- [15. Local Flying Area](#)
- [16. LZ Phoenix](#)
- [17. ARFF Vehicles](#)
- [18. Local Traffic Patterns and Altitudes](#)
- [19. ROZ Juliet](#)

1. Airfield Diagram

FIG C-1
Airfield Diagram



2. Location Identifiers

<i>TBL C-1</i>			
LOCATION IDENTIFIERS			
NAME	IDENTIFIER	NAME	IDENTIFIER
VORTACs			
POGGI	PGY	SANTA CATALINA	SXC
SAN MARCUS	RZS	PALMDALE	PMD
GOFFS	GFS	POMONA	POM
PASO ROBLES	PRB	PARADISE	PDZ
SAN LUIS OBISPO	SBP	OCEANSIDE	OCN
SANTA BARBARA	SBA	MISSION BAY	MZB
AVENAL	AVE	PALM SPRINGS	PSP
DAGGETT	DAG	THERMAL	TRM
SEAL BEACH	SLI	JULIAN	JLI
GORMAN	GMN	29 PALMS	TNP
LAKE HUGHES	LHS	IMPERIAL	IPL
FILLMORE	FIM	NEEDLES	EED
VENTURA	VTU	BLYTHE	BLH
LOS ANGELES	LAX	BARD	BZA
MARINE AIRFIELDS			
29 PALMS	NXP	MIRAMAR	NKX
YUMA	NYL	RED BEACH	NWL
NAVY AIRFIELDS			
SAN NICHOLAS ISLAND	NSI	EL CENTRO	NJK
POINT MUGU	NTD	SAN CLEMENTE	NUC
CHINA LAKE	NID	LEMOORE	NLC
NORTH ISLAND	NZY	FALLON	NFL
IMPERIAL BEACH	NRS		
AIR FORCE BASES			
MARCH ARB	RIV	DAVIS-MONTHAN	DMA
NELLIS AFB	LSV	EDWARDS AFB	EDW
TRAVIS AFB	SUU		
MISC AIRFIELDS			
LAGUNA FIELD	LGF	FRENCH VALLEY	F70
COAST GUARD HANGAR	DCG	OCEANSIDE	OKB
LOS ALAMITOS	SLI	FALLBROOK	L18
PALOMAR	CRQ	BURBANK	BUR

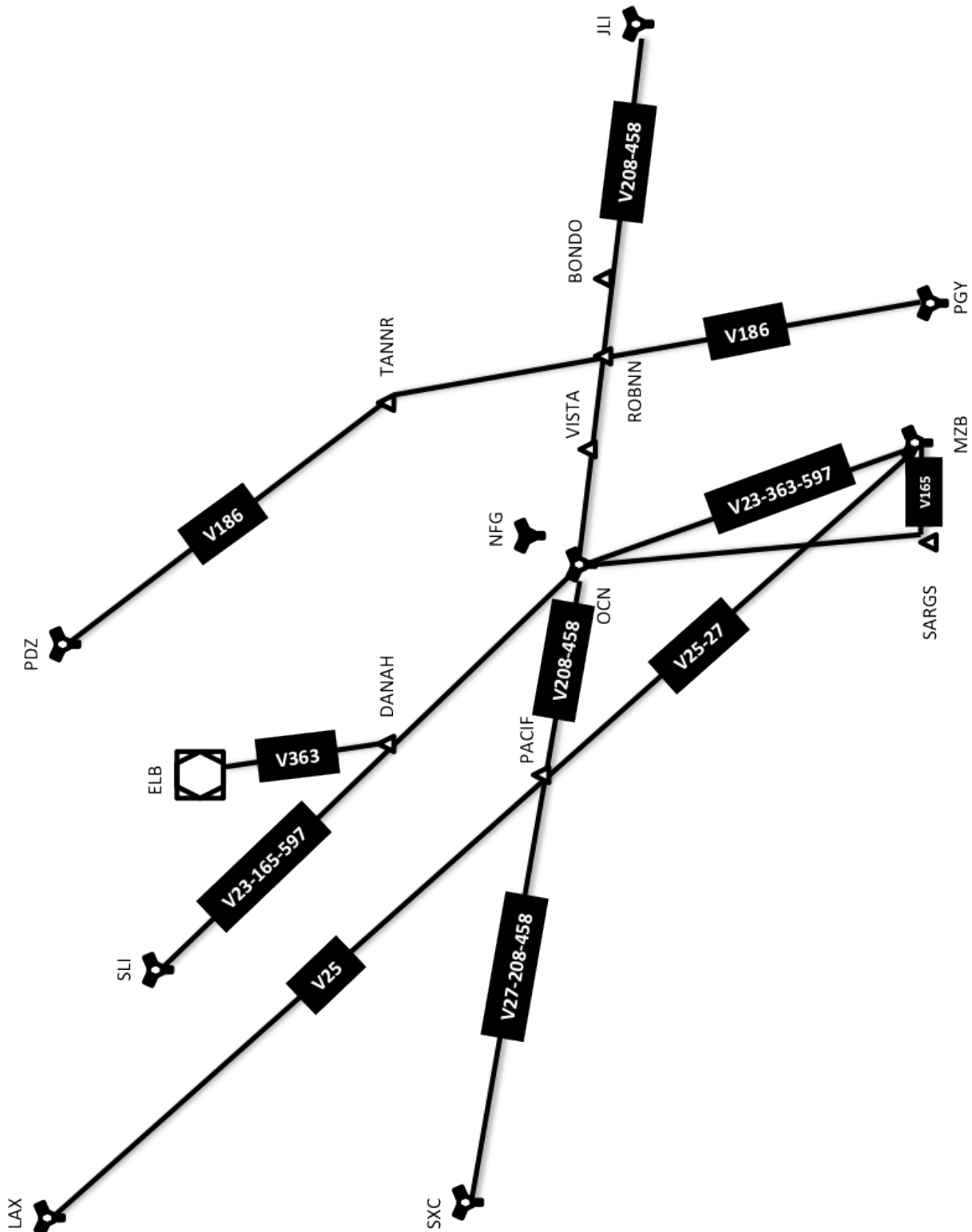
3. Squadrons

<i>TBL C-2</i> TENANT SQUADRONS				
CALLSIGN	SQUADRON	MODEX	STRIP MARKING ABBREVIATION	TYPE A/C
SCARFACE	HMLA 367	VT	SCA	H-1
VIPER	HMLA 169	SN	VIP	H-1
STINGER	HMLA 267	UV	STI	H-1
GUNFIGHTER	HMLA 369	SM	GUN	H-1
ATLAS	HMT 303	QT	ATL	H-1
KNIGHTRIDER	VMM 164	YT	KNI	MV-22
SWIFT	VMM 364	PF	SWI	MV-22
STAMPEDE	HHS		STMPD	B350

<i>TBL C-3</i> LOCAL NAVY / MARINE SQUADRONS		
CALLSIGN	SQUADRON	STRIP MARKING ABBREVIATION
ROCKY	VMM 166	ROC
THUNDER	HMH 462	THU
WARHORSE	HMH 465	WAR
LADY ACE	VMM 165	LAD
WOLFPACK	HMH 466	WOL
MOONLIGHT	VMM 764	MOO
ISLAND RULER	HSL 41	ISL
EVIL EYE	VMM 161	EVI

4. Low Altitude Airway Structure

FIG C-2
Low Altitude Airway Structure



5. Runway 3 SVFR Arrival & Departure Routes

FIG C-3
Runway 3 SVFR Arrival Routes

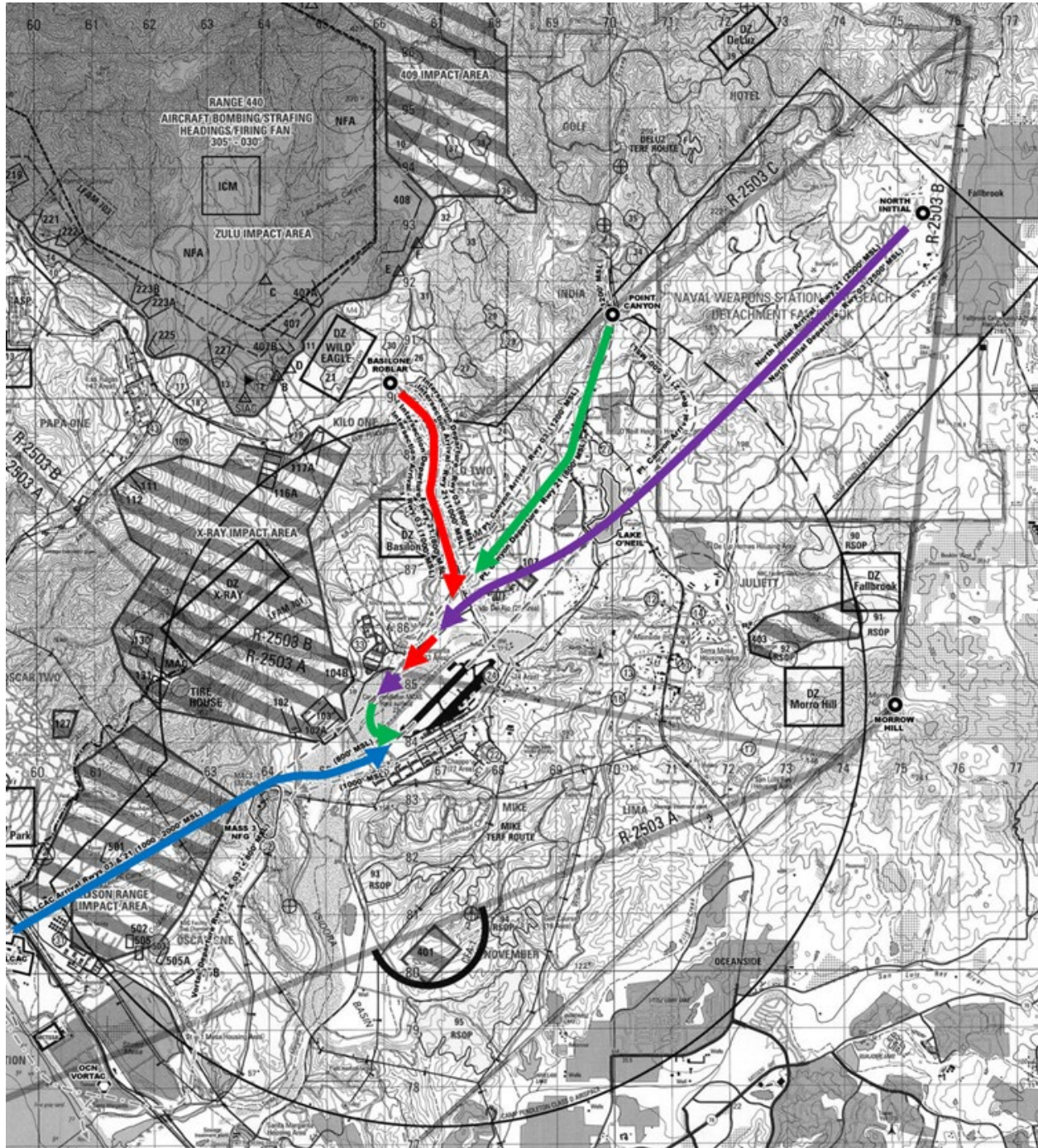
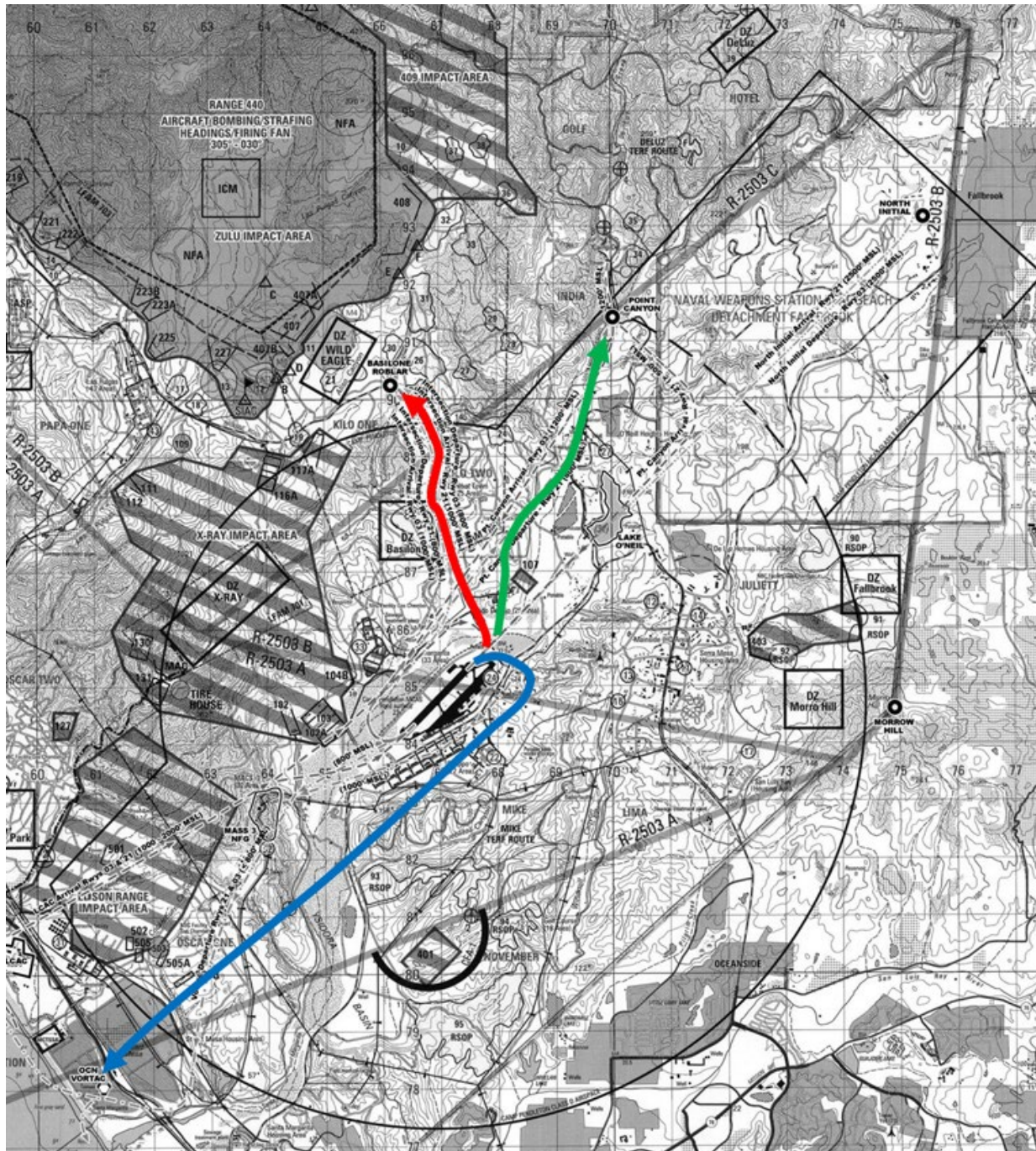


FIG C-4
Runway 3 SVFR Departure Routes



6. Runway 21 SVFR Arrival & Departure Routes

FIG C-5
Runway 21 SVFR Arrival Routes

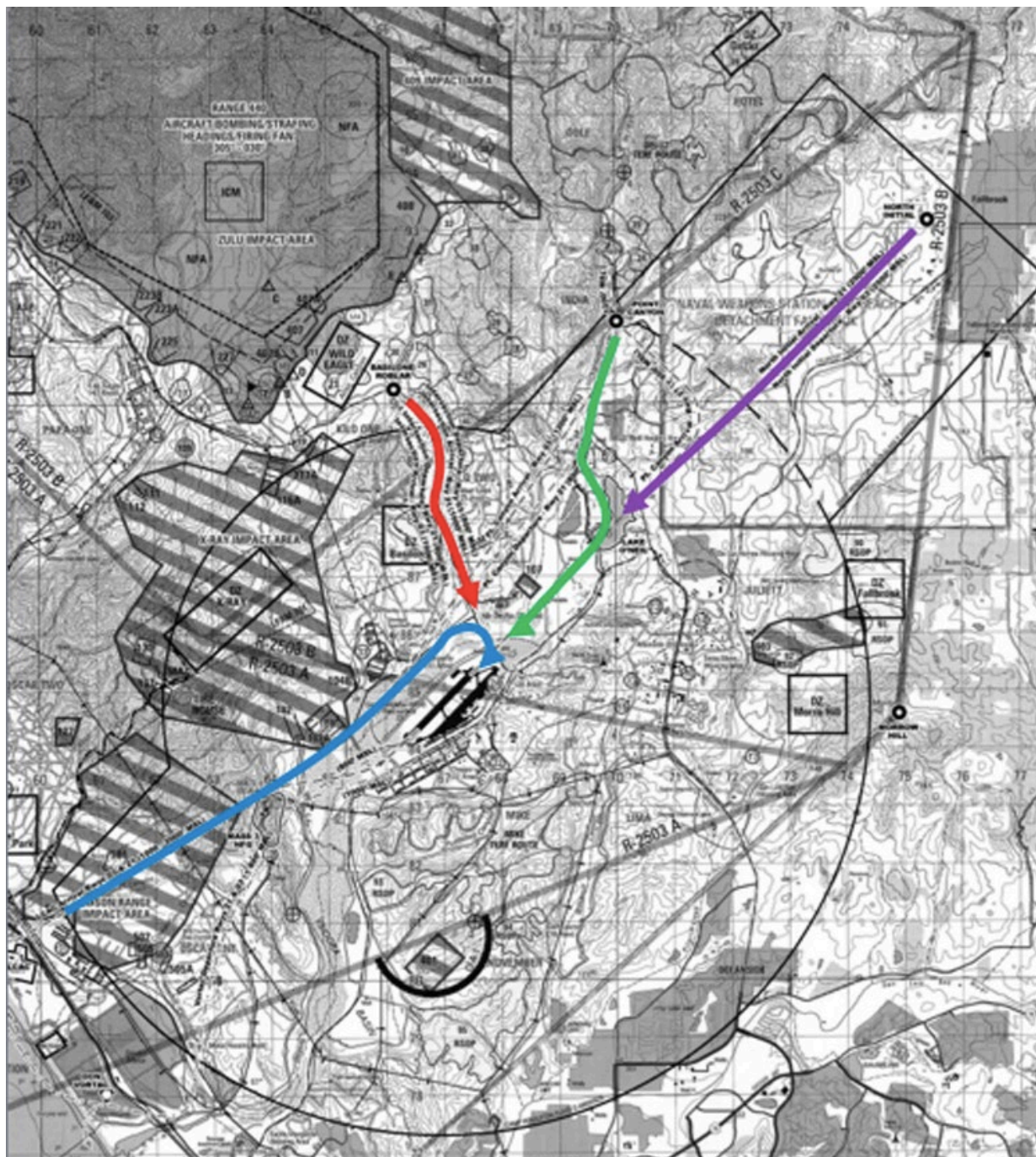
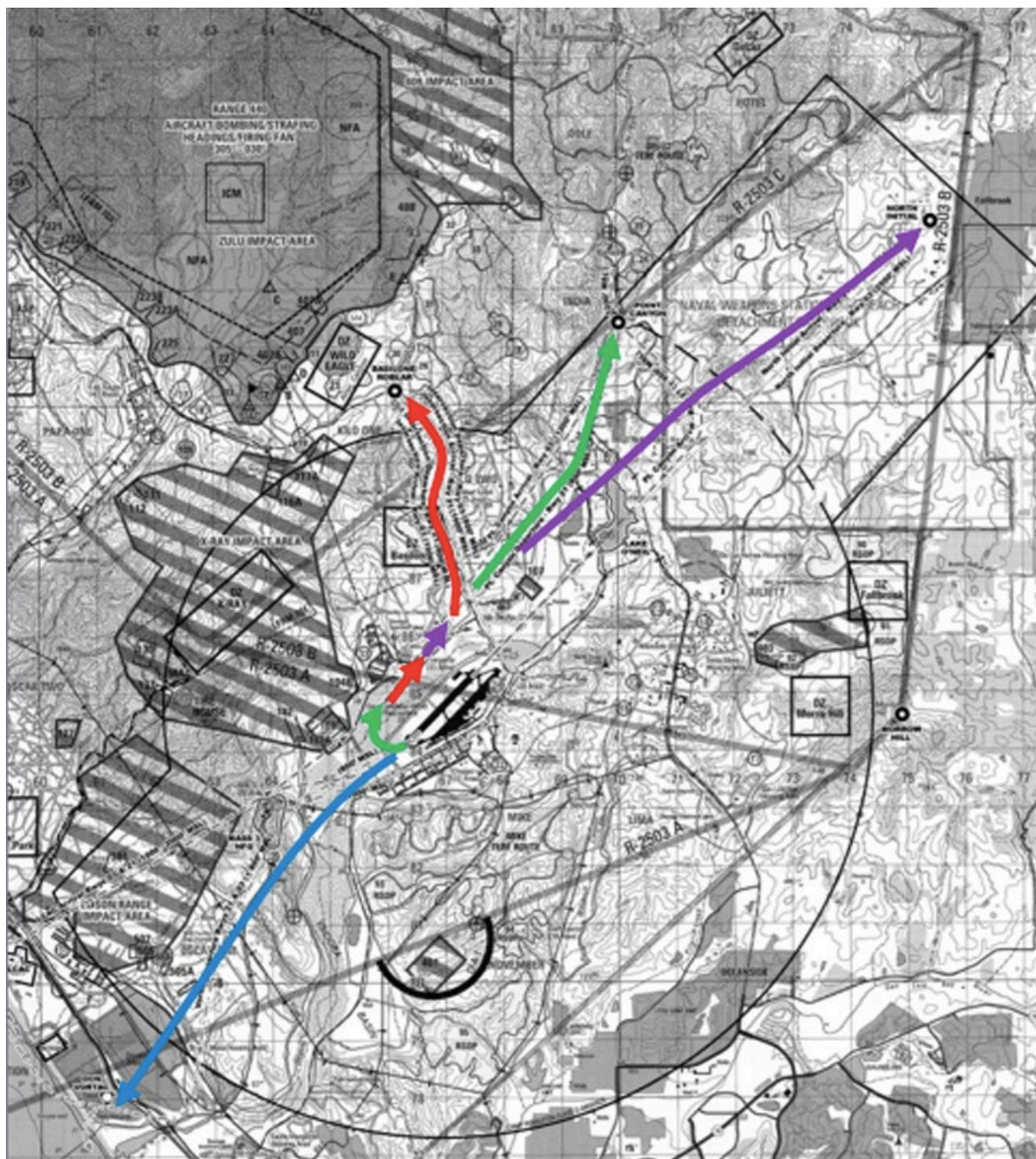


FIG C-6
Runway 21 SVFR Departure Routes



7. Special VFR Lost Communications

LOST COMMUNICATIONS

- a.** If a pilot loses radios prior to reporting clear of Class D airspace, they are expected to squawk 7600 and return to the airport for landing.
- b.** If a pilot loses radios after being cleared to enter Class D, they are expected to squawk 7600 and continue inbound for landing.
- c.** If a pilot loses radios outside Class D, they are expected to squawk 7600, climb to 4000, proceed to CORRI and execute the Copter TACAN approach.

AUTHORIZATION

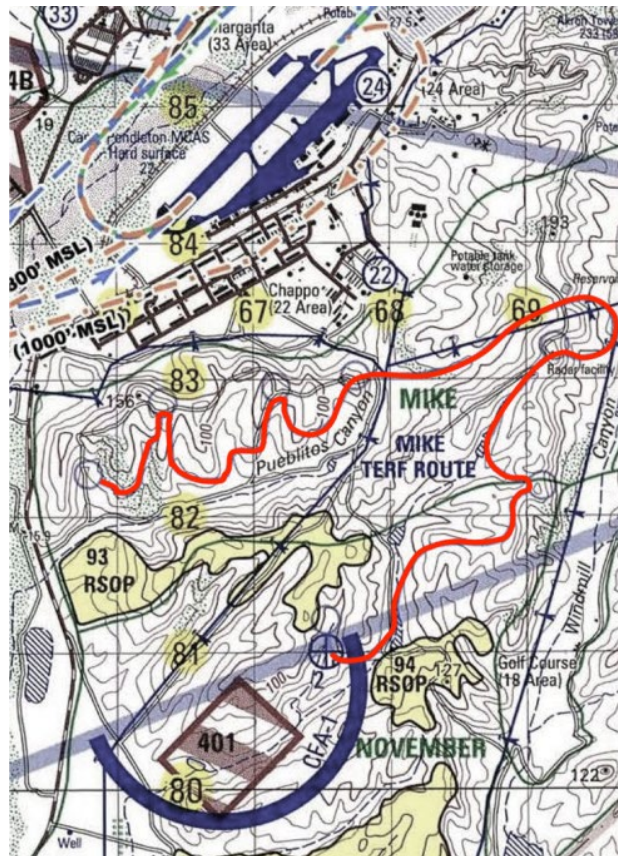
Special VFR is authorized only in that portion of Class D contained within the boundaries of Lima airspace.

8. Mike TERF Route

PROCEDURES

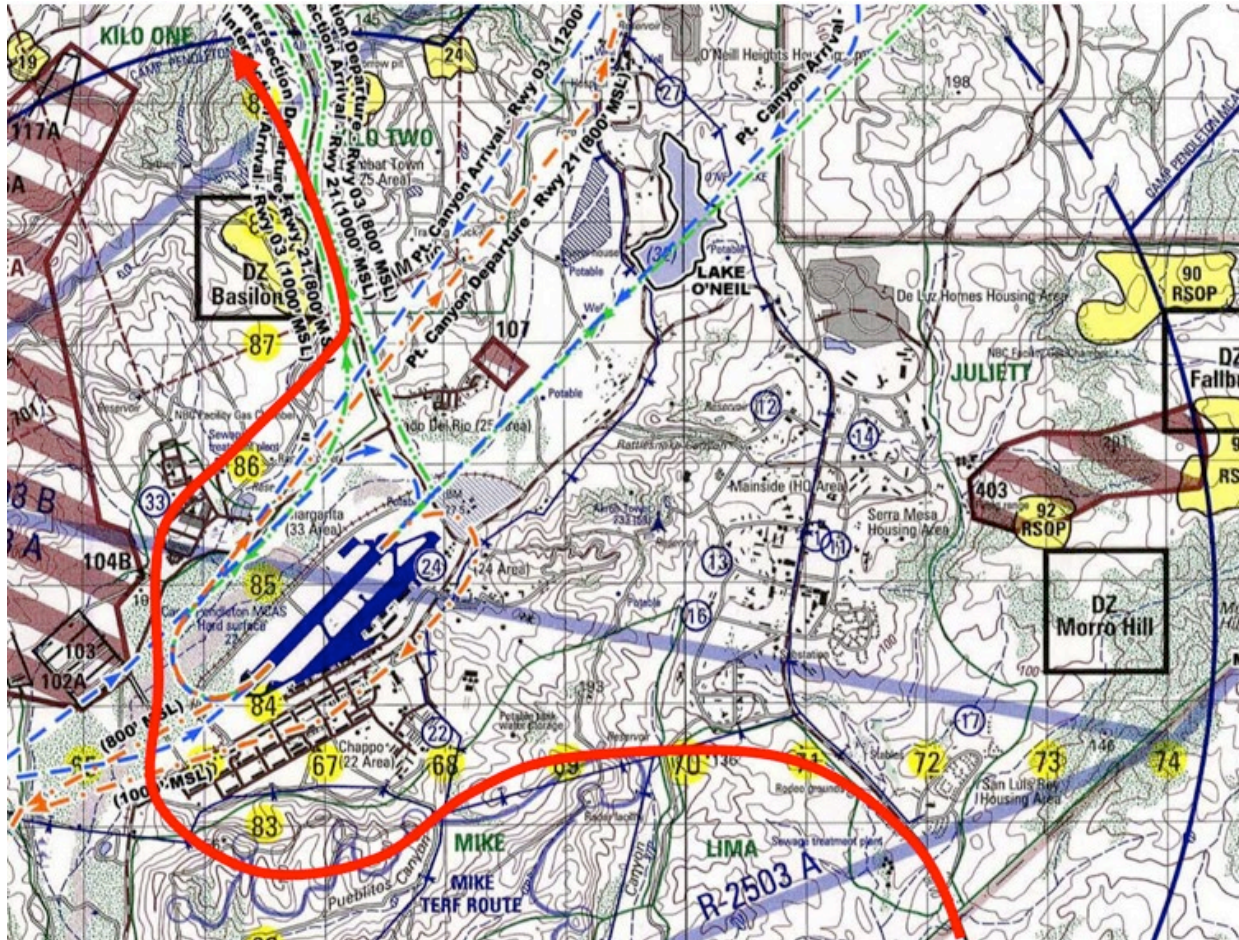
- a. Aircraft flying this route will usually have an escort aircraft flying above at 700 feet.
- b. The airport shall be VFR in order to use this area.
- c. Tower will advise GCA when aircraft are using this area.
- d. Tower will advise aircraft inbound to the TERF route if any other aircraft are known to be using the same area.
- e. Aircraft shall monitor the tower's frequency while in the Mike TERF.
- f. When exiting the area, aircraft shall establish communications with the tower before climbing above 500 feet.

FIG C-7
Mike TERF Route



9. Pipeline Patrol Route

FIG C-8
Pipeline Patrol Route



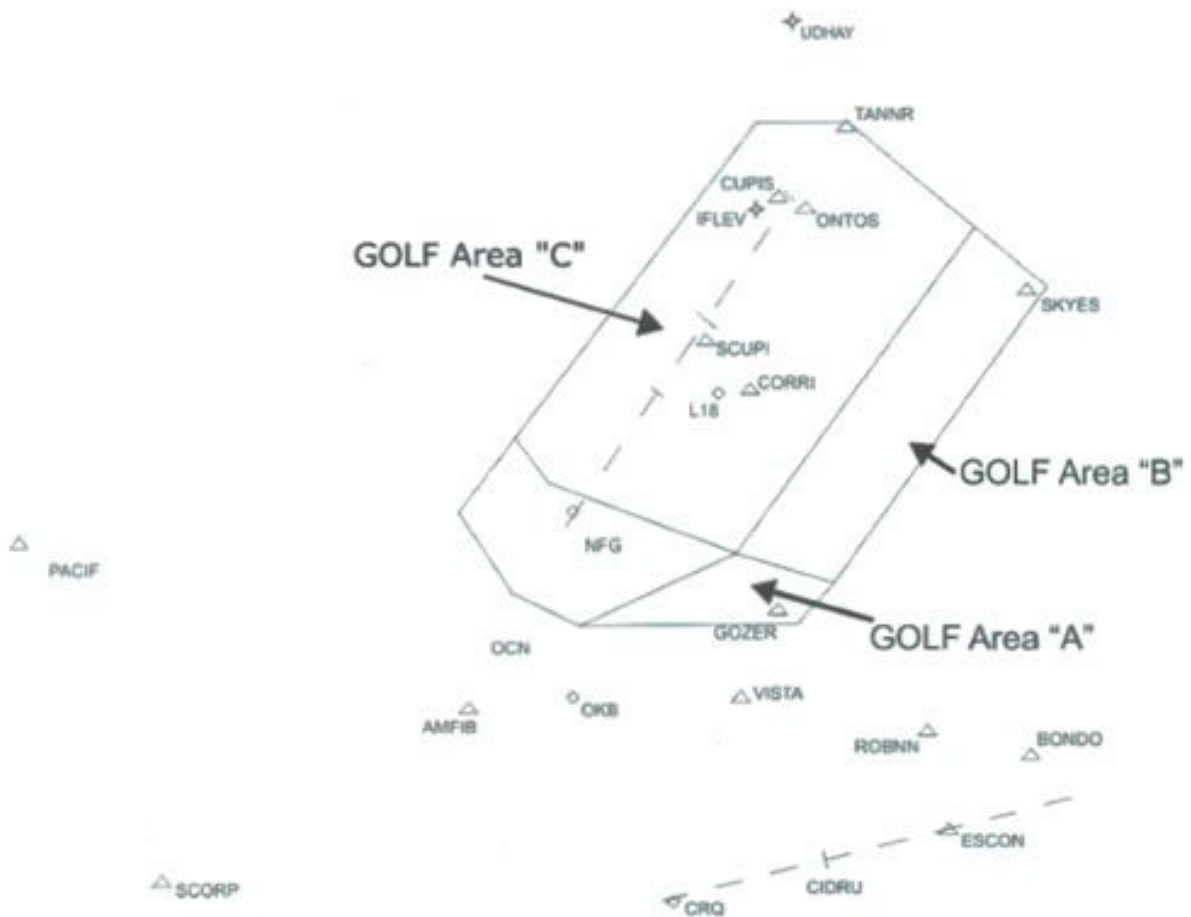
10. MVA Chart

FIG C-9
Minimum Vectoring Altitude Chart



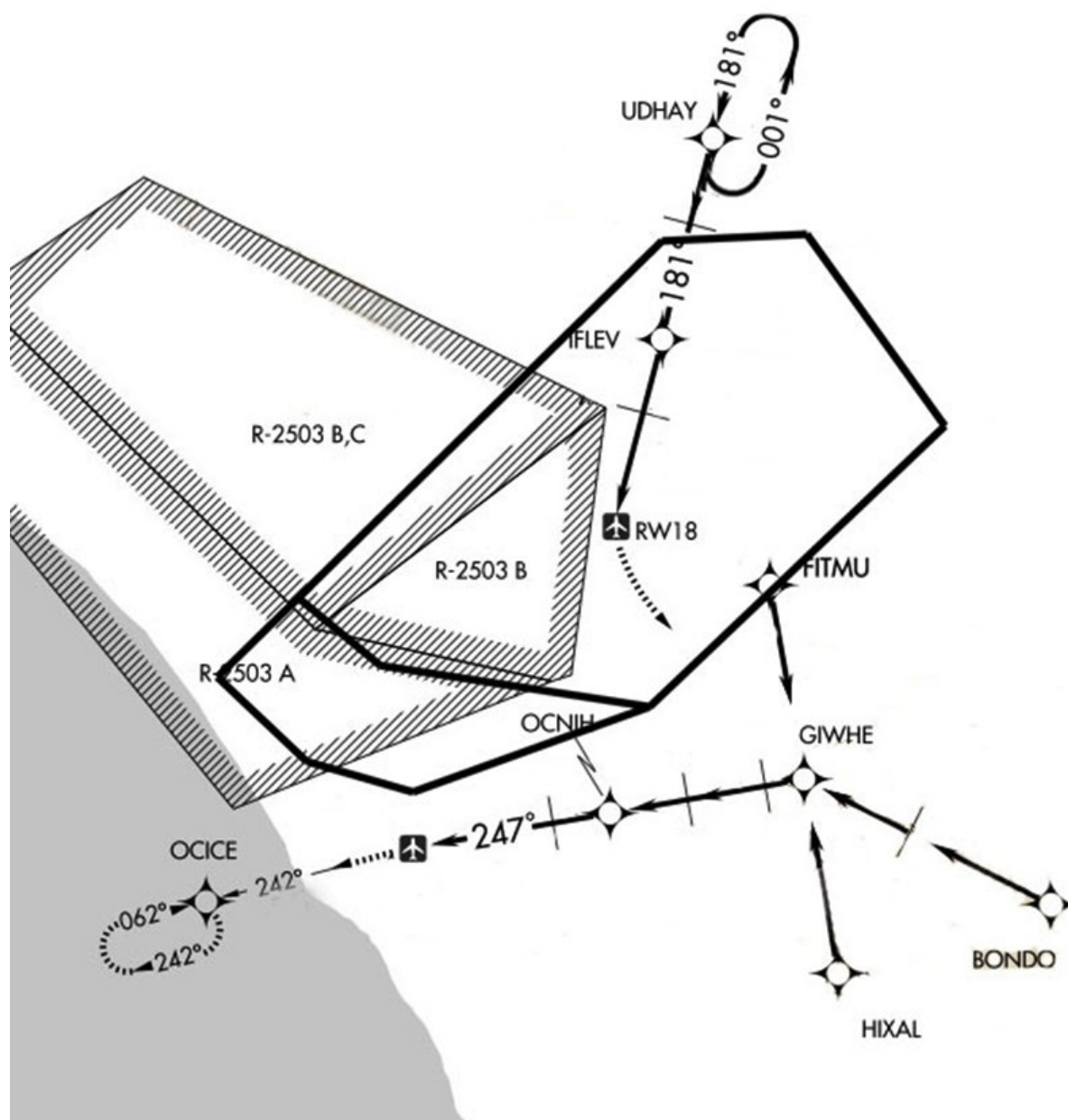
11. Lima and Golf Airspace

FIG C-10
Lima and Golf Airspace



12. Fallbrook and Oceanside GPS Approaches

FIG C-11
Fallbrook and Oceanside GPS Approaches



13. Class D and E Airspace

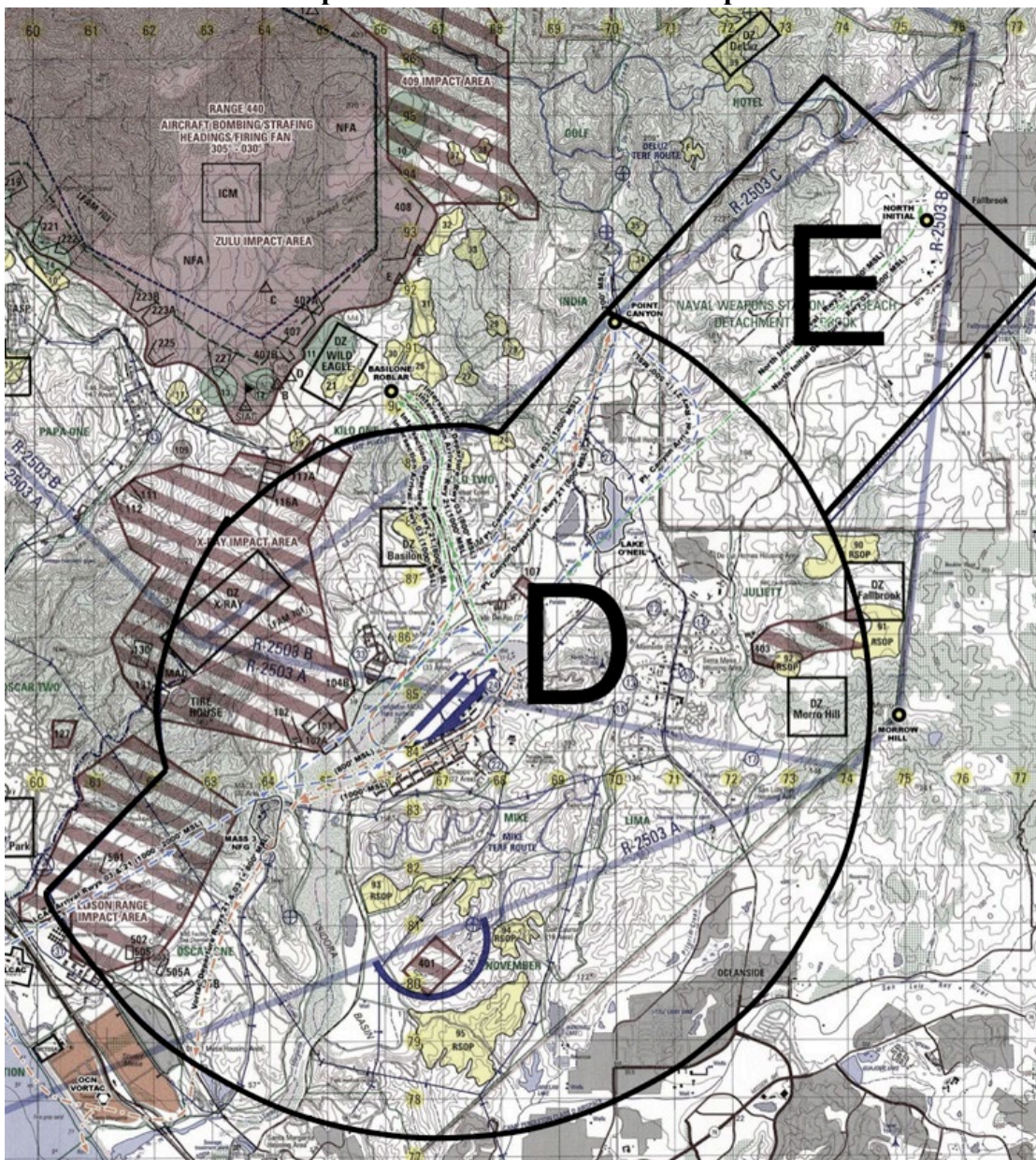
CLASS D

The circular airspace extending from the surface up to and including 2600 MSL from the Airport Reference Point (ARP) to 010° at 4NM, arcing southeast to 227° at 4NM, direct to 240° at 2.6NM, arcing northeast to 355° at 2.6NM, direct to 010° at 4NM.

CLASS E

The airspace extending from the surface up to and including 2600 MSL from the ARP to 010° at 4NM, direct to 018° at 6.9NM, direct to 041° at 6.9NM, direct to 050° at 4NM, direct to 010° at 4NM.

FIG C-12
Camp Pendleton Class D and E Airspace



14. Ranges and Impact Areas

GENERAL

The ranges and training facilities in this section impact Class D and E airspace.

RANGES

- a. Range 102 (Wilcox)
- b. Range 103 (Wilcox)
- c. Range 111
- d. Range 116A (SEAL complex)
- e. Range 127
- f. Range 130
- g. Range 131 (MOUT)
- h. Range 403
- i. Range 409 (MPRC)
- j. Range 501 (Edson)
- k. Range 503 (Edson)
- l. 25 Area Combat Town

ARTILLERY FIRING AREAS (AFAs)

- a. AFA 19
- b. AFAs 21 to 36
- c. AFA 39

IMPACT AREAS

- a. Edson Range (cross at or above 1000 AGL)
- b. X-Ray impact area

TERF ROUTES

- a. Mike TERF
- b. Deluz TERF

V/STOL PAD

- a. NFG 244 @ 3.8

LANDING ZONES

- a. LZ1. Northwest corner of 11 Area parade field, adjacent to MCB Headquarters
- b. LZ11. East side of 11 Area parade field
- c. LZ12. Behind building 1261, across from PMO
- d. LZ14. Near building 1491, 1st FSSG Headquarters
- e. LZ16. HQBN motor pool
- f. LZ17. North of San Luis Rey quarters (VIP only)
- g. LZ22. Northeast side of building 22172, across from chapel
- h. LZ27. East of Naval Hospital
- i. LZ31. Edson Range barracks blacktop
- j. LZ33. East of building 33458, Margarita area

CALSITES

- a. CAL site 1
- b. CAL site 2
- c. CAL site 3
- d. CAL site 4
- e. CAL site 5

DROP ZONES

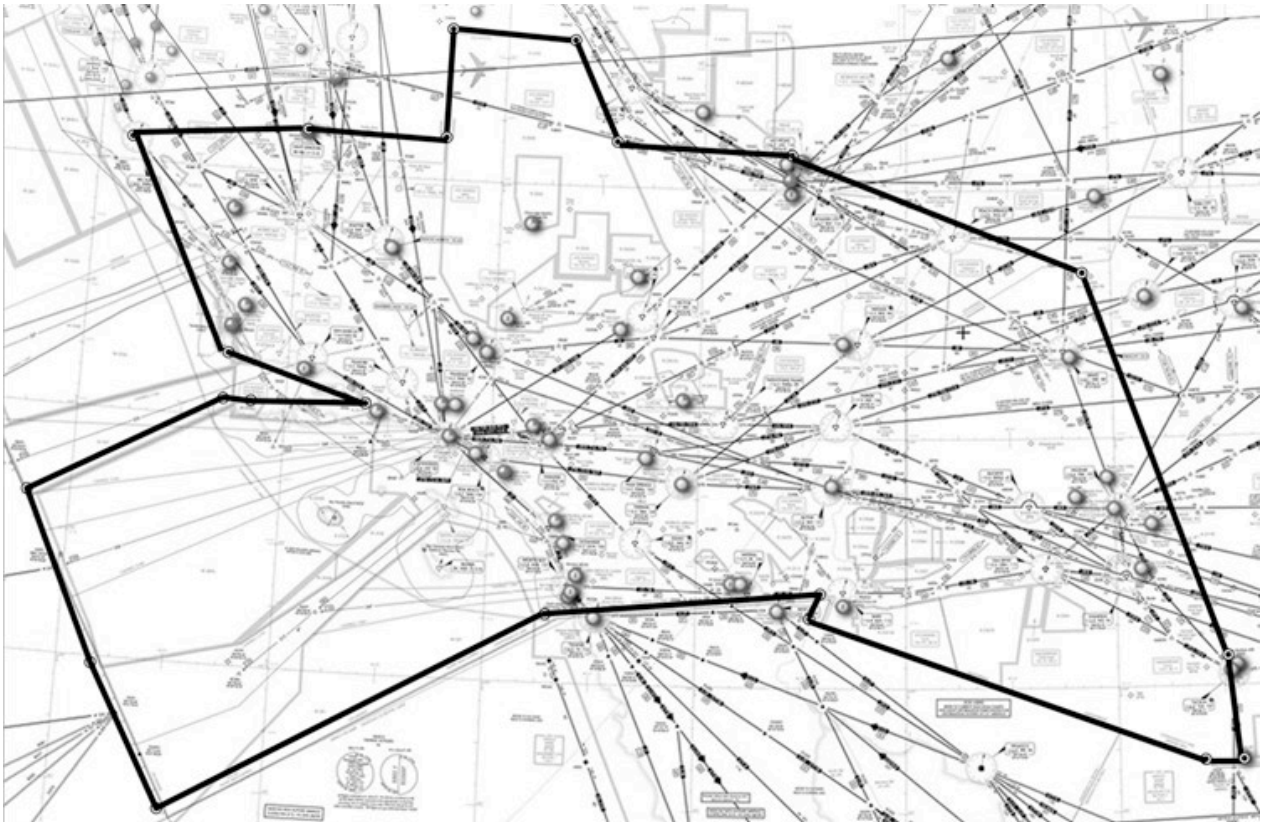
- a. DZ Basilone
- b. DZ Tank Park
- c. DZ Fallbrook
- d. DZ Morro Hill
- e. DZ Deluz

15. Local Flying Area

DEFINITION

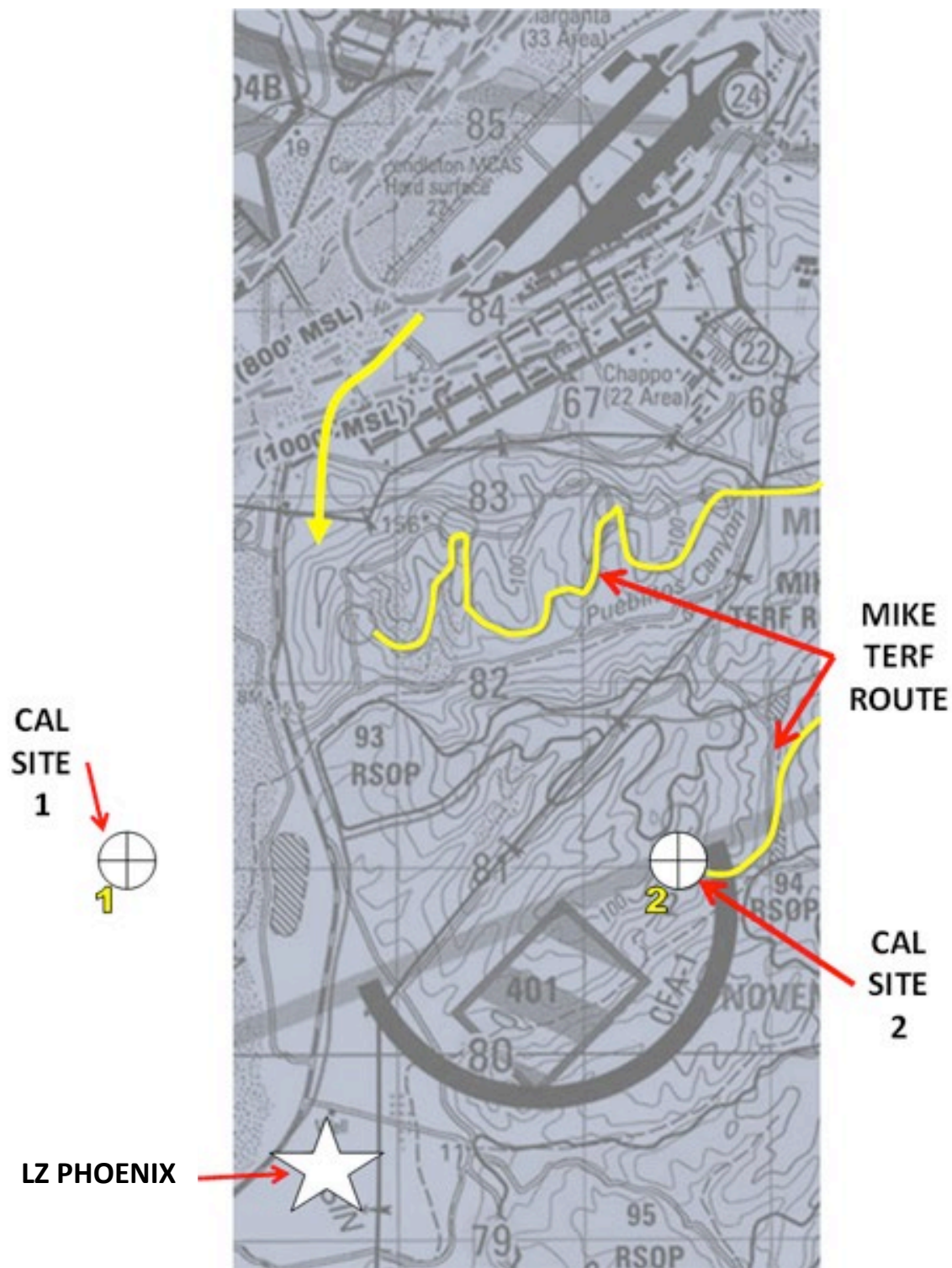
The local flying area for 3rd MAW aircraft is defined as the area bounded by a line from N31°20' W110°35' to Davis Monthan AFB to Williams Gateway Airport to Peach Springs VORTAC to Nellis AFB to NAS Lemoore to Big Sur VORTAC, and includes the airspace within R-2508, W-289, 290, and 291.

FIG C-13
Local Flying Area



16. LZ Phoenix

FIG C-14
LZ Phoenix

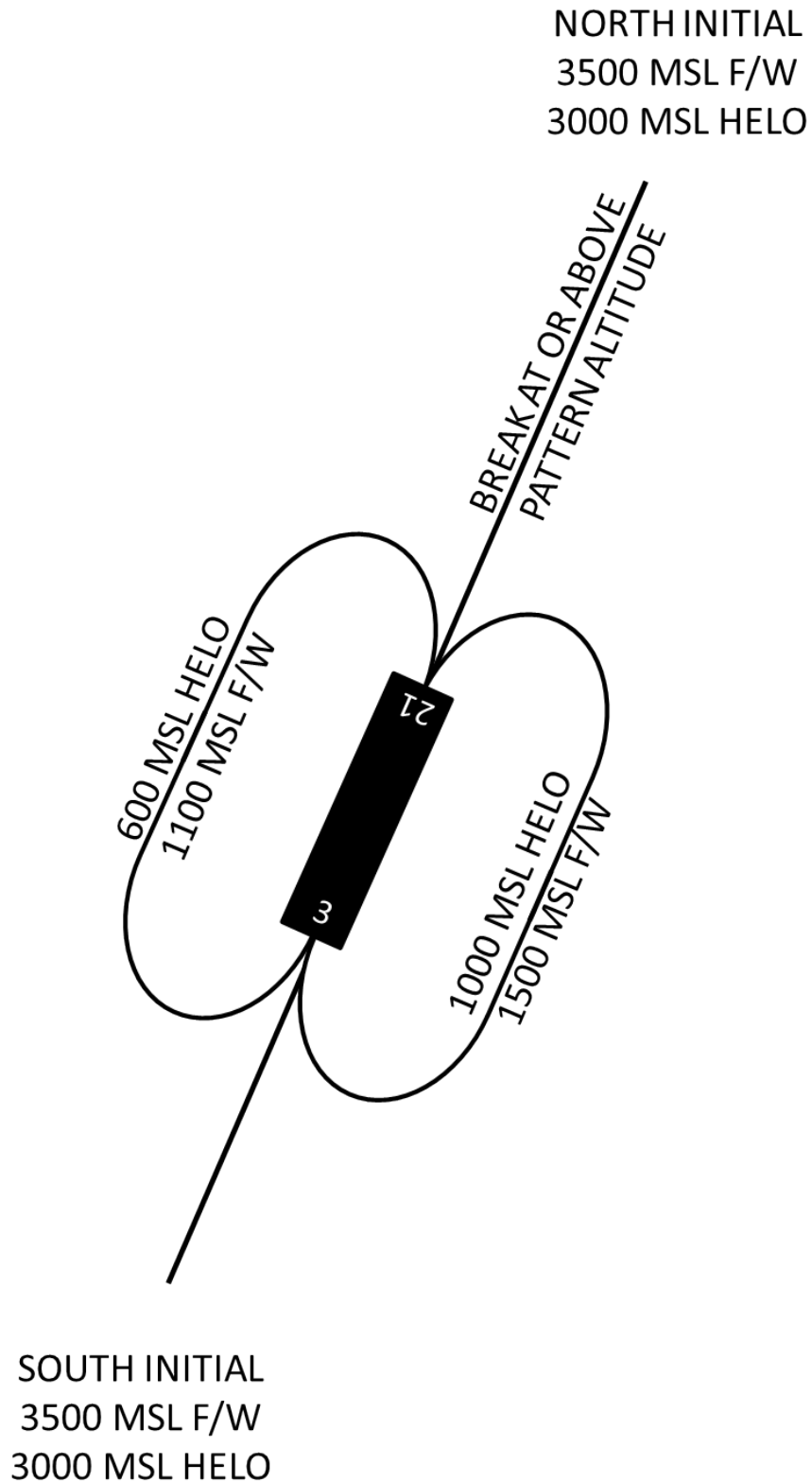


17. ARFF Vehicles

<i>TBL C-4</i> ARFF Vehicles	
CALLSIGN	TYPE VEHICLE
RESCUE 10	PICKUP TRUCK
RESCUE 11	P-10 RESCUE TRUCK (SECTION LEADER)
RESCUE 12	PICKUP TRUCK (OIC)
RESCUE 13	PICKUP TRUCK
RESCUE 14	SUBURBAN
RESCUE 28	WATER TANKER
RESCUE 29	CRANE
RESCUE 50	P-19 (MWSS-372)
RESCUE 56	P-19 (MWSS-372)
RESCUE 58	P-19 (MWSS-372)
RESCUE 59	P-19 (MWSS-372)
RESCUE 63	P-19 (MCAS)
RESCUE 64	P-19 (MCAS)
RESCUE 65	P-19 (MCAS)
RESCUE 66	P-19 (MCAS)
RESCUE 67	P-19 (MCAS)

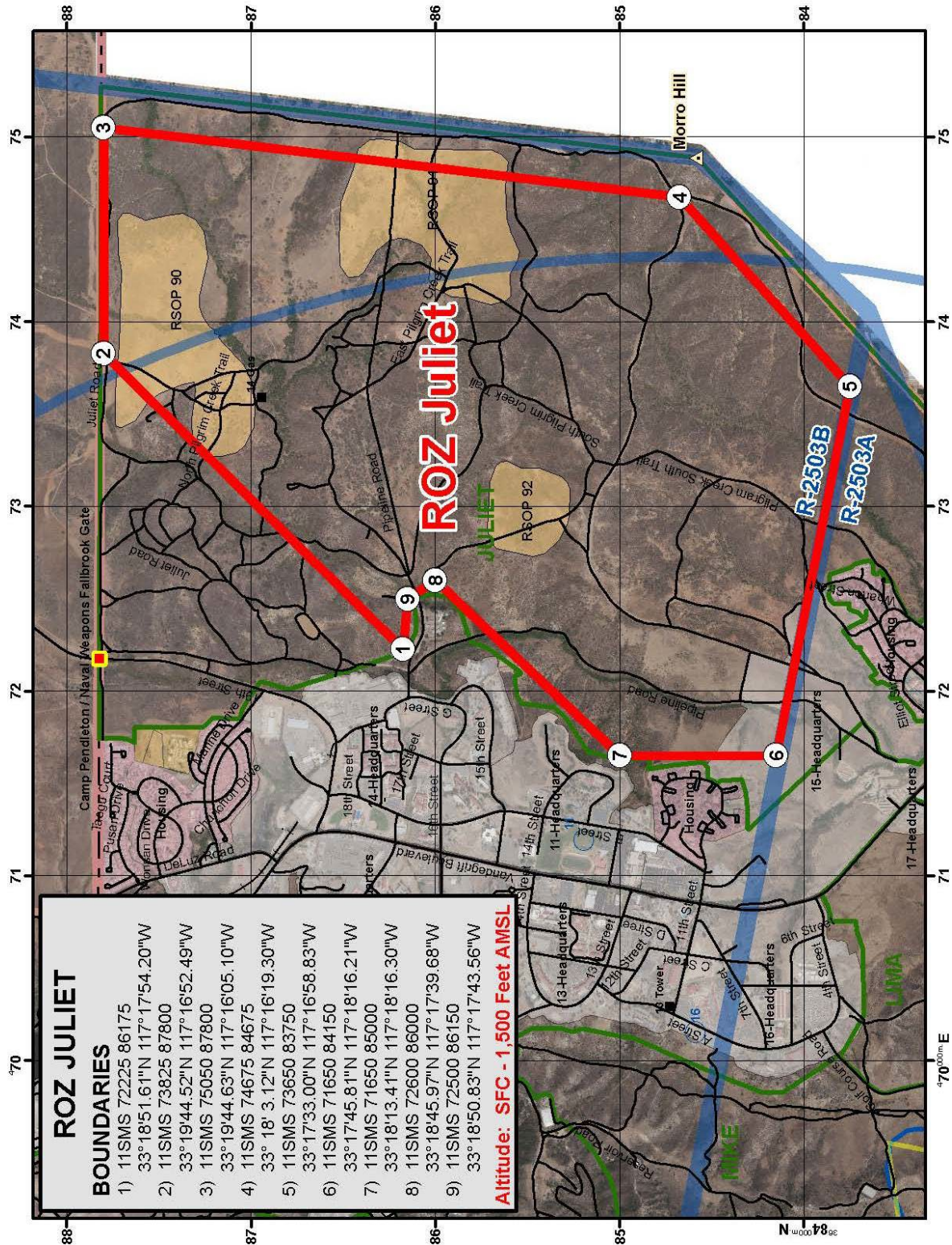
18. Local Traffic Patterns and Altitudes

FIG C-15
Traffic Patterns and Altitudes



19. ROZ Juliet

FIG C-16
ROZ Juliet



Appendix D. FLIGHT STRIPS

1. Control Tower Flight Strips

2. Radar Flight Strips

1. Control Tower Flight Strips

FIG D-1
Tower IFR Flight Strips

IFR DEPARTURE STRIP

CALLSIGN TYPE / EQPT	CODE	DEPT POINT	CLEARANCE LIMIT, ROUTE OF FLIGHT REMARKS	REQ RLS	RLS TIME	CLNC ISSUED
	P-TIME					DEPT TIME
	REQ ALT					FLT PLNG
ATLAS40 H/UH1Y/P	4774	NFG	NFG QUNTN1 QUNTN SKATE NZY ↑ 30 DEPARTURE FREQ IF OTHER THAN PUBLISHED	1900	02	B
	P1900					04
	60					CA

IFR EAST LOCAL VFR-ON-TOP STRIP - VISTA

ATLAS22 H/SUCO/P	7252	NFG	NFG BLDG2 VISTA ↑ $\frac{OTP}{40}$	2303	05	B
	P2300					05
	OTP/40					RP

IFR LIMA VFR-ON-TOP STRIP RWY 21 - CORRI

SWIFT13 G/V22/I	4540	NFG	NFG CORRI3 CORRI ↑ $\frac{OTP}{38}$	1558	1601	B
	P1600					02
	OTP/38					RP

IFR LIMA VFR-ON-TOP STRIP RWY 3 - CORRI

SWIFT22 G/V22/I	4543	NFG	NFG CORRI3 CORRI ↑ $\frac{OTP}{38}$	1732	35	B
	P1730					37
	OTP/38					RP

IFR ARRIVAL STRIP

CALLSIGN TYPE / EQPT	CODE	ETA AT COORD FIX	INTENTIONS REMARKS			
	PREVIOUS FIX					ARR TIME
	COORD FIX					FLT PLNG
STMPD19 H/B350/L	2252	1745	PAFS			
	JLI					1812
	OCN					RP

FIG D-2
Tower VFR Flight Strip

VFR DEPARTURE STRIP

CALLSIGN			REMARKS			
TYPE	P-TIME			DEST	TYPE STRIP	DEPT TIME
						FLT PLNG

STI76						
AH1	P1900			RIV	V D	1906
						CA

FIG D-3
Tower Special VFR Flight Strips

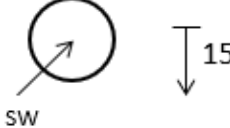
SVFR DEPARTURE STRIP

CALLSIGN TYPE			DEPARTURE ROUTE ALTITUDE RESTRICTIONS REMARKS			CLNC ISSUED
					INITIAL CONTACT	EXIT CLASS D

STI76						B
H/SUCO					1752	1759

SVFR ARRIVAL STRIP

CALLSIGN TYPE			ENTRY ROUTE ALTITUDE RESTRICTIONS REMARKS			CLNC ISSUED
					INITIAL CONTACT	ARR TIME

ATL40						B
H/UH1Y					1936	1945

SVFR PATTERN STRIP

CALLSIGN TYPE			LOCAL SVFR SYMBOL ALTITUDE RESTRICTIONS REMARKS			CLNC ISSUED
					INITIAL CONTACT	ARR TIME

ATL40						B
H/UH1Y					1658	1741

2. Radar Flight Strips

FIG D-4
Radar Departure Strips

IFR DEPARTURE STRIP

CALLSIGN TYPE / EQPT	CODE	DEPT POINT	CLEARANCE LIMIT, ROUTE OF FLIGHT REMARKS	RADAR CONTACT/HANDOFF (IF REQ) DEPARTURE FREQ IF OTHER THAN PUBLISHED	REQ	RLS	CLNC ISSUED
	P-TIME				RLS	TIME	DEPT TIME
	REQ ALT						
ATLAS40	4774	NFG	NFG BLDG2 VISTA V458 BONDO NKX ↑ 40				B
H/H1/P	P1900				1900	02	04
	50						

NOTE: ALTITUDE IS PLACED IN BLOCK 9 WHEN HAND WRITTEN

IFR EAST LOCAL VFR-ON-TOP STRIP - VISTA

CALLSIGN TYPE / EQPT	CODE	DEPT POINT	ROUTE OF FLIGHT, CLEARANCE LIMIT REMARKS	RADAR CONTACT & TERMINATION OR HANDOFF DEPARTURE FREQ IF OTHER THAN PUBLISHED	REQ	RLS	DEPT
	P-TIME				RLS	TIME	TIME
	REQ ALT						
ATLAS22	7252	NFG	NFG BLDG2 VISTA ↑ <u>OTP</u> 40				B
H/H1/P	P2300				2303	05	05
	40						

NOTE: ALTITUDE IS PLACED IN BLOCK 9 WHEN HAND WRITTEN

IFR VFR-ON-TOP STRIP RWY21 - CORRI

CALLSIGN TYPE / EQPT	CODE	DEPT POINT	ROUTE OF FLIGHT, CLEARANCE LIMIT REMARKS	RADAR CONTACT & TERMINATION OR HANDOFF DEPARTURE FREQ IF OTHER THAN PUBLISHED	REQ	RLS	DEPT
	P-TIME				RLS	TIME	TIME
	REQ ALT						
STI26	4543	NFG	CORRI3 CORRI ↑ <u>OTP</u> 38				
H/H1/P	P1815				2303	03	05
	OTP/38						

NOTE: ALTITUDE IS PLACED IN BLOCK 9 WHEN HAND WRITTEN

IFR VFR-ON-TOP STRIP RWY3 - CORRI

CALLSIGN TYPE / EQPT	CODE	DEPT POINT	ROUTE OF FLIGHT, CLEARANCE LIMIT REMARKS	RADAR CONTACT & TERMINATION OR HANDOFF DEPARTURE FREQ IF OTHER THAN PUBLISHED	REQ	RLS	DEPT
	P-TIME				RLS	TIME	TIME
	REQ ALT						
ATLAS22	4540	NFG	CORRI3 CORRI ↑ <u>OTP</u> 38				
H/H1/P	P2300				2303	05	05
	OTP/38						

NOTE: ALTITUDE IS PLACED IN BLOCK 9 WHEN HAND WRITTEN

FIG D-5
Radar Arrival Strips

IFR ARRIVAL STRIP

CALLSIGN TYPE / EQPT	CODE	ETA AT COORD FIX	ALTITUDE	RADAR CONTACT / HANDOFF	TYPE APCH	INTENT	BAY	
	PREVIOUS FIX	ACTUAL IFR						
	COORD FIX	INBOUND PASSED						
STMPD19	2252	1745	50 40 24	C_{SFA}	Ⓡ	TZ	FS	1
I/B350/L	SXC	I						
	OCN	✓		C₇				

PUT A CHECKMARK THROUGH THE TYPE APPROACH ONCE THE AIRCRAFT IS CLEARED.
FOR IFR AIRCRAFT, PLACE I IN BOX 8

IFR MULTIPLE GCA ARRIVAL STRIP - HANDWRITTEN

CALLSIGN TYPE / EQPT	CODE	TIME AT ID FIX	ALTITUDE	RADAR CONTACT / HANDOFF	TYPE APCH	INTENT	BAY	
		ACTUAL IFR			TYPE APCH	INTENT	BAY	
	RADAR ID FIX	INBOUND PASSED			TYPE APCH	INTENT	BAY	
ATLAS40	4534	2022	28 28 28 32 20 24 24	C_{SFA}	Ⓡ	PA	LAR	1
H/H1/P		I			Ⓡ	PA	LAR	2
	L18 3N	✓			R	SA	LAZ	1

PUT A 'C' AROUND THE FREQUENCY ONCE THE AIRCRAFT CHECKS IN ON THAT FREQUENCY.

NON-RADAR ARRIVAL STRIP

CALLSIGN TYPE / EQPT	CODE	TIME AT ID FIX	ALTITUDE	RADAR ID STATUS	TYPE APCH	INTENT	BAY	
	FREQ				TYPE APCH	INTENT	BAY	
	RADAR ID FIX	INBOUND TO TOWER			TYPE APCH	INTENT	BAY	
ATLAS40	4534	2022	50 40 28	CU 1910	✕	PA TZ	FS	1
H/H1/P	④			42 1915				
	L18 3N	✓						

NON-RADAR CONTROL MARKINGS SHOWN IN BLOCK 9 ARE OPTIONAL.

FIG D-6
Other Radar Strips

OVERFLIGHT STRIP

CALLSIGN	CODE	TIME AT ID FIX	ROUTE THROUGH LIMA AIRSPACE			
TYPE			RADAR CONTACT / HANDOFF			
	RADAR ID FIX		ALTITUDE		HANDOFF FACILITY	HANDOFF TIME
N23485	0252	1745	TANNR - CRQ	V R		
PA34						
	TANNR		45			RIV 1756

WRITE A LARGE 'V' IN BLOCK 9 IF THE OVERFLIGHT IS VFR. OTHERWISE, THE OVERFLIGHT IS AN IFR OPERATION.

POINT OUT STRIP

CALLSIGN	CODE	TIME ENTERING LIMA	ROUTE THROUGH LIMA AIRSPACE			
TYPE / EQPT			ALTITUDE		POINTOUT FACILITY	EXIT LIMA
	RADAR ID FIX					
N4282M	0255	2022	OKB - F70	V		
SR22						
	OCN		45			PSCT 2030

WRITE A LARGE 'V' IN BLOCK 9 IF THE POINT OUT IS VFR. OTHERWISE, THE POINT OUT IS AN IFR OPERATION.

FALLBROOK DEPARTURE STRIP

CALLSIGN	CODE	DEPT POINT	ROUTE OF FLIGHT, CLEARANCE LIMIT			
TYPE	P-TIME		CLEARANCE VOID TIME		DEP TIME	EXIT LIMA
	REQ ALT					
N234WC	5646	L18	ROBNN OCN V23 LAX V299 VTU			
2/AV8	P1800		VTU299R SBA		1811	1812
	60		V < 1810			